aselsan

STC-8255

Digital Tachograph



Operation Manual

DTVU-OPE

(76IE-8255-0001)

Revision A

October 2021

aselsan

Declaration of Conformity

This declaration, officially register the confirmation of the product named as

Definition : Aselsan Digital Tachograph Vehicle Unit

Model : STC-8255 v1.0

TOE Version : v1.0

TOE Software Version: v1.0

According to the Council Regulation (EEC) No. 3821/85 (Annex-1B) on recording equipment in road transport (EU) dated 20 December 1985 and its latest ammendments.

Product is evaluated according to the following European Union regulations:

1. Regulation (EC) No. 2135/98
2. Regulation (EC) No. 1360/2002
3. Regulation (EC) No. 561/2006
4. Regulation (EC) No. 1266/2009
5. AETR Agreement.

CE

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Legal Requirements

The use of digital tachographs is administrated by the latest consolidated valid version of the EC regulation (EEC) No. 3821/85 Annex-IB together with EC regulations (EC) no. 561/2006. These regulations assign a number of responsibilities to the driver and the owner of the vehicle (company). It is illegal to misrepresent, deactivate or delete recordings of the digital tachographs and the tachograph cards. Also, it is forbidden to falsify printouts. Modifications of the digital tachograph or its electronic signals or data links, especially if made with the intention of cheating, is forbidden and shall be punishable according to the legal stipulations.

Driver Responsibilities

As the driver, you must ensure proper use of the digital tachograph and the driver card. If the digital tachograph is malfunctioned, you must report the activities that is incorrectly recorded on a separate sheet or on the reverse side of the printout. The notes must be accompanied by your personal hand-written data (name, number of driver card, signature). If your driver card is lost or damaged, or if it malfunctions, you must create a daily report after your journey. This report is to be accompanied by your hand-written data and signature. If your driver card is damaged or malfunctions, submit it to the responsible authorities as soon as possible. Always notify the responsible authorities in your country of the loss of your driver card immediately.



Without inserting a valid workshop card, any disconnection of the vehicle's battery from the vehicle's electrical system will be recorded as a power supply interruption by the digital tachograph. Always have your vehicle battery disconnected or tachograph battery replaced by an authorised workshops and make sure the valid workshop card is inserted.

Company Responsibilities

As company representative, monitor the digital tachograph and the tachograph card to make sure that it functions properly, ensure that the vehicle and calibration data is entered in the digital tachograph by an authorised workshop, ensure that your drivers use the digital tachograph and the drivers' cards properly, instruct your drivers to comply with the driving times and the rest periods.

The company or the tachograph owner has to administrate a workshop inspection of the tachographs and store records from the inspection. It is the owner's or company's responsibility to renew this inspection within every two years.

The STC-8255 digital tachograph must be additionally inspected, when

- the vehicle's tyre circumference has been changed,
- the revolution ratio has been changed,
- the registration (VRN) or identification (VIN) number of the vehicle has been changed,
- the tachograph's UTC time differs more than 20 minutes from the correct UTC time.

Tachograph Version

The operating instructions presented in this operational manual are valid only for the **Aselsan STC-8255 (v1.0)** digital tachograph.

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Safety

Before using your STC-8255 digital tachograph, please read this operation manual.



General Precautions

ALWAYS use only ASELSAN authorized accessories (Download devices, calibration tools, cables and etc.).

Other unauthorized accessories have the risk of fire hazard, explosion, personal injury or damage to the STC-8255 digital tachograph.



Risk of Accident!

For traffic safety reasons, please do not attempt to insert or withdraw the driver card(s) while the vehicle is moving!



Caution!

Changes or modifications to your STC-8255 digital tachograph shall void its compliance with government laws/rules and make it illegal to use.

Avoid using the STC-8255 digital tachograph at temperatures below -25°C or above 80°C.

Avoid storing the STC-8255 digital tachograph at temperatures below -40°C or above 85°C.



Caution!

If the STC-8255 digital tachograph is stored more than 10 years on the shelf without installing to the vehicle, the battery in the tachograph is discharged and the Real Time Clock (RTC) data will be erased because of it. In this case, please have your internal battery re-charged in the authorised workshop and have set the RTC again.



Please read these operating instructions carefully before using your STC-8255 digital tachograph.

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INTRODUCTION

General

The ASELSAN STC-8255 digital tachograph (hereinafter abbreviated as STC-8255) is a secure recording device which records the driving times, work periods, availability periods and rest periods of the driver and co-driver. Additionally, STC-8255 automatically records the speed and distance covered by the vehicle. The securely recorded vehicle and driver related data by the digital tachograph is served as proof for the controlling bodies (e.g. police). The digital tachograph STC-8255 complies with the technical specifications as specified in the Commission regulation (EC) No. 1360/2002 of 13 July 2002 and Commission regulation (EC) No. 1266/2009 of 16 December 2009, the 7th and 10th adaptations of Council regulation (EEC) No. 3821/85 to the technical progress of Recording Equipment in Road Transport respectively.

This user guide is for all STC-8255 users (especially for drivers and company representatives also for workshop staff and control bodies). The user guide describes the proper use of the digital tachograph STC-8255 as specified in the regulation. Read the corresponding sections carefully and familiarise yourself with the operation of the STC-8255 digital tachograph.

Notational Conventions

The following notational conventions are used in this document.



Warning Message

A warning message indicates potential risk of injury or accident.



Attention!

The text nearby this symbol contains critical information to prevent damage to the device, to avoid loss of data or to obey legal obligations.



Hint

The hint sign will give you some guidance, information or clues which is beneficial for you if it is taken into account.



Book

The book sign indicates to a reference to another documentation.

Symbols

* The asterisk mark indicates a special feature or option.

➔ The reference sign indicates that the related issue will be explained detailly in the given pages.

Steps

1. The numbered steps describe actions for the given step by step instructions.
2. Subsequent actions are numbered.

CONFIGURATION

The parts that make up the configuration of STC-8255 Digital Tachograph Vehicle Unit:

1. STC-8255 Digital Tachograph,
2. Printer paper,
3. Quick start guide,
4. Operation manual.

GETTING STARTED



In order to use your STC-8255 digital tachograph, ensure that it is installed, activated and calibrated according to your vehicle specific parameters at the certified workshops.

For the Driver and Co-Driver

1. Before starting your travelling, you must insert your driver card (with chip side up) in to the driver slot (the left-hand slot).
➔ *Please refer to "Inserting driver and co-driver card(s)" on page 19.*

2. Using the "Manual entry" you can add activities to your driver card.
➔ *Please refer to "Manual entries" on page 20.*

3. Use the activity selection button to adjust the activity you want to carry out at that moment.
➔ *Please refer to "Setting activities" on page 23.*

4. Adjust the time to the current local time in your region.
➔ *Please refer to "Local Time" on page 86.*

5. **Important!** During a rest period or break time, always set the activity to "⊢"

6. In case of any faults or events in the device or system components, a relevant warning message will appear on the display. Acknowledge the warning message.
➔ *Please refer to "Messages, Warnings, Faults and Pictogram Combinations" on pages 33, 49, 63, 71 and 87 .*

7. You can print or display the activities from preceeding days as well as saved events, by using the menu functions.
➔ *Please refer to "Displaying and Printing Card Data" on page 31-32.*

8. At the end of the driving or working day, you can withdraw your driver card from the card slot.
➔ *Please refer to "Withdrawing driver card and co-driver cards" on page 40.*

For the Company

1. Register your company within the STC 8255. Insert the company card into any of the card slots.
   ➔ *Please refer to "Registration of the Company (Lock-in)" on page 42.*



Attention!

The company cards must not be used for driving!

2. Withdraw the company card from the card slot at the end of the registration or data download.
   ➔ *Please refer to "De-registration of the Company (Lock-out)" on page 48.*

OVERVIEW

Front Panel

The STC-8255 digital tachograph has the following display and control elements.



Figure 1: STC-8255 front view

1	Alphanumeric display
2	Service (calibration and data download) interface (6-Pin)
3	Keypad **#1** for the driver (driver#1), short push for adjusting the driver activity, long push for driver card withdrawal.
4	Backspace key (backspace & clear or cancel)
5	Card slot #1 for the driver's tachograph card
6	Control key (backward / previous entry)
7	Control key (forward / next entry)
8	**OK** Key (call menu, confirm)
9	Sealing cap (for warranty and security)
10	Keypad **#2** for the co-driver (driver#2), short push for adjusting the co-driver activity, Long push for co-driver card withdrawal.
11	Card slot #2 for the co-driver's tachograph card
12	Printer drawer (for inserting a new roll of paper see section "Paper Roll")

USER INTERACTION

Display

Display used in the STC-8255 is a Liquid Crystal Display (LCD) with backlight. Display has 2 lines with 16 characters shown at each line. Backlight color choice is optional and can be selected prior to ordering and only be assembled at the factory during production.

The display may be used in different ways:

Basically, the display is used to show information to the user. The display is used to navigate in a menu system together with the four buttons (not the numbered ones).

The display is also used as a feedback when entering information into the tachograph or changing settings.

Display Modes

If the tachograph is not in sleep mode and no need to display warnings or no other menu is selected, pressing the control keys ‹ or › changes the default display into following screens.

1. Default display (Speed and odometer)
2. Driver 1 and driver 2 DAGS times at driving state
3. Driver 1 DAGS times
4. Driver 2 DAGS times
5. Driver 1 weekly DAGS times
6. Driver 1 daily DAGS times
7. DAGS extended informations
8. Driver and co-driver activities and local time
9. UTC time

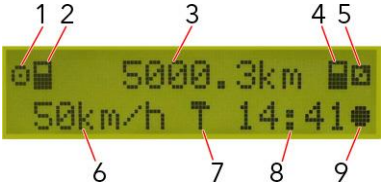
While the vehicle is moving, only the displays #1, 2, 8 and 9 are available for selection.

When the vehicle is stationary, only the displays #1, 3, 4, 8 and 9 are available for selection.

When the vehicle is stationary and the valid card is inserted into the slot-1, all displays, #1, 2, 3, 4, 5, 6, 7, 8 and 9, are available for selection. If reading process of the driver card is not completed, on the display #3, 5, 6 and 7, “-” symbol is shown instead of durations.

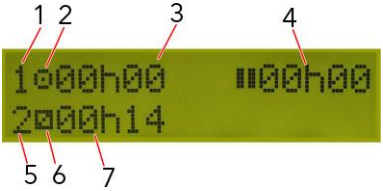
Default Display (Speed & Odometer) (#1)

Normally, (when the ignition is OFF) the tachograph is in sleep (power saving) mode and the display does not show any information in the display window, but a touch of any button, moving of vehicle or turning ignition ON will wake it up (both illumination and text).



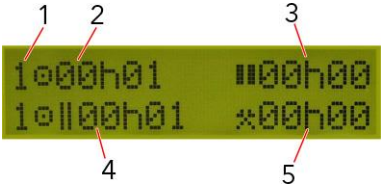
1	Driver 1 activity	6	Speed
2	Driver 1 card symbol	7	Mode of operation
3	Total odometer value	8	Time (UTC time if "●" symbol is not present)
4	Driver 2 card symbol	9	Local time symbol "●"
5	Driver 2 activity		

Driver 1 and Driver 2 DAGS Times at Driving State (#2)



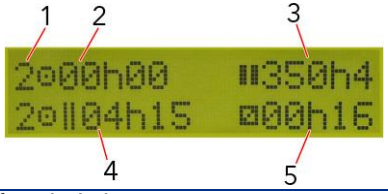
1	Indicates the driver information is shown
2	Driver 1 activity (driving)
3	Driver 1 driving time since last break
4	Valid rest/break time in break periods of at least 15 min. followed by 30 min.
5	Indicates the driver-2 information is shown
6	Driver 2 activity (available)
7	Driver 2 current activity time

Driver 1 DAGS Times (#3)



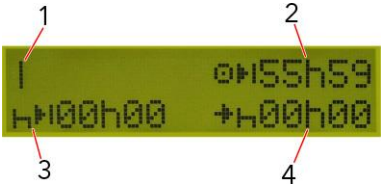
1	Indicates the driver information is shown
2	Driving time since last break/rest
3	Cumulative break/rest time
4	Cumulative driving time during the current and previous week
5	Current activity time

Driver 2 DAGS Times (#4)



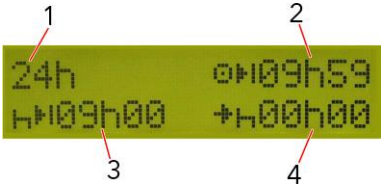
1	Indicates the driver-2 information is shown
2	Driving time since last break/rest
3	Cumulative break/rest time
4	Cumulative driving time during the current and previous week
5	Current activity time

Driver 1 Weekly DAGS Times (#5)



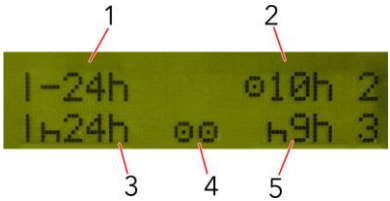
1	Weekly pictogram icon
2	Driver 1 weekly remaining driving time
3	Driver 1 weekly remaining rest time
4	Driver 1 remaining time to next weekly rest period beginning

Driver 1 Daily DAGS Times (#6)



1	Daily pictogram icon
2	Driver 1 daily remaining driving time
3	Driver 1 daily remaining rest time
4	Driver 1 remaining time to daily shift period ending

DAGS Extended Informations (#7)



1	Weekly and daily pictogram icon
2	Number of remaining daily extended driving time
3	Weekly rest limit
4	Crew driving
5	Number of remaining daily reduced rest time

Note That!

DAGS informations mentioned above are in accordance with Regulation (EC) no. 561/2006. For detailed information about rules on driving times, breaks and rest periods for drivers, please refer to Regulation (EC) no. 561/2006.

Driver and Co-driver Activities and Local Time (#8)



In this display mode, following information is shown (first line) from left to right:

- Symbol for driver activity (driving is shown in this case).
- Card symbol indicates that card is inserted to the card slot#1.
- Current local time.
- Symbol for co-driver activity (driving is shown in this case).

In the second line the following information is displayed:

- Current date.
- Symbol for the current mode of operation (in this case OPERATIONAL mode is shown).

UTC Time (#9)



In this display mode, following information is shown (first line) from left to right:

- Pictogram for UTC time.
- Current date

In the second line the following information is displayed:

- Current UTC time.
- Time offset set for the local time.

Special Displays

Activation status

If the STC-8255 has not been activated then the symbol "⊟" will be shown in the display. The STC-8255 will not accept any tachograph cards except the workshop cards.



In order to use your STC-8255 digital tachograph, ensure that it is installed, activated and calibrated according to your vehicle specific parameters at the certified workshops.

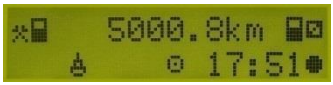
Out of scope

When the vehicle is being driven in countries where tachograph legislation does not exist "Out of Scope" mode could be activated. When it is activated from the menu the display shows the **"OUT"** symbol on the screen.



➔ *Please refer to "Out of Scope Condition" on page 26.*

Ferry transfer / Train crossing



When the vehicle is located on a ferry or on a train, the display shows the symbol "⛴". User action is required to set this functionality from the menu.

➔ *Please refer to "Ferry or Train Condition" on page 40.*

Using the Keypad

OK	OK key	:	Calls the menu, confirms a selection, acknowledges a message or warning.
⟨I	Backspace key	:	Moves back in a menu, clears or cancels the entry, returns to the default display when pressed repeatedly.
⟨	Backward key	:	Moves right in a menu to select the desired function or selection. Decreases a value. Toggles options.
⟩	Forward key	:	Moves left in a menu to select the desired function or selection. Increases a value. Toggles options.
1	Keypad #1	:	Short push for adjusting the driver activity, long push for driver card withdrawal.
2	Keypad #2	:	Short push for adjusting the co-driver activity, long push for co-driver card withdrawal.

Card Slots

There are two card slots where you insert your tachograph cards.



Figure 2: STC-8255 card slots

The card slot for the driver is on the left and card slot for the co-driver is on the right. Each card must be inserted all the way to the stop. The cards are locked mechanically and normally they can only be removed electrically with the help of the user action (long pressing the **keypad #1** and **keypad #2**) when the vehicle is stationary.

Printer

The printer is used to provide printed record of information stored in the tachograph.



Figure 3: STC-8255 printer drawer lid (closed)

Inside the printer drawer there is a printer mechanism that holds printer and a roll of paper. The printed paper will come out from a small opening at the bottom of the printer drawer.

Paper Roll

In order to print the stored information in the tachograph a valid type approved paper rolls (recommended by Aselsan Inc.) should be inserted into the printer drawer. To replace the new paper roll, please obey the following instructions.

1. Press on the upper part of the printer drawer lid as shown in the Figure *4*.



Figure 4: STC-8255 printer drawer lid (closed)

2. The bottom side of the printer drawer will be raised as shown in the Figure 5.



Figure 5: STC-8255 printer drawer lid (opened)

3. Carefully pull out the printer drawer by grabbing from its lower part until it is entirely detached from the tachograph body as shown in the Figure 6. Check if any plastic piece exists in the drawer from the old paper roll. If so, remove this plastic piece from the paper holder.



Figure 6: STC-8255 printer drawer (pulled outside)

4. Insert new roll of paper as shown in the figures below. Keep the feeding part of the paper to be on the upper side as shown in Figure 7 and forward it to pass the lower edge of the drawer as shown in Figure 8.



Figure 7: STC-8255 replacing paper roll-1



Figure 8: STC-8255 replacing paper roll-2

5. Hold the drawer to be the same level with the printer slot and push it gently into the tachograph to close the drawer as shown in the Figure 9, Figure 10 and Figure 11.



Figure 9: STC-8255 replacing paper roll-3



Figure 10: STC-8255 replacing paper roll-4



Figure 11: STC-8255 replacing paper roll-5

6. Pull the paper upwards and tear it off.

Service (Calibration and Data Download) Interface (6-Pin)

The service interface (6-Pin) which is located on the front panel of the STC-8255 digital tachograph is covered with a silicon cover.



Figure 12: STC-8255 6-pin interface

The port has two independent components; a calibration interface and a download interface.

The calibration interface serves the calibration, configuration and diagnostics of the STC-8255 digital tachograph.

The download interface is used to download the data stored in mass memory or on the inserted driver or co-driver smartcard.

Vehicle Dashboard Integration

Some vehicles can display some of the tachograph information on an integrated display. Here are the examples on the type of information that can be displayed:

- Speed
- Total odometer
- Travelled (trip) distance
- Date and Time
- Daily Driving Time
- Driving Time Since Last Break
- Messages, Warnings, System Events and Faults.

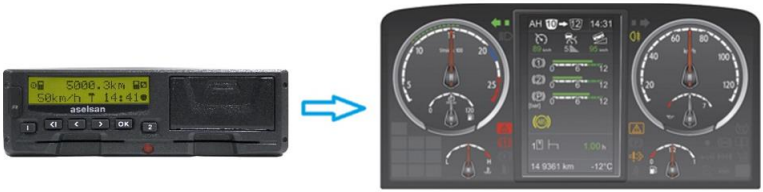


Figure 13: Vehicle dashboard integration

Basic Display Symbols

The following table lists the details of the pictograms in the STC-8255.

Table 1: List of Pictograms

Symbol	Meaning / Description
	Company, company mode
	Controller/inspector, control mode
	Driver, driving, continuous driving time, operational mode
	Workshop/test station, calibration mode
	Manufacturer, tachograph not activated
	Available, current availability period
	Rest, current rest period
	Work, current work period
	Break, Cumulative break time
	Unknown
	Driver card slot
	Co-driver card slot
	Smartcard
	Clock
	Display
	External storage
	Power supply
	Printer/printout
	Motion sensor
	Tyre size
	Vehicle/vehicle unit (tachograph)
	Out of scope (digital tachograph not required)
	Ferry/train crossing
	Events
	Faults
	Start of daily work period
	End of daily work period
	Location
	Manual entry of driver activities
	Security
	Speed
	Time
	Total/summary
	Daily
	Weekly
	Two weeks
	From or to

Menu Tree

Menu structure of the STC-8255 is shown in Figure 14: Menu structure shows the STC-8255 menu structure.

Depending on the mode of the tachograph (which is set when certain tachograph cards are inserted), STC-8255 menu items that can be accessed in the Main menu are changed.

The menu items which are appended "*" symbol before the following number in the figure requires some pre-requisities to be accessed.

*1) Only available in the active tachograph device.

*2) Only available with insertion of driver or workshop card to the driver slot.

*3) Only available with insertion of driver or workshop card to the co-driver slot.

*4) Only available when language change by card and manual is selected from "General configuration".

*5) Only available in the active tachograph device. OUT Begin is only available when out of scope is close, OUT End is only available when out of scope is open.

*6) Only available in the active tachograph device and when out of scope option is close.

*7) Only available if backlight is selected to be arranged by menu.

*8) Only available with existence of status inputs D1-D2 as a hardware configuration.

*9) Only available with existence of GNSS module as a hardware configuration.

*10) Only available in the COMPANY Mode.

*11) Only available (once) in the COMPANY mode if VRN is not entered to the tachograph while tachograph device is activated.

*12) Only available in the CALIBRATION mode.

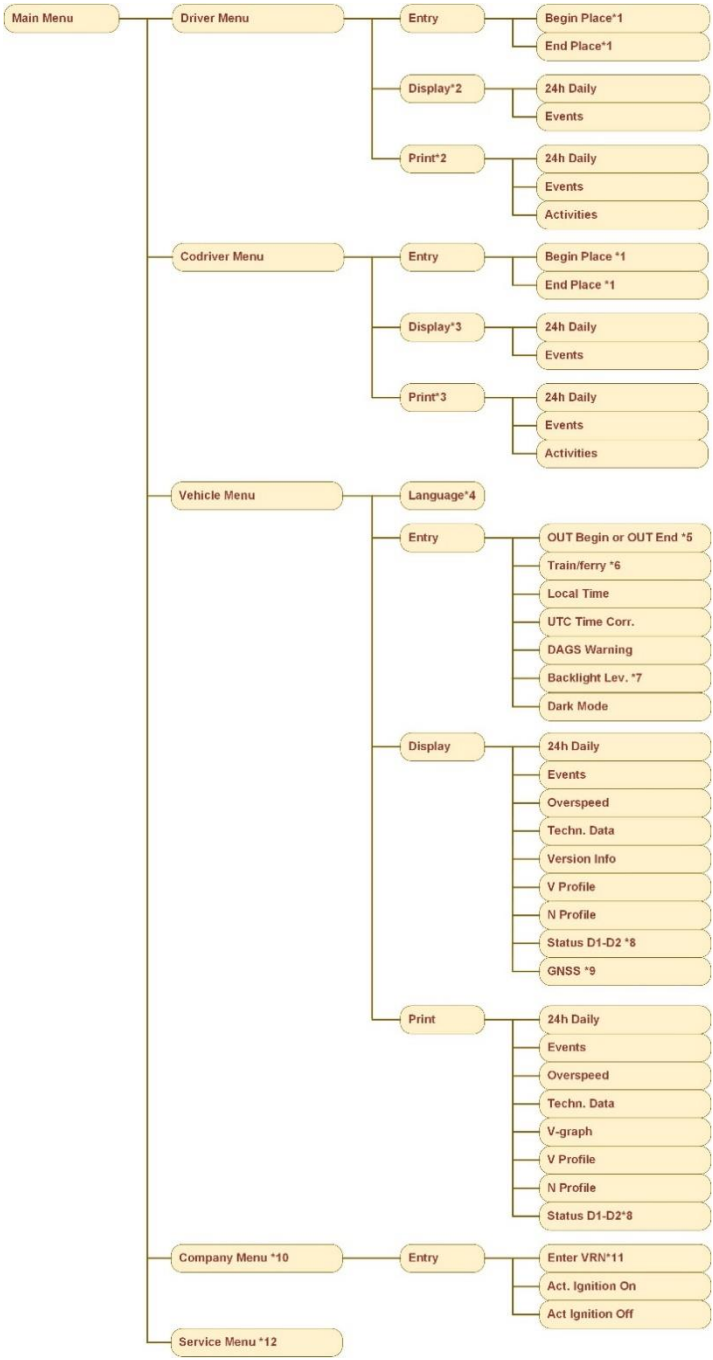


Figure 14: Menu structure

MODE OF OPERATIONS

Users and mode of operations can be found on the following table:

Users	Mode of Operation
Driver and codriver	Operational mode
Workshops	Calibration mode
Controller	Control mode
Company Representative	Company mode
Unknown user	Unknown mode

Operational (Driver and Co-Driver) Mode (Role)

When no card is inserted into the tachograph the mode of operation is still operational mode. However, a driver card must be inserted into the tachograph in order to identify the driver. The driver card is personal and may not be used by anyone else but the rightful card holder.

Inserting driver and co-driver cards



Risk of Accident!

For traffic safety reasons, please do not attempt to insert the driver card(s) while the vehicle is moving!

It is possible to insert the driver card(s) while the vehicle is moving, but this action will be shown on the display and saved as an event to the STC-8255.

➔ *Please refer to "Messages, Warnings and Faults of Operational Mode" on page 33.*

1. Before starting the trip, the driver must insert his/her driver card (with the chip side up and arrow pointing forward) into to driver card slot (left-hand slot). Push the card into the slot until it is locked in the slot (click sound is heard).
2. After the card is read from the tachograph; your name, the date and the time of your last card withdrawal will be displayed on the displayer for a moment.
3. Next, the display shows the addition of manual entry. If so, select **YES** with control keys, press **OK** and follow the manual entry procedure in the following section.
4. If you do not want to enter activities manually, select **NO** by using control keys and press **OK** or press backspace button for shortly.
5. Subsequently, select the country where your working day begins. Use the control keys to select the country (and region if applicable) codes and press **OK.**

➔ *Please refer to "National / Regional abbreviations and European time zones" on page 106."*

6. If a co-driver is present, he/she must insert his/her card into the co-driver card slot (right hand slot) and conduct the same procedure. When the co-driver takes over the vehicle, the cards must be exchanged so that the driver card of the person operating the vehicle is always in the left-hand slot.



Note that.

Menu guidance is carried out according to the language stored on the driver card. You can individually set the preferred language from the settings.

➔ *Please refer to "Language" on page 85.*

Manual Entries

According to the EC regulations, activities, which can not be registered on the driver cards, are to be registered by using manual entry.

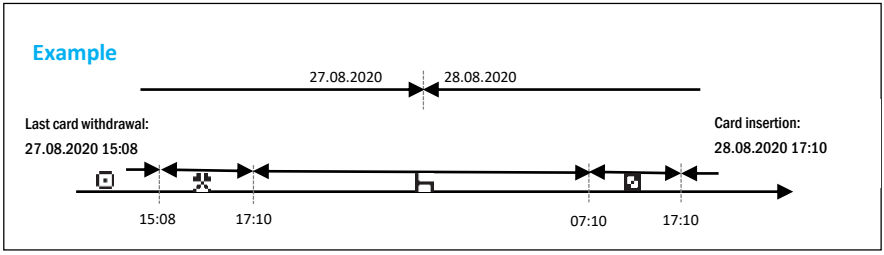
The activities that are not registered to the cards can be registered next time when the driver inserts the driver card into one of the slots.

Working with manual Entry

- Select the desired activity (, , or), function or numeric value with control keys / on the cursor field.
- Confirm the selection with the **OK** button.

Note that.

All activities in the manual entry procedure are entered in local time.

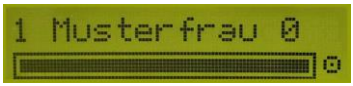


When you insert the driver card into the driver card slot follow the instructions in order to enter manual entries. For the scenario shown on the above example, to register manual entries do as follows:

1. Insert the driver card into driver slot. Greeting text message shown for approximately 3 seconds. Display also shows the current UTC (17:10) and local time (17:10, time offset = 0 hour).



2. The driver's last name is shown on the display. A progress bar indicates that the driver card is being read.

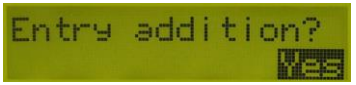


3. Date and time of the last card withdrawal of the driver card is displayed in local time for approximately 3 seconds.



4. Use control keys ◄ / ► to select **YES** then press **OK.**

Note: If you do not want to add any manual entries select **NO** and press **OK.**



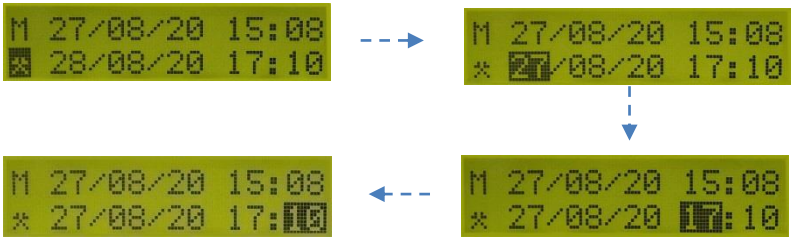
5. The following screen shown on the display is the first manual entry input screen. The first line indicates the date & time of card last withdrawal and the second line indicates the date & time of current card insertion.



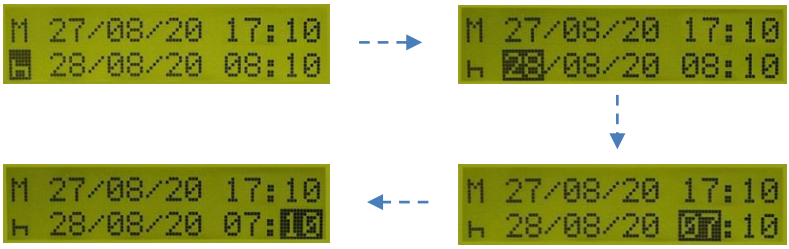
You can successively enter the logically possible variables in the following order:
"Activity – Day – Month – Year – Hour – Minute".
The process ends when the card insertion time is reached.

6. The 1st step of the manual entry example (work from 15:08 to 17:10) is then completed with the following steps. Use control keys ◄ / ► to select the activity (e.g. ⚒ – working) and date & time, then press **OK** to confirm.

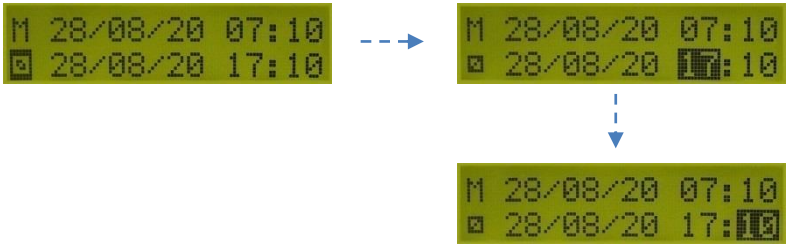
Correct the entries with backspace ◄| button, use control keys ◄ / ► to select the desired activity or value, confirm with **OK** after correction and proceed to the next entry field.



7. The 2nd step of the manual entry example (resting time from 17:10 to 07:10) is then completed with the following steps. Use control keys ◄ / ► to select the activity (e.g. ⊢ – rest) and date & time, then press **OK** to confirm and proceed to the next entry field.
Correct the entries with backspace ◄| button, use control keys ◄ / ► to select the desired activity or value, confirm with **OK** after correction and proceed to the next entry field.



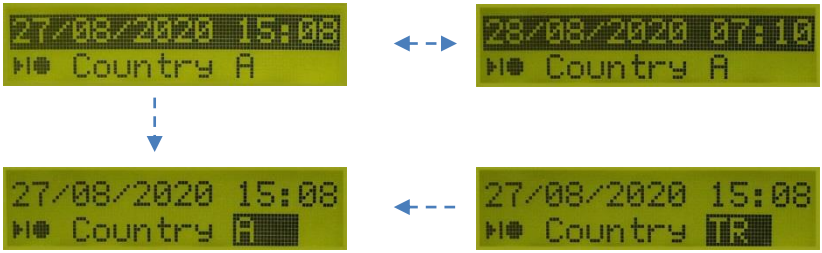
8. The 3rd step of the manual entry example (Availability from 07:10 to 17:10) is then completed with the following steps. Use control keys ◄ / ► to select the activity (e.g. ⧈ – availability) and date & time, then press **OK** to confirm and proceed to the next entry field.
Correct the entries with backspace ◄| button, use control keys ◄ / ► to select the desired activity or value, confirm with **OK** after correction and proceed to the next entry field.



9. Use control keys ◄ / ► (select **YES** than press **OK**) to enter end country during the manual entry (if end country is entered while the previous card withdrawal, following screen is not shown). If selected **YES,** then continue to manual entry procedure with the following step, otherwise jump to the **step 11** for manual entry confirmation.



10. The following screen is shown on the display after selecting **YES** to the previous end place entry question. Select the shift end time by using control keys ◄ / ►, then press **OK** button. After the shift selection, end country can be selected by the using control keys ◄ / ►, then press **OK** button.



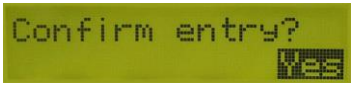
11. Use control keys ◄ / ► (select **YES** than press **OK**) to enter begin country during the manual entry. If selected **YES,** then continue to manual entry procedure with the following step, otherwise jump to the step **13** for manual entry confirmation.



12. The following screen is shown on the display after selecting **YES** to the previous begin place entry question. Select the shift begin time by using control keys ◄ /►, then press **OK** button. After the shift selection, begin country can be selected by using the control keys ◄ /►, then press **OK** button.

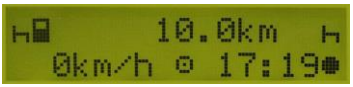


13. Confirm the manual entries with selecting the **YES** by using control keys ‹ / › and press the **OK** button.



It is not possible to change the manual entries after confirmation. If you want to make a correction, please select **NO** and press **OK**. Then, you could start the manual entry procedure from the beginning (**step 5**).

14. After confirmation of the manual entries, the data has been written to the driver cards and card pictogram on the first slot does not blink. The following default display will be shown. It is seen from the display that mode of operation is operational ( ) mode. Now, it is ready to drive.



Setting Activities

Available activities are;

= Driving (Automatically changed while driving the vehicle)

= Work (while working when the vehicle is stationary, e.g. when loading the vehicle)

= Available (waiting times of the co-driver)

= Rest (break times and rest periods)

= Unknown activity (no activity type is recorded, especially used in manual entry)

Manually setting activities



Risk of Accident!

For traffic safety reasons, the activities may be set only if the vehicle is stationary.



Figure 15: Manual activity selection

1. For the driver-1, press shortly to **keypad #1** until the desired activity (, , or), is displayed.
2. For the driver-2, press shortly to **keypad #2** until the desired activity (, , or), is displayed.



Note

Driving is always selected automatically and cannot be changed manually by the drivers.

Automatic setting activities

The tachograph switches to the following activities automatically.

When the vehicle is **moving**:

Driving "⊙" activity is selected for the Driver, and **Available** "⊠" activity is selected for the co-driver.

When the vehicle **stops**:

Work "⚒" activity is selected for the Driver, and **Available** "⊠" is selected for the co-driver.

Important!

At the end of a shift or working day, always set the activity to Rest " ⊢ " after vehicle stops. Otherwise, the STC-8255 will save working time " ⚒ " for driver and available " ⊠ " for the co-driver while the ignition is ON!

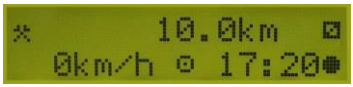
Automatic setting of activity after ignition OFF/ON

The STC-8255 can be configured to switch to a pre-defined activity (e.g. Rest "⊢") automatically after ignition ON/OFF. You can change the settings both for ignition ON and OFF cases for the driver and co-driver.

➔ *Please refer to "Automatic Setting of Activity after Ignition OFF/ON" section, on page 45.*

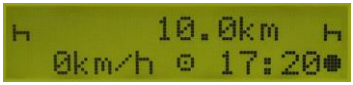
When the vehicle **stops**:

Work "⚒" activity is selected for the Driver, and **Available** "⊠" activity is selected for the co-driver automatically and display will be as follows:



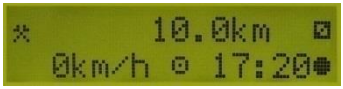
After igniton is **Off**:

The activities can be set automatically to a pre-defined activity (e.g. **Rest** "⊢"). The display shows the activities as shown below.



After igniton is **On** again:

The activities can be set automatically to a pre-defined activity (e.g. **Work** "⚒" and **Available** "⊠"). The display shows the activities as shown below.



- Change the set activity according to your current task, as necessary.

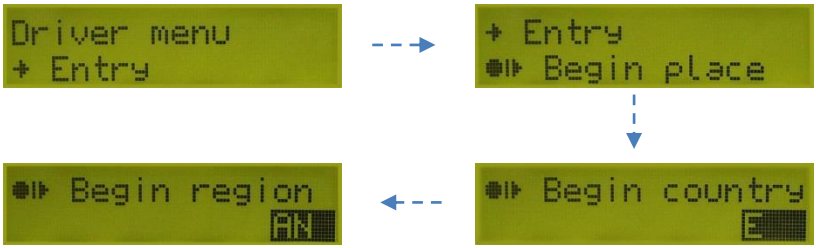
Begin and End Place

When you start and end your working shift tachograph needs to register both begin place or end place of your trip.

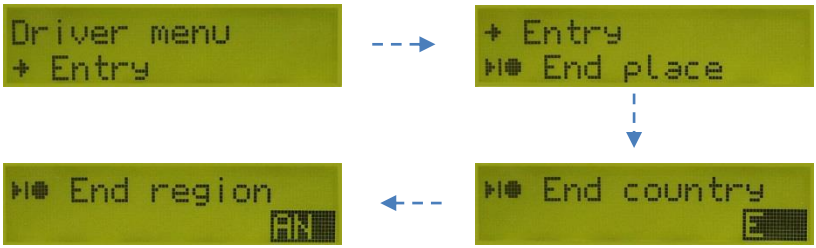
It is possible to enter begin place after card insertion to the slots (during manual entry procedure) or any time during the day. Registering begin place during the manual entry is explained detailly in the "*Operational Mode – Working with Manual entry*" section on page 20.

➔ *Please refer to page 20-23 for the visual explanations.*

1. For the later begin/end place entries during your trip (after manual entry process), follow the below steps. For entering the **Begin Place**:
   - Press **OK** to enter Main Menu. Select **Driver menu>Entry** by using control ◄ / ► keys. Press **OK** and navigate to **Begin place**. Press **OK** to enter the menu. Select the place and press **OK** to confirm. If region place entry is possible, select the region. Press **OK** to confirm.



2. For entering the **End Place**:
   Press **OK** to enter Main Menu. Select **Driver menu>Entry** by using control ◄ / ► keys. Press **OK** and navigate to **End place**. Press **OK** to enter the menu. Select the place and press **OK** to confirm. If region place entry is possible, select the region. Press **OK** to confirm.



- To register begin or end place for the co-driver, perform the same procedure again but navigate to co-driver menu not the driver menu.
- It is possible to enter several begin or end places during the trip. STC-8255 records all of the entries unless they are entered on the same minute. If you enter several begin places at the same minute, STC-8255 will register only the last one to that minute. For some countries (e.g. Spain), STC-8255 may require to enter region, please also give the relevant information by means of control keys and acknowledge it by pressing **OK**.

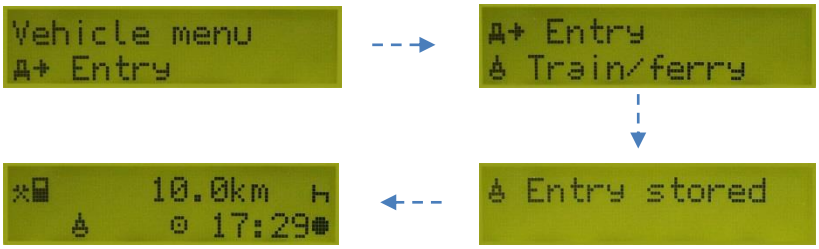
Ferry or Train Condition

When the vehicle is located on a train or on a ferry, entering these special condition times can be registered to the tachograph.

To activate the **ferry/train** condition:

1. Press **OK** to enter Main Menu. Select **Vehicle menu>Entry** by using control ◄ / ► keys. Press **OK** and navigate to **Train/Ferry**. As soon as you pressed **OK**, the train/ferry condition is entered. **Entry**

stored message is displayed. Press the backspace button several times to return standart display.



Train/ferry condition ends when driving activity is stored, driver card is ejected from slot 1, DAGS Ferry/Train rules accordance with regulation (EC) no. 561/2006 are broken or daily rest time is completed while ferry/train condition is active.

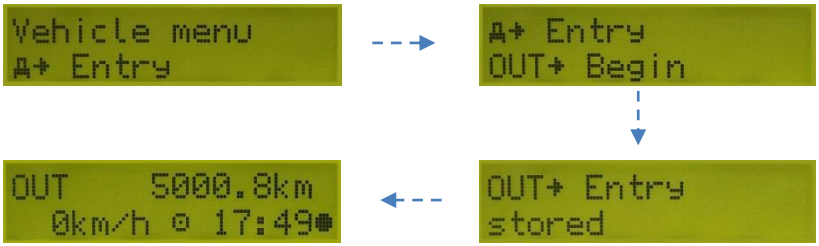
➔ *Please refer to "Ferry transfer / Train crossing display explanations" on page 10.*

Out of Scope Condition

Some driving conditions do not require recording of time on the tachograph, for example driving in countries where no such legislation exists. This driving mode is called **Out of Scope**.

To activate the **Out of Scope** condition:

1. Press **OK** to enter Main Menu. Select **Vehicle menu>Entry** by using control / keys. Press **OK** and navigate to **OUT -> Begin**. As soon as you pressed **OK**, the **Out of Scope Begin** condition is entered. **Entry stored** message is displayed. Press the backspace button several times to return standart display.

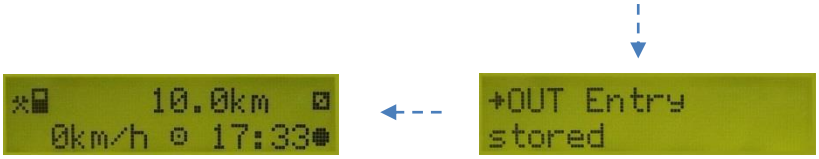


➔ *Please refer to "Out of Scope display explanations" on page 10.*

To de-activate the **Out of Scope** condition:

1. Press **OK** to enter Main Menu. Select **Vehicle menu>Entry** by using control / keys. Press **OK** and navigate to **->OUT End**. As soon as you pressed **OK**, the **Out of Scope End** condition is entered. **Entry stored** message is displayed. Press the backspace button several times to return standart display.





- The **Out of Scope** condition also ends automatically with driver card insertion into card slot or driver card withdrawal from the card slot.

- Exceeding drive time warning messages will not display after 4h15m and 4h30m when the **Out of Scope** condition is activated.

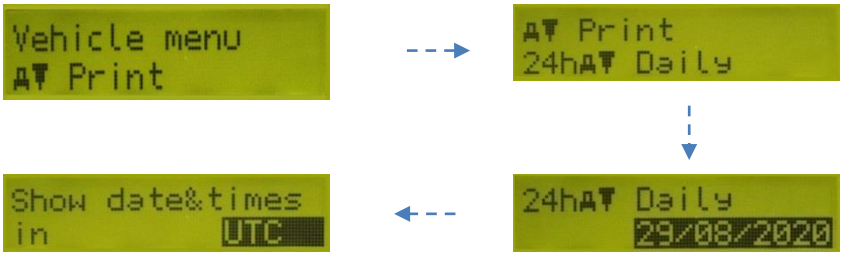
Printing Stored Data

Operator (driver) can print the 24h daily activities and the events data related to the driver card inserted to the first slot. In addition to this, the driver can also print the stored 24h daily activities, events & faults, overspeed, technical data, v-graph, v-profile, n-profile and status D1-D2 of the vehicle.

Here, printing of the vehicle data will be demonstrated.

1. To print **Vehicle menu >24h Daily Data,** perform as follows:

- Press **OK** to enter Main menu. Select **Vehicle menu>Print** by using control ◄ / ► keys. Press **OK** and navigate to **Print 24h Daily**. Press **OK** to enter the menu. Select the date you want to take the printout and press **OK** to confirm. Select the date and time format (UTC or Local) to be printed and press **OK** to confirm. Printing message appears on the display and meanwhile printing job starts. Press the backspace ◄| button several times (press once to return one level back) to return standard display.



➔ *Please refer to "24h (Daily) Vehicle Unit Printout" explanations given on page 92.*

2. To print **Vehicle menu > Events&Faults Data,** perform as follows:

- Press **OK** to enter Main menu. Select **Vehicle menu>Print** by using control ◄ / ► keys. Press **OK** and navigate to **Print Events**. Press **OK** to enter the menu. Select the date and time format (UTC or Local) to be printed and press **OK** to confirm. Printing message appears on the display and meanwhile printing job starts. Press the backspace ◄| button several times (press once to return one level back) to return standard display.

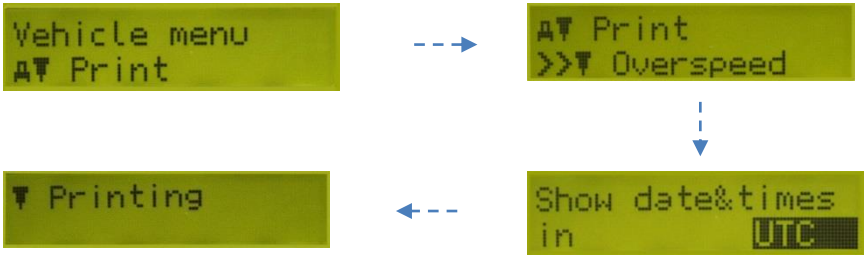




➔ *Please refer to the "Events & Faults Vehicle Unit Printout" explanations given on page 94-105.*

3. To print **Vehicle menu > Overspeed Data,** perform as as follows:

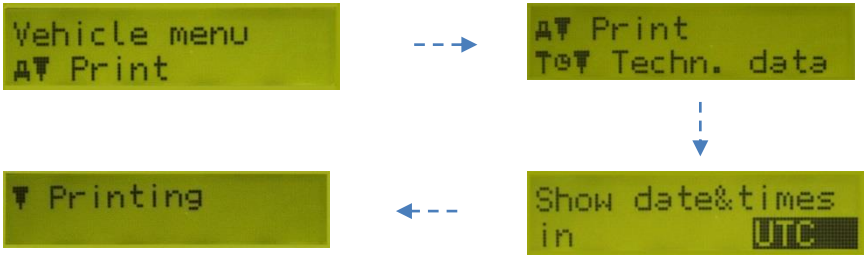
- Press **OK** to enter Main menu. Select **Vehicle menu>Print** by using control ◄ / ► keys. Press **OK** and navigate to **Print Overspeed**. Press **OK** to enter the menu. Select the time format (UTC or Local) to be printed and press **OK** to confirm. Printing message appears on the display and meanwhile printing job starts. Press the backspace ◄I button several times (press once to return one level back) to return standard display.



➔ *Please refer to "Vehicle Overspeed Printout" explanations given on page 98.*

4. To print **Vehicle menu > Technical Data,** perform as as follows:

- Press **OK** to enter Main menu. Select **Vehicle menu>Print** by using control ◄ / ► keys. Press **OK** and navigate to **Print Technical Data**. Press **OK** to enter the menu. Select the date/time format (UTC or Local) to be printed and press **OK** to confirm. Printing message appears on the display and meanwhile printing job starts. Press the backspace ◄I button several times (press once to return one level back) to return standard display.

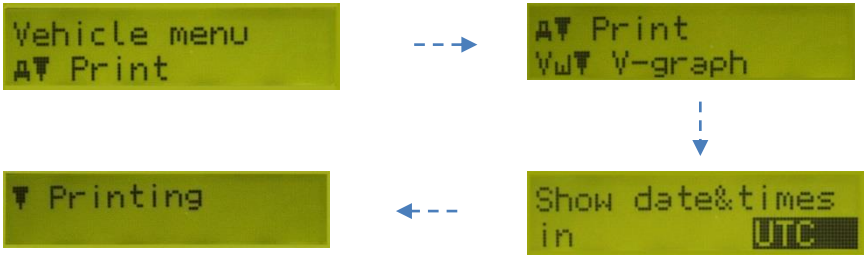


➔ *Please refer to "Vehicle Technical Data Printout" explanations given on page 104.*

5. To print **Vehicle menu > V-graph,** perform as as follows:

- Press **OK** to enter Main menu. Select **Vehicle menu>Print** by using control ◄ / ► keys. Press **OK** and navigate to **Print V-graph**. Press **OK** to enter the menu. Select the date/time format (UTC or

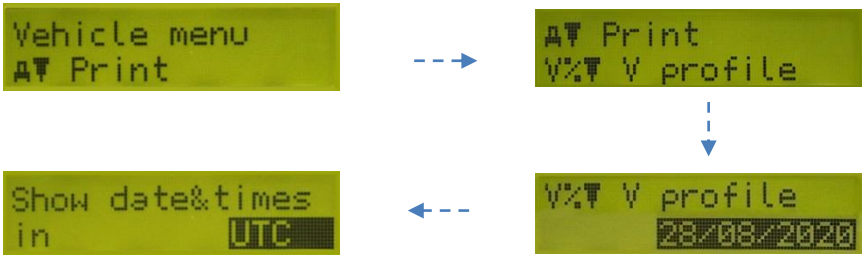
Local) to be printed and press **OK** to confirm. Printing message appears on the display and meanwhile printing job starts. Press the backspace button several times (press once to return one level back) to return standard display.



➔ *Please refer to "V-graph Printout" explanations given on page 103.*

6. To print **Vehicle menu > V-profile,** perform as follows:

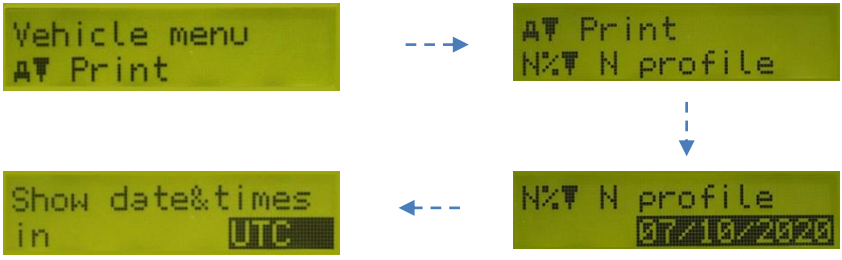
- Press **OK** to enter Main menu. Select **Vehicle menu>Print** by using control / keys. Press **OK** and navigate to **Print V-profile**. Press **OK** to enter the menu. Select the date/time format (UTC or Local) to be printed and press **OK** to confirm. Printing message appears on the display and meanwhile printing job starts. Press the backspace button several times (press once to return one level back) to return standard display.



➔ *Please refer to "V-profile printout" explanations given on page 100.*

7. To print **Vehicle menu > N-profile,** perform as follows:

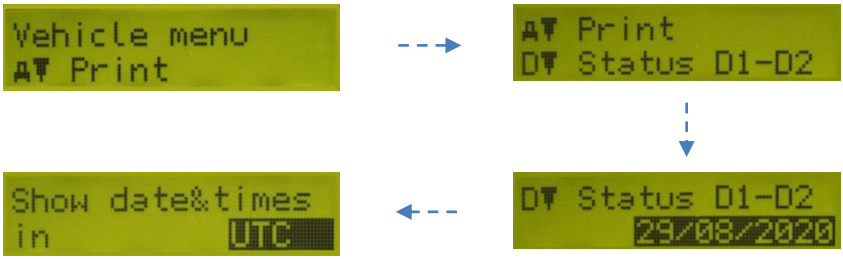
- Press **OK** to enter Main menu. Select **Vehicle menu>Print** by using control / keys. Press **OK** and navigate to **Print N-profile**. Press **OK** to enter the menu. Select the date/time format (UTC or Local) to be printed and press **OK** to confirm. Printing message appears on the display and meanwhile printing job starts. Press the backspace button several times (press once to return one level back) to return standard display.



➔ *Please refer to "N-profile printout" explanations given on page 99.*

8. To print **Vehicle menu > Status D1-D2,** perform as follows:

- Press **OK** to enter Main menu. Select **Vehicle menu>Print** by using control ‹ / › keys. Press **OK** and navigate to **Print Status D1-D2**. Press **OK** to enter the menu. Select the date/time format (UTC or Local) to be printed and press **OK** to confirm. Printing message appears on the display and meanwhile printing job starts. Press the backspace ◄ button several times (press once to return one level back) to return standard display.



➔ *Please refer to "Status D1-D2 printout" explanations given on page 101.*

Downloading Driver Card Data

Driver cards hold driver specific information (e.g. driver daily activities, total km driven for each day, vehicles used and etc.) on the driver cards. During normal driving operation, the driver's daily activities are saved on the driver card for at least 28 days. When the memory of the card is full, the information will be overwritten by new information and thereby it will be lost forever. To avoid this and to secure card data a frequent download of information is needed. It means that data will be transferred from the card to a data storage outside the vehicle. Data has to be downloaded regularly.

Note that!

Downloading will not destroy or erase any data of the cards. Data is only erased from cards when it is overwritten by new data.

Tachograph driver cards may be downloaded in two ways.

1. Downloading the driver cards over the tachograph device with data download equipment (sold separately in the market and compliant with the protocol defined in the latest amended legislative document 3821/ 85 Annex-1B, Appendix-7). In this mode, the driver cards should be in the first slot and the mode of the tachograph should be operational mode.
2. Directly download the cards with data download equipments (available on the market). Many types of equipments can download cards directly by inserting them in a card slot on the equipment in question.

To download the driver cards over the STC-8255 is the issue of this document. For the second method, please refer to the data download equipment's user guide.

To explain card download over the STC-8255;

1. Open the silicon cover in front of the tachograph.
2. Connect the download equipment's 6-pin connector to the tachograps's 6-pin interface behind the silicon cover.
3. Start downloading data according to the instructions on the download equipment.
4. When downloading starts a flashing animation is played on the second line of the display, showing the downloading is in progress.



5. When the downloading is finished animation on the display stops.
6. Get the digitally signed data which identifies the driver card and store it in a data storage media (flash disk, hard disk, CD-ROM or SD-card).

Please Note That!

Only the driver card in the first slot will be downloaded if both slots are occupied with driver cards. Please do not disconnect the data connection to the digital tachograph before the download is complete.

Note That!

For detailed information about the card data download, please refer to the download equipment's appropriate documentation.

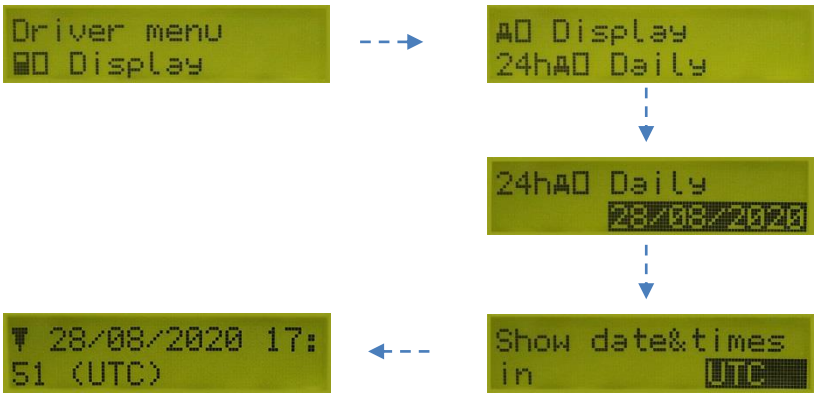
Note That!

To download card data over tachograph, the STC-8255 should not be in sleep (power saving) mode. If so, press any button firstly to wake the STC-8255 up or switch the ignition key ON before the process.

Displaying Card Data

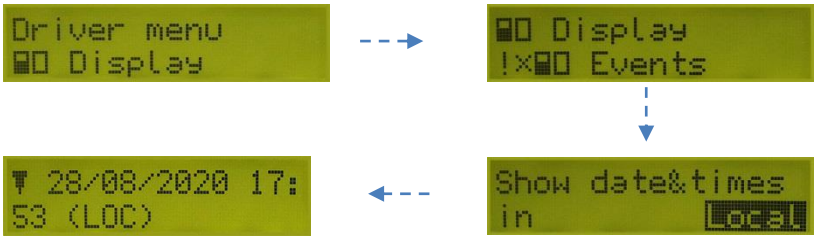
The driver can view his/her inserted driver card information on the display. Both drivers can display their cards' information under the relevant menus. Driver can view the 24h Daily activities data and the events data related to his/her card.

1. To display card daily activity data, perform as follows:
   - Press **OK** to enter Main menu. Select **Driver menu>Display** by using control ‹ / › keys. Press **OK** and navigate to **Display 24h Daily**. Press **OK** to enter the menu. Select the date to be displayed and press **OK** to confirm. Select the time format (UTC or Local) to be displayed and press **OK** to confirm. Scroll with control ‹ / › keys to view all of the data. Press the backspace ‹| button several times (press once to return one level back) to return standard display.



2. To display card events data, perform as follows:

- Press **OK** to enter Main menu. Select **Driver menu>Display** by using control ◄ / ► keys. Press **OK** and navigate to **Display Events**. Press **OK** to enter the menu. Select the time format (UTC or Local) to be displayed and press **OK** to confirm. Scroll with control ◄ / ► keys to view all of the data. Press the backspace ◄ button several times (press once to return one level back) to return standard display.



Note That!

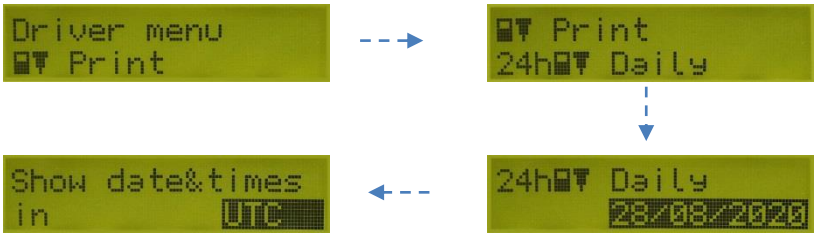
The data appear on the display is similar to how it appears on the printout; however one printout line (24 characters) will be divided into two lines.

Displaying the co-driver's card information is similar to the driver one, therefore will not be explained again. You can display the co-driver's card information under the co-driver display menu.

Printing Card Data

The driver can print out the data of his/her inserted driver card. Both drivers can print their cards' data under the relevant menus. Driver can print the 24h Daily activities data, events data and activity graph related to his/her card.

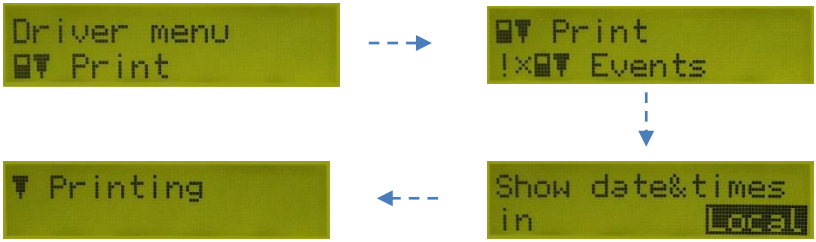
1. To print card daily activity data, perform as follows:
   - Press **OK** to enter Main menu. Select **Driver menu>Print** by using control ◄ / ► keys. Press **OK** and navigate to **Print 24h Daily**. Press **OK** to enter the menu. Select the date you want to take the printout and press **OK** to confirm. Select the time format (UTC or Local) to be printed and press **OK** to confirm. Printing message appears on the display and meanwhile printing job starts. Press the backspace ◄ button several times (press once to return one level back) to return standard display.



➔ *Please refer to "24h (Daily) Driver Card Printout" explanations given on page 90.*

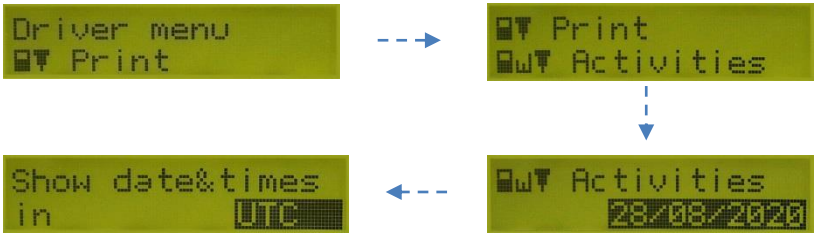
2. To print card events data, perform as follows:
   - Press **OK** to enter Main menu. Select **Driver menu>Print** by using control ◄ / ► keys. Press **OK** and navigate to **Print Events**. Press **OK** to enter the menu. Select the time format (UTC or

Local) to be printed and press **OK** to confirm. Printing message appears on the display and meanwhile printing job starts. Press the backspace button several times (press once to return one level back) to return standard display.



➔ *Please refer to Events & Faults Driver Card Printout explanations given on page 97.*

3. To print card activity graph data, perform as follows:
   - Press **OK** to enter Main menu. Select **Driver menu>Print** by using control / keys. Press **OK** and navigate to **Print Activities**. Press **OK** to enter the menu. Select the date of the activity you prefer then select time format (UTC or Local) to be printed and press **OK** to confirm. Printing message appears on the display and meanwhile printing job starts. Press the backspace button several times (press once to return one level back) to return standard display.



➔ *Please refer to Activitiy Printout explanations given on page 102.*

Printing the co-driver's card data is similar to the driver one, therefore will not be explained again. You can print the co-driver's card data under the co-driver print menu.

Acknowledge Event or Fault Messages

When certain events or faults occur, STC-8255 displays a warning message with an alert tone at the beginning of the warning message. While the warning message appears on the display, plain text of the message will flash for about **30 seconds** or until they are confirmed witk **OK** button by the driver. After confirmation with any button on the front panel or 30 seconds timeout the flashing of the text message will stop immediately. Press **OK** button again to acknowledge the message then the message disappears and the standard display will appear again.

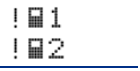
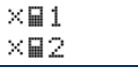
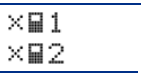
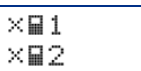
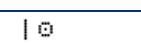
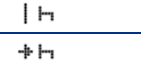
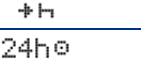
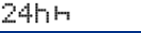
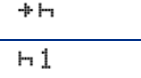
Messages, Warnings and Faults of Operational Mode

There are four types of messages that can be shown on the display.

- **Messages –** give information on the process or remind the driver. Messages can not be stored and printed. Press the backspace button or **OK** button to clear the message.

- **Pre-warnings –** appear as early reminders to the warnings. Pre-warnings are not stored and can not be printed. Press **OK** button twice to clear the Pre-warning.
- **Warnings (Events) –** appear in the event of e.g. overspeed or violation of the law or if the tachograph can not be recording. Warnings are stored and can be printed. Press **OK** button twice to acknowledge the Warning.
- **Faults –** are more critical than warnings and are displayed if there is a fault detected in the tachograph, in the sensor or cards. On the other hand, fault messages are shown if the device detects a tampering. Faults are stored and can be printed. Press **OK** button twice to acknowledge the fault.

Table 2: List of Operational Mode Messages

Pictogram & Message ID	Message	Explanation	User Action
 1	"non-valid card"	A non-valid or expired card has been inserted to one of the slots.	Acknowledge the message. Eject the non-valid or expired card if it is not ejected automatically.
 134	"Wrong card type"	Displayed if the inserted card into the tachograph is not a tachograph card. The card inserted is not accepted and ejected automatically.	Please insert a valid tachograph card.
 135	"PIN is blocked"	Displayed when workshop card is blocked because of wrong card pin entries.	Acknowledge the message. Apply for the new service card to "STAUM". https://staum.org.tr.
 136	"Recording Inconsistent"	Displayed if there is an inconsistency with the driver or workshop card records while inserting the cards to any slot.	Acknowledge the message.
 144	"Upgrade failed"	Displayed when software upgrade is failed.	Acknowledge the message.
 	"Drive Cont." xxhxx	Displayed if the weekly allowable driving time is exceeded.	Acknowledge the message. Give a break.
 	"Rest Cont." xxhxx	Displays the remaining time for the upcoming weekly rest period.	Acknowledge the message. Prepare for the rest period.
 	"Drive Cont." xxhxx	Displayed if the daily allowable driving time is exceeded.	Acknowledge the message. Give a break.
 	"Rest Cont." xxhxx xxhxx	Displays the remaining time for the upcoming daily rest period.	Acknowledge the message. Prepare for the rest period.
 	"Give a break" xxhxx xxhxx	Displayed after 4h15m or 4h30m continuous driving time.	Acknowledge the message. Give a break.
	"Drive Cont."		Acknowledge the message. Give a break.

Pictogram & Message ID	Message	Explanation	User Action
	“xxhxx”	Displayed if the 2 weekly allowable driving time is exceeded.	
171	“No personalization”	Displayed if the card is inserted into the not personalized tachograph.	Acknowledge the message. Have the STC-8255 checked by authorized workshop.
172	“Tachograph is not ready”	Displayed if the tachograph is not ready when the card is inserted into the tachograph.	Acknowledge the message. Try again after 1 minute. Have the STC-8255 checked by authorized workshop if this message is displayed frequently.
174	“Eject driver card”	Displayed if there is a valid driver card in any slots while the ignition is turning to OFF.	Acknowledge the message. Do not forget to withdraw your driver card from the tachograph.
174	“Veh. is not rdy to driv.”	Displayed if the ignition is ON and there is no valid driver card in the first slot or there is a smart card rather than driver card in the second slot.	Acknowledge the message. Do not forget to insert your driver card to the first slot before you start your trip.
	“Power is insufficient”	Displayed if the power is insufficient when the printing process is started or during the printing.	Acknowledge the message. Have your vehicle battery checked in the authorized services.
	“Card is not ready”	Displayed if the card reading is not finished for both card slots before the printing process is started.	Acknowledge the message. Wait for a while for the card reading process.
	“Stop to print”	Appears when the vehicle moves while the printer is busy with printing.	Stop the vehicle and try to print again when the vehicle is stationary.
	“No paper”	Displayed when the printer is out of paper at the beginning of the printing task.	Load new roll of paper to the printer drawer.
	“No data”	Displayed when there is no data for the requested printout type.	Acknowledge the message.
	“Thermal stress”	Displayed if the printer is too hot and the user requests to print any report.	Acknowledge the message. Please wait for a while for the printer head to cool down and try again later.
	“Thermal stress” “Cancel”	Displayed if the printer has continuously printed and the printer head is too hot to continue. Disappears when the user pushes the backspace button or the print head cools down.	Wait for the printer head cools down and printer continues printing job or press backspace to cancel the job. Please do not touch the printer head with your hand when this message appears.
	“Load paper” -Progress Bar-	Displayed when the printer is out of paper while printing and displayed until new printer paper roll is inserted (45 sec period).	Insert new printer paper roll.
	“Printer is ready” -Progress Bar- “Cancel printing”	Displayed after the new roll of paper is loaded to the printer drawer.	Load new roll of paper to the printer drawer.

Pictogram & Message ID	Message	Explanation	User Action
	Yes – No Dialog	Displayed when the printer is not loaded with a new roll of paper in a certain period (45sec).	Press YES to cancel the printing job.
	“not possible”	Displayed when the user is trying to enter the menu while the vehicle is moving.	Stop the vehicle if necessary to navigate in the menus.
	“Printing”	Appears when the printer is busy with printing.	No user action required.
	“Ejection is not possible”	Displayed if the cards inserted to the tachograph slots are requested while the battery supply level is too low.	Acknowledge the message. Have your vehicle battery checked in the authorized services.
	“Ejection is not possible”	Displayed if the cards inserted to the tachograph slots are requested while the vehicle is moving.	Please stop the vehicle and then request the card withdrawal from the tachograph.
	“UTC correct. not possible”	Displayed if you are trying to correct the UTC time which has already been corrected within the last 7 days	No user action required.
	“Wait for ejection”	Displayed if the cards inserted to the tachograph slots are requested while the inserted activity is recorded to the tachograph cards.	Acknowledge the message. Please try again one minute later.

Table 3: List of Operational Mode Pre-warnings

Pictogram & Message ID	Pre-Warning Messages	Explanation	User Action
>> 129	“Oversp. prev. warning”	The vehicle has exceeded the overspeed limit which is set for that vehicle. If exceeding continues for 30 sec. (or pre-defined value) the overspeed event will be recorded into tachograph memory.	Acknowledge the message. Reduce the vehicle speed.
130	“Card will expire soon”	Means expiration prewarning day limit of card inserted to the tachograph slot is reached and this card will be expired soon.	Acknowledge the message. Contact the responsible authority to get new one.
131	“Time for service”	Displayed if the calibration pre-warning date is passed and reminds the driver.	Acknowledge the message. Have the STC-8255 checked and calibrated by authorized workshop.
146	“Download card”	Displayed if the card download pre-warning date is passed and reminds the driver to download his/her card to external media.	Acknowledge the message. Download the driver card to any external media.
147	“Download vehicle unit”	Displayed if the tachograph download pre-warning date is passed and reminds the driver to download his/her tachograph to external media.	Acknowledge the message. Download the tachograph mass memory to any external media.

Table 4: List of Operational Mode Warnings

Pictogram & Message ID	Warning (Event) Messages	Explanation	User Action
2	“Card conflict”	Displayed if two tachograph cards, incompatible with each other, are inserted to the tachograph.	Acknowledge the message. Remove one of the cards (which is useless) from the card slot.
3	“Time overlap”	The last withdrawal date/time of the inserted driver card from previous tachograph is later than the date/time of recently inserted tachograph.	Acknowledge the message. Check the date/time of the tachographs used and determine the tachograph with the incorrect UTC time, make the tachograph time to be corrected by authorized workshop. Or, wait for the overlap period to be elapsed.
4	“Driving w/o valid card”	Driving started with an inappropriate card or with an inappropriate card combination.	Acknowledge the message. Stop vehicle and insert a valid driver card or remove inappropriate card from co-driver slot.
5	“Card insertion while driving”	A driver card was inserted after driving has begun.	Acknowledge the message. No further action required.
6	“Card closed incorrectly”	Card was not removed properly from the last tachograph or data could not be saved to the card correctly.	Acknowledge the message. No further action required.
7	“Over speeding”	Driver exceeded the permissible max speed limit for longer than 1 minute.	Acknowledge the message. Reduce vehicle speed.
8	“Power supply interruption”	The power supplied to the STC-8255 is below or above the certain limit more than 200 miliseconds or has been disconnected for more than 200 miliseconds. The event is not triggered in Calibration Mode.	Acknowledge the message. Have the STC-8255 checked by authorized workshop.
9	“Motion sensor data error”	The data link failure occurred between the motion sensor and the tachograph.	Acknowledge the message. Have the STC-8255 checked by authorized workshop.
10	“Motion conflict”	Displayed if the second source of motion data (GNNS or CAN-Bus) is not consistent with the primarily used motion sensor data.	Acknowledge the message. Have the STC-8255 checked by authorized workshop.
16	“No further details”	Displayed if an unexpected error occurs while authenticating the motion sensor.	Acknowledge the message. Have the STC-8255 checked by authorized workshop.
17	“Motion sensor auth. failure”	Displayed if the tachograph does not authenticate with the motion sensor	Acknowledge the message. Have the STC-8255 checked by authorized workshop.
	“Card auth. Failure”		

Pictogram & Message ID	Warning (Event) Messages	Explanation	User Action
18		Displayed if the tachograph does not authenticate with the smart card.	Acknowledge the message. Have the STC-8255 checked by authorized workshop.
!  19	“Unauthorised change of MS”	Displayed if unpaired motion sensor is linked to the tachograph paired with another motion sensor.	Acknowledge the message. Have the STC-8255 checked by authorized workshop.
!  20	“Card data integ. err.”	Displayed if smart card data is corrupted.	Acknowledge the message. Have the STC-8255 checked by authorized workshop.
!  21	“Data integ. err.”	Displayed if the tachograph data is corrupted.	Acknowledge the message. Have the STC-8255 checked by authorized workshop.
!  24	“Hardware sabotage”	Displayed if card(s) ejection by force is detected.	Acknowledge the message. Have the STC-8255 checked by authorized workshop.
!  32	“No further details”	Displayed if a generic error occurs in the motion sensor.	Acknowledge the message. Have the STC-8255 checked by authorized workshop.
! 33	“Auth. Failure”	Displayed when error related to authentication occurs in the motion sensor.	Acknowledge the message. Have the STC-8255 checked by authorized workshop if this fault occurs frequently.
! 34	“Data integ. err.”	Displayed when error related to non-volatile memory occurs in the motion sensor.	Acknowledge the message. Have the STC-8255 checked by authorized workshop if this fault occurs frequently.
!  37	“Hardware sabotage”	Displayed when error related to sensor element or temperature occurs in the motion sensor and the tachograph is not in calibration mode.	Acknowledge the message. Have the STC-8255 checked by authorized workshop if this fault occurs frequently.

Table 5: List of Operational Mode Faults

Pictogram & Message ID	Fault Messages	Explanation	User Action
50	“Printer fault”	Displayed when unexpected interrupt is received from printer.	Acknowledge the message. Repeat print request and if necessary switch off / on the ignition.
51	“Display fault”	Displayed if the display self-test fails.	Acknowledge the message. Have the STC-8255 checked by authorized workshop.
64	"No further details"	Displayed because of generic smart card fault.	Acknowledge the message. Check if the card contact surface is clean. If it is not, clean the contacts and insert it again.
157	“Internal fault”	Displayed if there is problem while the tachograph is rebooting.	Acknowledge the message. Have the STC-8255 checked by authorized workshop.
158	“Sensor fault”	Displayed when error related to controller memory occurs in the motion sensor.	Acknowledge the message. Have the STC-8255 checked by authorized workshop if this fault occurs frequently.
159	“Sensor fault”	Displayed when error related to controller instruction occurs in the motion sensor.	Acknowledge the message. Have the STC-8255 checked by authorized workshop if this fault occurs frequently.
160	“Sensor fault”	Displayed when error related communication occurs in the motion sensor.	Acknowledge the message. Have the STC-8255 checked by authorized workshop if this fault occurs frequently.
161	“Internal fault”	Displayed when there is motion sensor authentication failure.	Acknowledge the message. Have the STC-8255 checked by authorized workshop if this fault occurs frequently.
162	“Internal fault”	Displayed when the smart card power-up voltage is invalid.	Acknowledge the message. Have the STC-8255 checked by authorized workshop if this fault occurs frequently.
163	“Internal fault”	Displayed if the smart card protocol is unsupported.	Acknowledge the message. Have the STC-8255 checked by authorized workshop if this fault occurs frequently.
164	“Internal fault”	Displayed if the smart card status is invalid slot.	Acknowledge the message. Have the STC-8255 checked by authorized workshop if this fault occurs frequently.
165	“Internal fault”	Displayed if the smart card status is unexpected value.	Acknowledge the message. Have the STC-8255 checked by authorized workshop if this fault occurs frequently.
166	“Internal fault”	Displayed if the user data storage is insufficient.	Acknowledge the message. Have the STC-8255 checked by authorized workshop.
169	“Internal fault”	Displayed if the internal battery voltage is below critical level.	Acknowledge the message. Have the STC-8255 checked by authorized workshop.

173	"Clock Failure"	Displayed when there is a fault in the real time clock (RTC) module of the tachograph and recorded to the device as a DTC record.	Acknowledge the message. Have the STC-8255 checked by authorized workshop.
175	"Internal fault"	Displayed if software integrity failure occurs.	Acknowledge the message. Have the STC-8255 checked by authorized workshop.
176	"Internal fault"	Displayed if software integrity failure occurs.	Acknowledge the message. Have the STC-8255 checked by authorized workshop.
177	"Internal fault"	Displayed if software integrity failure occurs.	Acknowledge the message. Have the STC-8255 checked by authorized workshop.

Acoustic Alerts

STC-8255 generates acoustic alerts (beep tone) to warn user about some warning situations (events) or irrelevant user entries. Acoustic alerts can be heard on these events:

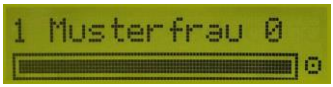
1. Overspeed,
2. DAGS warning,
3. Pre-warnings,
4. Irrelevant user entries,
5. Buzzer test.

Withdrawing Driver Card and Co-Driver Cards

At the end of the work shift / day or when driver and co-driver change places, the cards will be ejected from the tachograph.

In order to withdraw (eject) driver or co-driver card, do as follows:

1. Be sure that the vehicle is stopped (not moving).
2. Enter the desired activities at the end of the work shift, (e.g **Rest** " ⊢ ") by using the **keypad#1** or **keypad#2**.
3. After setting the latest activities, ***long*** **press** to **keypad#1** or **keypad#2** for ejection of the driver and co-driver cards, respectively.
4. Driver's last name appears as shown below. Progress bar in the second line indicates that STC-8255 is transferring data to the driver/co-driver card.

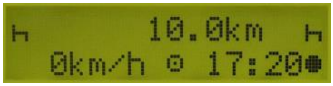


5. Select your end country by using control ‹ / › and press **OK**. If required, select the end region and confirm by pressing **OK**.



If you do not want to enter end place, you can abort the end place entry by pressing the backspace ⌫ button.

6. Card(s) will be ejected from the requested slot. Take the card(s) and keep it in a secure environment.
7. After the driver card(s) is released, display turns back to standard display as below.



Risk of Accident!

You must withdraw the driver and co-driver cards from the card slots when the vehicle is stationary. Otherwise, the STC-8255 will not eject the cards with displaying an "Ejection is not possible" message.

Ejection is also not possible when the data on the card is being processed.

Driver/Vehicle Change During Operation

When the crew exchange their positions, withdraw driver and co-driver cards from the related slots and exchange them by inserting into the other card slots. Co-driver (now driver) should first insert his/her driver card into card slot 1 and driver (now co-driver) should insert his/her driver card into card slot 2. Set the activity you wish.

When the driver or co-driver who leaves the current vehicle should request and takes his/her card out from the STC-8255. New vehicle crew (driver or co-driver) should insert their new driver or co-driver cards into the related card slots.

Power Saving Mode

When the ignition key is switched OFF the STC-8255 will turn to power saving mode to preserve the vehicle's battery. The STC-8255 will turn to power saving mode (sleep mode) in approximately 1 minute after the last interaction to reduce the power consumption.

The display will turn ON again when:

- Switching the ignition key ON,
- Pressing any button once (turns back to power saving mode again when no activity),
- Inserting or withdrawing any cards into/from card slots (turns back to power saving mode again after cards data is being processed by STC-8255),
- When the vehicle is moving (e.g by tow).

Company Mode (Company Representative Role)

This part of the documentation contains information especially for the vehicle owners and transport companies who have certain responsibilities such as:

1. Company Inspection,
2. Workshop Inspection.

Company Inspection

The company is obligated to carry out inspections of the tachographs and has to keep records.

The company inspection shall ensure that:

- Type approval number on the tachograph is correct.

- Tachograph's UTC time is accurate by less than 20 minutes.
- The tachograph is within the correct calibration interval.
- The installation plaque is time valid and not broken.
- The sealing label is not torn apart.
- The stored calibration parameters agree with what is recorded on the installation plaque.
- The tachograph's internally stored vehicle paramaters such as VIN (Vehicle Identification Number) and VRN (Vehicle Registration Number) is correct.
- The tachograph does not have any visual damage.

The company has to keep a record of inspection.

Workshop Inspection

The company has to administrate a workshop inspection of the tachographs and store records from the inspection. It is the company's responsibility to renew this inspection within every two years. The installation plaque, which is fixed near the tachograph, indicates the validity dates of the inspection.

Note that!

Next time when the tachograph is at the inspection, the installation plaque must be valid and not broken.

Functions of the Company Card

The company cards identify the company and register it into the STC-8255, when the first time the company card is inserted. Therefore, access rights to the data assigned to the company will be ensured.

The company has to take care of the correct use of the company card(s).

By inserting the company card, you can gain the following capabilities.

1. Registeration and de-registration of the company,
2. Entering vehicle registration number,
3. Download and access to mass storage of vehicle,
4. Download and access to card data,
5. Display and print out of the driver card and vehicle data.

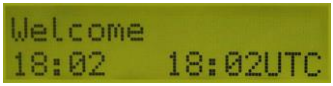
The company card is designed for the data management of the company only and must not be used for driving! If the company card is used for driving, a warning message will be displayed and recorded as an event to the STC-8255.

Registration of the Company (Lock-in)

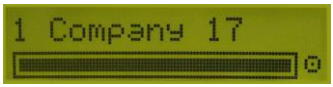
The company owners can registrate (lock-in) themselves to the tachograph by inserting their company cards in order to protect data of the tachograph from unauthorized persons. Before the tachograph is being used, it should be lock-in to the company otherwise all data which is unlocked until lock-in is available to everyone.

In order to lock-in your tachograph to your company, perform the steps below:

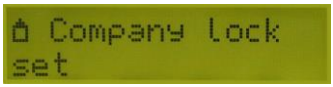
1. Insert your company card (chip side is up and arrow pointing forward) into one of the card slots of the STC-8255. Make sure that the other slot has driver card only or no card inside.
2. Greeting text message is shown for approximately 3 seconds. Display also shows the current UTC (18:02) and local time (18:02, time offset = 0 hour).



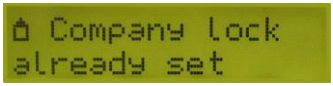
3. The Company's name is shown on the display. A progress bar indicates that the company card is being read.



4. If the company card is inserted for the first time, the company will automatically be lock-in the STC-8255. "Company lock set" message appears on the display as shown below.



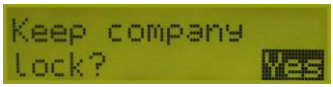
5. If the inserted company card already has an active lock, the following message appears on the display.



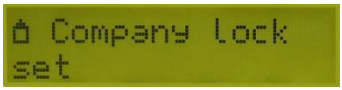
6. After company card is being read, the following default screen will be displayed showing the company mode is active.



Withdraw the company card from slot in which it is inserted. Then, select **YES** to the question (Keep company lock) and press **OK** to perform the lock-in.



7. "Company lock set" message appears on the display as shown below.



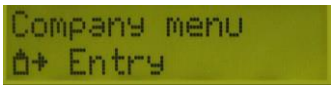
Entering Vehicle Registration Number

If the Vehicle Registration Number (VRN) field of a new tachograph is not entered in the workshops it can be entered only once in Company Mode.

To do so, perform the following steps:

1. Make sure that the STC-8255 is in Company mode (Company card is inserted).

2. Press **OK** to enter Main Menu. Select **Company menu>Entry** by using control ◄ / ► keys. Press **OK** and navigate to **Enter VRN**. Press **OK** to enter the menu.





3. The following display will be as shown below.



4. Select RMS (Registering Member State) by using control ◄ / ► keys and press **OK.**

➔ *Please refer to "National/Regional abbreviation and European time zones", on page 106.*



5. Select CP (Code Page) by using control ◄ / ► keys and press **OK.** Code Page brings different alphabet sets to enter the VRN.



6. Use control ◄ / ► keys to select the desired character and confirm your selection with **OK** button. After pressing **OK** button pass to the next character and select and confirm with **OK**. Repeat process until the final "**?**" character is cleared with a text or space. When it is ready to confirm, the diamond symbol at the end of the second line should become carriage return symbol. Press **OK** button to confirm the whole VRN entry.



When you need a correction use backspace ◄ button and delete your entry and after selecting desired character repeat the above steps.

After confirmation of VRN with **OK** button, display shown above will be closed and VRN Entry menu (under the Vehicle Menu) will disappear.

You can print a technical data found under the Vehicle Menu Print, and check that the VRN entered above is written on the printout.

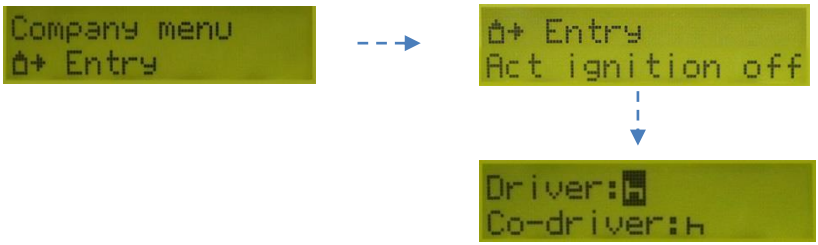
Automatic Setting of Activity after Ignition OFF/ON

The STC-8255 can be configured to switch to a pre-defined activity (e.g.“ , and ”) automatically after ignition On/Off. You can change the settings both for ignition OFF and ON cases for the driver and co-driver. These settings can be adjusted when the STC-8255 is in the Company Mode or in the Calibration Mode.

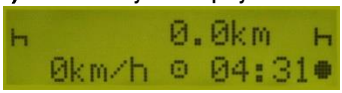
Ignition OFF Activities for Drivers

To adjust the Ignition OFF activities in Company Mode;

1. Insert your company card into one of the card slots. Make sure that the other slot has the driver card or no card is inserted (STC-8255 is in Company Mode).
2. Press **OK** to enter Main Menu. Select **Company menu>Entry** by using control / keys. Press **OK** to enter the menu. Select **Act. ignition off** menu by using the control / keys. Press **OK** to enter the menu. Adjust the activities that you desire to change automatically when the ignition off.



For example: Rest “” activities are selected both for the Driver and co-driver. When the ignition becomes off, the activities of driver and co-driver will both change to pre-defined activity (Rest “ ”) automatically and display will be as follows:



Note That!

Adjusting the activities to “?” both for driver/co-driver maintains the activities of driver and co-driver when the ignition is turned off/on again.

Igniton ON Activities for Drivers

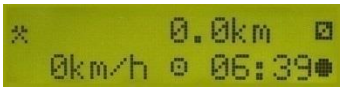
To adjust the Ignition ON activities in Company Mode;

1. Insert your company card into one of the card slots. Make sure that the other slot has the driver card or no card is inserted (STC-8255 is in Company Mode).
2. Press **OK** to enter Main Menu. Select **Company menu>Entry** by using control / keys. Press **OK** to enter the menu. Select **Act. ignition on** menu by using the control / keys. Press **OK** to enter the menu. Adjust the activities that you desire to change automatically when the ignition on.





After igniton is **On** again:
The activities can be set automatically to a pre-defined activity (e.g. **Work "⚒"** and **Available "⧈"**).
The display shows the activities as shown below.



- Change the set activity according to your current task, as necessary.

Downloading Mass Storage (Vehicle Data)

Tachograph data memory records and holds the relevant data (e.g. driver and co-driver activities, country entries for both drivers, special entries, recording control activities, recordings of time adjustments, Lock-in / Lock-out procedures by the company card, identification of the STC-8255, device calibrations, system events & faults data, installation data and speed informations over 168 hours and etc.) over a time period of at least 365 calendar days. When the memory of the tachograph is full, the information will be overwritten by new information and thereby it will be lost forever. To avoid this and to secure vehicle data, a frequent download of information is needed. It means that data will be transferred from the tachograph to data storage outside the vehicle. Data has to be downloaded regularly.

Note that!

Downloading will not destroy or erase any data from the memory. Data is only erased from mass storage when it is overwritten by new data.

Tachograph mass storage may be downloaded in the following modes:

1. **Company Mode:** In this mode company card should be in one of the card slots. Driver card or no cards should be in the other slot. If the company card is locked-in to the tachograph, only that company card which has locked can access to stored data (card insertion/withdrawals, driver activities and places) in company mode. With using different company cards, one cannot access (display, print or download) to these data, which is previously locked by another company.

2. **Calibration Mode:** In this mode workshop card should be in one of the card slots. Driver card or no cards should be in the other slot.

3. **Control Mode:** In this mode control card should be in one of the card slots. Driver card or no cards should be in the other slot. Whether it is locked by any company or not, control cards can access (display, print or download) to whole stored data (card insertion/withdrawals, driver activities and places) which is previously locked by any company.

Note That!

Downloading of mass storage data of the tachograph requires data download equipment (sold separately in the market and compliant with the protocol defined in the latest amended legislative document 3821/85 Annex-1B Appendix-7).

To explain mass storage download of the STC-8255, perform the following steps:

1. Open the silicon cover in front of the tachograph.
2. Connect the download equipment's 6-pin connector to the tachograps's 6-pin interface behind the silicon cover.
3. Start downloading data according to the instructions on the download equipment.
4. When downloading starts a flashing animation is played on the second line of the display, showing the downloading is in progress.



5. When the downloading is finished animation on the display stops.
6. Get the digitally signed data which identifies the digital tachograph and store it in a data storage media (flash disk, hard disk, CD-ROM or SD-card)..

Note that!

Do not disconnect the data connection to the digital tachograph before the download is complete.

Note That!

For detailed information about the data download, please refer to the download equipment's appropriate documentation.

Note That!

To download card data over tachograph, the STC-8255 should not be in sleep (power saving) mode. If so, press any button at first to wake the STC-8255 up or switch the ignition key ON before the process.

Downloading Card Data

Downloading procedure of driver card data in company mode is similar to the procedure explained in the operational mode; therefore, will not be explained again.

➔ Please *refer to "Operational mode – Downloading Driver Card Data" part on page 30.*

Displaying and printing driver card and vehicle data

Displaying and printing driver card and vehicle data procedure in company mode is similar to the procedure explained in the operational mode and in the control mode and will not be explained again.

➔ *Please refer to "Operational mode – Displaying Card Data and Printing Card Data" part on page 31-32 and "Control mode – Displaying Stored and Card Data, and Printing Stored Data" part on page 55-59.*

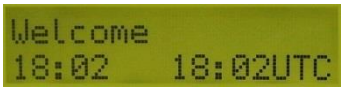
De-registration of the Company (Lock-out)

The company owners can de-registrate (lock-out) itself from the tachograph by withdrawing their company cards from the tachograph or by inserting different company card to the tachograph which is previously locked-in by another company card. The company lock will be removed automatically when you insert a different company card. Before the tachograph is being used by another company (e.g, when you sell your vehicle), it should be locked-out from the previous company otherwise next company's data will be recorded (appended) to the previous company's data.

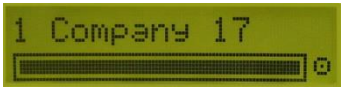
In order to Lock-out your tachograph from your company, perform the steps below:

1. Insert the same company card (if it has not been inserted already) into one of the card slots of the STC-8255. Make sure that the other slot has driver card only or no cards inside.

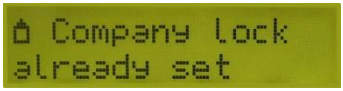
2. Greeting text message is shown for approximately 3 seconds. Display also shows the current UTC (18:02) and local time (18:02, time offset = 0 hour).



3. The Company's name is shown on the display. A progress bar indicates that the company card is being read.



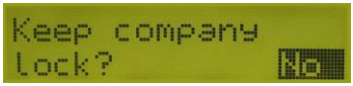
4. If the inserted company card already has an active lock, the following message appears on the display.



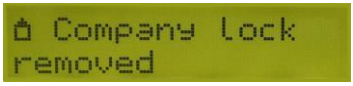
5. After company card is read, the following default screen will be displayed showing the company mode is active.



6. Withdraw the company card from the slot it is inserted. Then, select **NO** to the question (Keep company lock) and press **OK** to perform the lock-out. If you do not press the **OK** button in 10 seconds, the desired choice will be saved.



7. "Company lock removed" message appears on the display as shown below.



Acknowledge Event or Fault messages

When certain events or faults occur, STC-8255 displays a warning message with an alert tone at the beginning of the warning message. While the warning message appears on the display, plain text of the message will flash for about **30 seconds** or until they are confirmed with **OK** button by the driver. After confirmation with **OK** button or 30 seconds timeout the flashing of the text message will stop immediately. Press **OK** button again to acknowledge the message then the message disappears and the standard display will appear again.

Messages, Warnings and Faults of Company Mode/Role

There are four types of messages that can be shown on the display.

- **Messages –** give information on the process or remind the driver. Messages can not be stored and printed. Press the backspace button or **OK** button to clear the message.
- **Pre-warnings –** appear as early reminders to the warnings. Pre-warnings are not stored and can not be printed. Press **OK** button twice to clear the Pre-warning.
- **Warnings (Events) –** appear in the event of e.g. overspeed or violation of the law or if the tachograph can not be recording. Warnings are stored and can be printed. Press **OK** button twice to acknowledge the Warning.
- **Faults –** are more critical than warnings and are displayed if there is a fault detected in the tachograph, in the sensor or cards. On the other hand, fault messages are shown if the device detects a tampering. Faults are stored and can be printed. Press **OK** button twice to acknowledge the fault.

Table 6: List of Company Mode Messages

Pictogram & Message ID	Message	Explanation	User Action
 1	"non-valid card"	A non-valid or expired card has been inserted to one of the slots.	Acknowledge the message. Eject the non-valid or expired card if it is not ejected automatically.
 134	"Wrong card type"	Displayed if the inserted card into the tachograph is not a tachograph card. The card inserted is not accepted and ejected automatically.	Please insert a valid tachograph card.
 136	"Recording Inconsistent"	Displayed if there is an inconsistency with the driver or workshop card records while inserting the cards to any slot.	Acknowledge the message.
	"Drive Cont."	Displayed if the weekly allowable driving time is exceeded.	Acknowledge the message. Give a break.
	xxhxx		
	"Rest Cont."	Displays the remaining time for the upcoming weekly rest period.	Acknowledge the message. Prepare for the rest period.
	xxhxx		
	"Drive Cont."	Displayed if the daily allowable driving time is exceeded.	Acknowledge the message. Give a break.
	xxhxx		
	"Rest Cont."		

Pictogram & Message ID	Message	Explanation	User Action
	xxhxx  xxhxx	Displays the remaining time for the upcoming daily rest period.	Acknowledge the message. Prepare for the rest period.
	“Give a break”	Displayed after 4h15m or 4h30m continuous driving time.	Acknowledge the message. Give a break.
	xxhxx  xxhxx		
	“Drive Cont.”	Displayed if the 2 weekly allowable driving time is exceeded.	Acknowledge the message. Give a break.
	“xxhxx”		
 171	“No personalization”	Displayed if the card is inserted into anunpersonalised tachograph.	Acknowledge the message. Have the STC-8255 checked by authorized workshop.
 172	“Tachograph is not ready”	Displayed if the tachograph is not ready when the card is inserted into the tachograph.	Acknowledge the message. Try again after 1 minute. Have the STC-8255 checked by authorized workshop if this message is displayed frequently.
 174	“Eject driver card”	Displayed if there is a valid driver card in any slots while the ignition is turning to OFF.	Acknowledge the message. Do not forget to withdraw your driver card from the tachograph.
 174	“Veh. is not rdy to driv.”	Displayed if the ignition is ON and there is no valid driver card in the first slot or there is a smart card rather than driver card in the second slot.	Acknowledge the message. Do not forget to insert your driver card to the first slot before you start your trip.
	“Power is insufficient”	Displayed if the power is insufficient when the printing process is started or during the printing.	Acknowledge the message. Have your vehicle battery checked in the authorized services.
	“Card is not ready”	Displayed if the card reading is not finished for both card slots before the printing process is started.	Acknowledge the message. Wait for a while for the card reading process.
	“Stop to print”	Appears when the vehicle moves while the printer is busy with printing.	Stop the vehicle and try to print again when the vehicle is stationary.
	“No paper”	Displayed when the printer is out of paper at the beginning of the printing task.	Load new roll of paper to the printer drawer.
	“No data”	Displayed when there is no data for the requested printout type.	Acknowledge the message.
	“Thermal stress”	Displayed if the printer is too hot when the user makes a request to print any report.	Acknowledge the message. Please wait for a while for the printer head to cool down and try again later.
	“Thermal stress”	Displayed if the printer has continuously printed and the printer head is too hot to continue. Disappears when the user pushes the backspace button or the printer head cool down.	Wait for the printer head cools down and printer continues printing job or press backspace to cancel the job. Please do not touch the printer head
	“Cancel”		

Pictogram & Message ID	Message	Explanation	User Action
			with your hand when this message appears.
	"Load paper" -Progress Bar-	Displayed when the printer is out of paper while printing and displayed until new printer paper roll is inserted (45 sec period).	Insert new printer paper roll.
	"Printer is ready" -Progress Bar-	Displayed after the new roll of paper is loaded to the printer drawer.	Load new roll of paper to the printer drawer.
	"Cancel printing" Yes – No Dialog	Displayed when the printer is not loaded with a new roll of paper in a certain period (45sec).	Press YES to cancel the printing job.
	"not possible"	Displayed when the user is trying to enter the menu while the vehicle is moving.	Stop the vehicle if necessary to navigate in the menus.
	"Printing"	Appears when the printer is busy with printing.	No user action required.
	"Ejection is not possible"	Displayed if the cards inserted to the tachograph slots are requested while the battery supply level is too low.	Acknowledge the message. Have your vehicle battery checked in the authorized services.
	"Ejection is not possible"	Displayed if the cards inserted to the tachograph slots are requested while the vehicle is moving.	Please stop the vehicle and then request the card withdrawal from the tachograph.
	"UTC correct. not possible"	Displayed if you are trying to correct the UTC time which has already been corrected within the last 7 days	No user action required.
	"Wait for ejection"	Displayed if the cards inserted to the tachograph slots are requested while the inserted activity is recorded to the tachograph cards.	Acknowledge the message. Please try again one minute later.

Table 7: List of Company Mode Pre-warnings

Pictogram & Message ID	Pre-Warning Messages	Explanation	User Action
>> 129	"Oversp. prev. warning"	The vehicle has exceeded the overspeed limit which is set for that vehicle. If exceeding continues for 30 sec. (or pre-defined value) the overspeed event will be recorded into tachograph memory.	Acknowledge the message. Reduce the vehicle speed.
 130	"Card will expire soon"	Means expiration prewarning day limit of card inserted to the tachograph slot is reached and this card will be expired soon.	Acknowledge the message. Contact the responsible authority to get new one.
 131	"Time for service"	Displayed if the calibration pre-warning date is passed and reminds the driver.	Acknowledge the message. Have the STC-8255 checked and calibrated by authorized workshop.
 146	"Download card"	Displayed if the card download pre-warning date is passed and reminds the driver to download his/her card to external media.	Acknowledge the message. Download the driver card to any external media.

147	“Download vehicle unit”	Displayed if the tachograph download pre-warning date is passed and reminds the driver to download his/her tachograph to external media.	Acknowledge the message. Download the tachograph mass memory to any external media.

Table 8: List of Company Mode Warnings

Pictogram & Message ID	Warning (Event) Messages	Explanation	User Action
2	“Card conflict”	Displayed if the two tachograph cards incompatible with each other are inserted to the tachograph.	Acknowledge the message. Remove one of the cards (which is useless) from the card slot.
3	“Time overlap”	The last withdrawal date/time of the inserted driver card from previous tachograph is later than the date/time of recently inserted tachograph.	Acknowledge the message. Check the date/time of the tachographs used and determine the tachograph with the incorrect UTC time, make the tachograph time to be corrected by authorized workshop. Or, wait for the overlap period to be elapsed.
4	“Driving w/o valid card”	Driving started with an inappropriate card or with an inappropriate card combination.	Acknowledge the message. Stop vehicle and insert a valid driver card or remove inappropriate card from co-driver slot.
5	“Card insertion while driving”	A driver card was inserted after driving has begun.	Acknowledge the message. No further action required.
6	“Card closed incorrectly”	Card was not removed properly from the last tachograph or data could not be saved to the card correctly.	Acknowledge the message. No further action required.
7	“Over speeding”	Driver exceeded the permissible max speed limit for longer than 1 minute.	Acknowledge the message. Reduce vehicle speed.
8	“Power supply interruption”	The power supplied to the STC-8255 is below or above the certain limit more than 200 miliseconds or has been disconnected for more than 200 miliseconds. The event is not triggered in Calibration Mode.	Acknowledge the message. Have the STC-8255 checked by authorized workshop.
9	“Motion sensor data error”	The data link failure occurred between the motion sensor and the tachograph.	Acknowledge the message. Have the STC-8255 checked by authorized workshop.
10	“Motion conflict”	Displayed if the second source of motion data (GNSS or CAN-Bus) is not consistent with the primarily used motion sensor data.	Acknowledge the message. Have the STC-8255 checked by authorized workshop.
16	“No further details”	Displayed if an unexpected error occurs while authenticating the motion sensor.	Acknowledge the message.

Pictogram & Message ID	Warning (Event) Messages	Explanation	User Action
			Have the STC-8255 checked by authorized workshop.
! 17	“Motion sensor auth. failure”	Displayed if the tachograph does not authenticate with the motion sensor	Acknowledge the message. Have the STC-8255 checked by authorized workshop.
! 18	“Card auth. Failure”	Displayed if the tachograph does not authenticate with the smart card.	Acknowledge the message. Have the STC-8255 checked by authorized workshop.
! 19	“Unauthorised change of MS”	Displayed if unpaired motion sensor is linked to the tachograph paired with another motion sensor.	Acknowledge the message. Have the STC-8255 checked by authorized workshop.
! 20	“Card data integ. err.”	Displayed if smart card data is corrupted.	Acknowledge the message. Have the STC-8255 checked by authorized workshop.
! 21	“Data integ. err.”	Displayed if the tachograph data is corrupted.	Acknowledge the message. Have the STC-8255 checked by authorized workshop.
! 24	“Hardware sabotage”	Displayed if card(s) ejection by force is detected.	Acknowledge the message. Have the STC-8255 checked by authorized workshop.
! 32	“No further details”	Displayed if a generic error occurs in the motion sensor.	Acknowledge the message. Have the STC-8255 checked by authorized workshop.
! 33	“Auth. Failure”	Displayed when error related to authentication occurs in the motion sensor.	Acknowledge the message. Have the STC-8255 checked by authorized workshop if this fault occurs frequently.
! 34	“Data integ. err.”	Displayed when error related to non-volatile memory occurs in the motion sensor.	Acknowledge the message. Have the STC-8255 checked by authorized workshop if this fault occurs frequently.
! 37	“Hardware sabotage”	Displayed when error related to sensor element or temperature occurs in the motion sensor and the tachograph is not in calibration mode.	Acknowledge the message. Have the STC-8255 checked by authorized workshop if this fault occurs frequently.

Table 9: List of Company Mode Faults

Pictogram & Message ID	Fault Messages	Explanation	User Action
50	“Printer fault”	Displayed when unexpected interrupt is received from printer.	Acknowledge the message. Repeat print request and if necessary switch off / on the ignition.

 51	“Display fault”	Displayed if the display self-test fails.	Acknowledge the message. Have the STC-8255 checked by authorized workshop.
 64	"No further details"	Displayed because of generic smart card fault.	Acknowledge the message. Check if the card contact surface is clean. If it is not, clean the contacts and insert it again.
 157	“Internal fault”	Displayed if there is problem while the tachograph is rebooting.	Acknowledge the message. Have the STC-8255 checked by authorized workshop.
 158	“Sensor fault”	Displayed when error related to controller memory occurs in the motion sensor.	Acknowledge the message. Have the STC-8255 checked by authorized workshop if this fault occurs frequently.
 159	“Sensor fault”	Displayed when error related to controller instruction occurs in the motion sensor.	Acknowledge the message. Have the STC-8255 checked by authorized workshop if this fault occurs frequently.
 160	“Sensor fault”	Displayed when error related communication occurs in the motion sensor.	Acknowledge the message. Have the STC-8255 checked by authorized workshop if this fault occurs frequently.
 161	“Internal fault”	Displayed when there is motion sensor authentication failure.	Acknowledge the message. Have the STC-8255 checked by authorized workshop if this fault occurs frequently.
 163	“Internal fault”	Displayed if the smart card protocol is unsupported.	Acknowledge the message. Have the STC-8255 checked by authorized workshop if this fault occurs frequently.
 164	“Internal fault”	Displayed if the smart card status is invalid slot.	Acknowledge the message. Have the STC-8255 checked by authorized workshop if this fault occurs frequently.
 165	“Internal fault”	Displayed if the smart card status is unexpected value.	Acknowledge the message. Have the STC-8255 checked by authorized workshop if this fault occurs frequently.
 166	“Internal fault”	Displayed if the user data storage is insufficient.	Acknowledge the message. Have the STC-8255 checked by authorized workshop.
 169	“Internal fault”	Displayed if the internal battery voltage is below critical level.	Acknowledge the message. Have the STC-8255 checked by authorized workshop.
 173	“Clock Failure”	Displayed when there is a fault in the real time clock (RTC) module of the tachograph and recorded to the device as a DTC record.	Acknowledge the message. Have the STC-8255 checked by authorized workshop.
 175	“Internal fault”	Displayed if software integrity failure occurs.	Acknowledge the message. Have the STC-8255 checked by authorized workshop.
 176	“Internal fault”	Displayed if software integrity failure occurs.	Acknowledge the message. Have the STC-8255 checked by authorized workshop.

177	"Internal fault"	Displayed if software integrity failure occurs.	Acknowledge the message. Have the STC-8255 checked by authorized workshop.

Acoustic Alerts

STC-8255 generates acoustic alerts (beep tone) to warn user about some warning situations (events) or irrelevant user entries. Acoustic alerts can be heard on these events:

1. Overspeed,
2. DAGS warning,
3. Pre-warnings,
4. Irrelevant user entries,
5. Buzzer test.

Control Mode (Controller Role)

Inserting a control card into one of the card slots changes the mode of operation of the tachograph to CONTROL mode. However, make sure that the other slot has the driver card or no card is inserted. A valid control card is required to display, to print or to download driving data stored on either an inserted driver card or in the tachographs data memory. The control card provides read-access to the entire tachograph data memory contents whether or not the data has been company locked using a valid company card.

Displaying Stored and Card Data

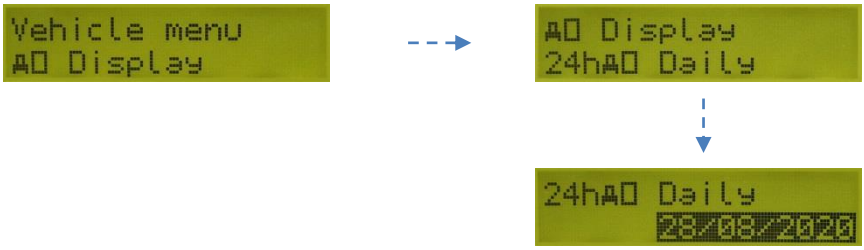
With a valid control card, enforcement authorities can view inserted driver card data and stored data on the display. In this scenario, the driver card should be in the first slot and the control card should be in the second slot. Controller can view the 24h daily activities, the events data related to the driver card inserted to the first slot. In addition to this, the controller can also view the stored 24h daily activities, events & faults, overspeed, technical data, V profile, N profile and GNSS data of the vehicle.

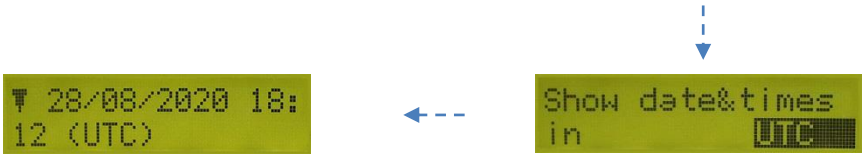
On the operational mode (driver part) card display is explained detailly, therefore will not be explained again. Here, displaying of the vehicle data will be demonstrated.

The data appear on the display is similar to how it appears on a printout; however one printout line (24 characters) will be divided into two lines.

1. To display **Vehicle menu>24h Activities,** perform as follows:

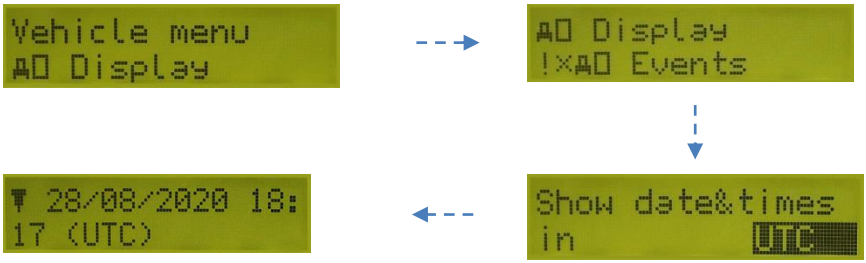
- Press **OK** to enter Main menu. Select **Vehicle menu>Display** by using control ‹ / › keys. Press **OK** and navigate to **Display 24h Daily**. Press **OK** to enter the menu. Select the date and time format (UTC or Local) to be displayed and press **OK** to confirm. Scroll with control ‹ / › keys to view all of the data. Press the backspace ⌫ button several times (press once to return one level back) to return standard display.





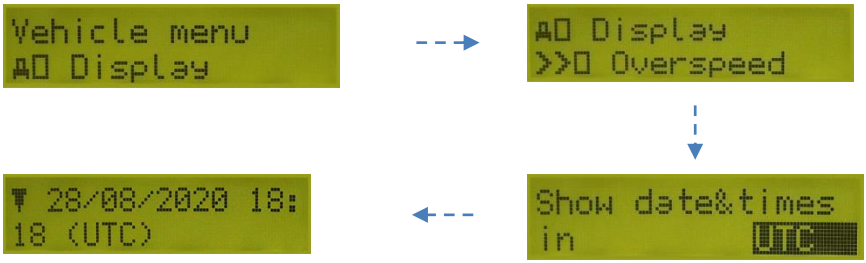
To display **Vehicle menu> Events&Faults Data** perform as follows:

- Press **OK** to enter Main menu. Select **Vehicle menu>Display** by using control ◄ / ► keys. Press **OK** and navigate to **Display Events**. Press **OK** to enter the menu. Select the date and time format (UTC or Local) to be displayed and press **OK** to confirm. Scroll with control ◄ / ► keys to view all of the data. Press the backspace ◄| button several times (press once to return one level back) to return standard display.



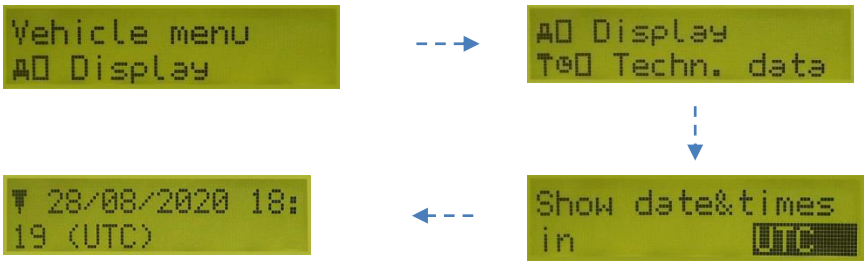
2. To display **Vehicle menu> Overspeed Data** perform as follows:

- Press **OK** to enter Main menu. Select **Vehicle menu>Display** by using control ◄ / ► keys. Press **OK** and navigate to **Display Overspeed**. Press **OK** to enter the menu. Select the date and time format (UTC or Local) to be displayed and press **OK** to confirm. Scroll with control ◄ / ► keys to view all of the data. Press the backspace ◄| button several times (press once to return one level back) to return standard display.



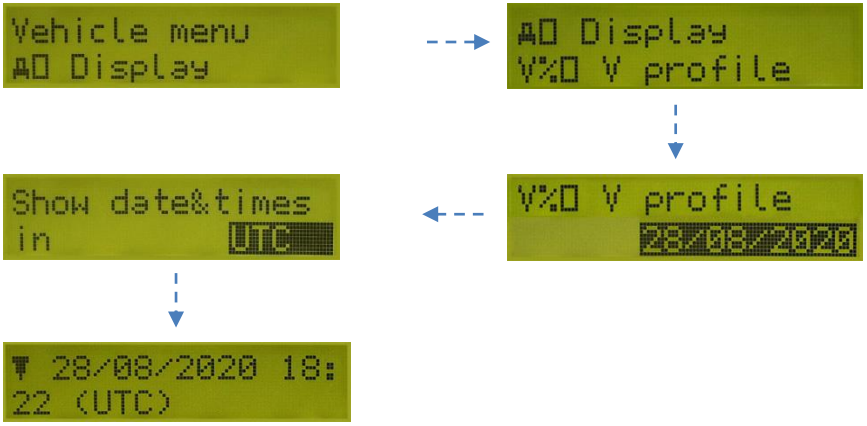
3. To display **Vehicle menu> Technical Data** perform as follows:

- Press **OK** to enter Main menu. Select **Vehicle menu>Display** by using control ◄ / ► keys. Press **OK** and navigate to **Display Techn. data**. Press **OK** to enter the menu. Select the date and time format (UTC or Local) to be displayed and press **OK** to confirm. Scroll with control ◄ / ► keys to view all of the data. Press the backspace ◄| button several times (press once to return one level back) to return standard display.



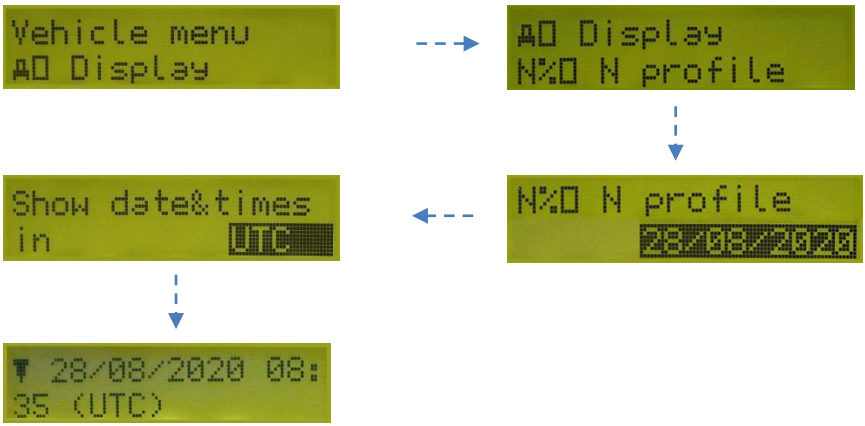
4. To display **Vehicle menu> V Profile** perform as as follows:

- Press **OK** to enter Main menu. Select **Vehicle menu>Display** by using control / keys. Press **OK** and navigate to **Display V profile**. Press **OK** to enter the menu. Select the date and time format (UTC or Local) to be displayed and press **OK** to confirm. Scroll with control / keys to view all of the data. Press the backspace button several times (press once to return one level back) to return standard display.



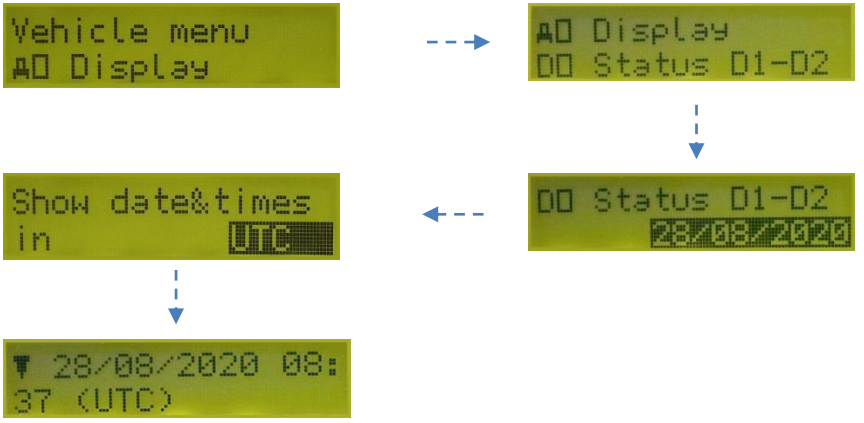
5. To display **Vehicle menu> N Profile** perform as as follows:

- Press **OK** to enter Main menu. Select **Vehicle menu>Display** by using control / keys. Press **OK** and navigate to **Display N profile**. Press **OK** to enter the menu. Select the date and time format (UTC or Local) to be displayed and press **OK** to confirm. Scroll with control / keys to view all of the data. Press the backspace button several times (press once to return one level back) to return standard display.



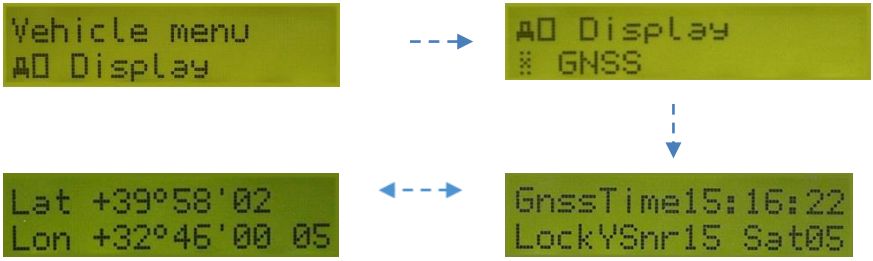
6. To display **Vehicle menu> Status D1-D2** perform as follows:

- Press **OK** to enter Main menu. Select **Vehicle menu>Display** by using control ◄ / ► keys. Press **OK** and navigate to **Display Status D1-D2**. Press **OK** to enter the menu. Select the date and time format (UTC or Local) to be displayed and press **OK** to confirm. Scroll with control ◄ / ► keys to view all of the data. Press the backspace ◄| button several times (press once to return one level back) to return standard display.



7. To display **Vehicle menu> GNSS** perform as follows:

- Press **OK** to enter Main menu. Select **Vehicle menu>Display** by using control ◄ / ► keys. Press **OK** and navigate to **Display GNSS**. Press **OK** to enter the menu. Scroll with control ◄ / ► keys to view all of the data. Press the backspace ◄| button several times (press once to return one level back) to return standard display. In this menu, GNSS signal levels and number of satellites' signals received can be viewed.



Printing Stored Data

Controller can print the 24h daily activities and the events data related to the driver card inserted to the first slot. In addition to this, the controller can also print the stored 24h daily activities, events & faults, overspeed, technical data, v-graph, v-profile, n-profile and status D1-D2 of the vehicle.

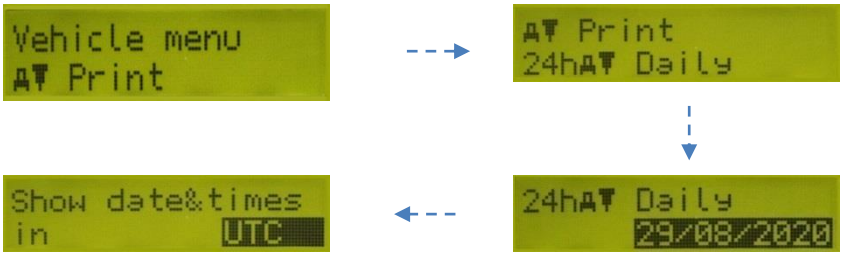
On the operational mode (driver part) printout of the card related data is explained detailly, therefore will not be explained again.

➔ *Please refer to "Operational mode – Printing Card data" on page 32.*

Here, printing of the vehicle data will be demonstrated.

1. To print **Vehicle menu >24h Daily Data,** perform as follows:

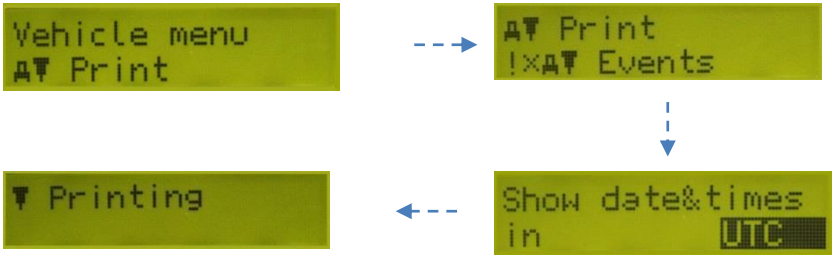
- Press **OK** to enter Main menu. Select **Vehicle menu>Print** by using control ‹ / › keys. Press **OK** and navigate to **Print 24h Daily**. Press **OK** to enter the menu. Select the date you want to take the printout and press **OK** to confirm. Select the date and time format (UTC or Local) to be printed and press **OK** to confirm. Printing message appears on the display and meanwhile printing job starts. Press the backspace ⌫ button several times (press once to return one level back) to return standard display.



➔ *Please refer to "24h (Daily) Vehicle Unit Printout" explanations given on page 92.*

2. To print **Vehicle menu > Events&Faults Data,** perform as follows:

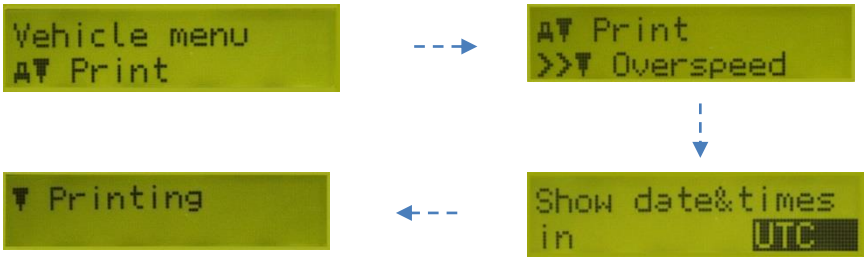
- Press **OK** to enter Main menu. Select **Vehicle menu>Print** by using control ‹ / › keys. Press **OK** and navigate to **Print Events**. Press **OK** to enter the menu. Select the date and time format (UTC or Local) to be printed and press **OK** to confirm. Printing message appears on the display and meanwhile printing job starts. Press the backspace ⌫ button several times (press once to return one level back) to return standard display.



➔ *Please refer to the "Events & Faults Vehicle Unit Printout" explanations given on page 94-105.*

3. To print **Vehicle menu > Overspeed Data,** perform as follows:

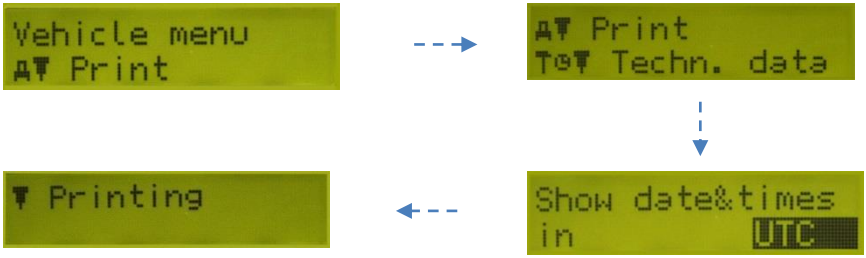
- Press **OK** to enter Main menu. Select **Vehicle menu>Print** by using control ◄ / ► keys. Press **OK** and navigate to **Print Overspeed**. Press **OK** to enter the menu. Select the time format (UTC or Local) to be printed and press **OK** to confirm. Printing message appears on the display and meanwhile printing job starts. Press the backspace ◄ button several times (press once to return one level back) to return standard display.



➔ *Please refer to "Vehicle Overspeed Printout" explanations given on page 98.*

4. To print **Vehicle menu > Technical Data,** perform as follows:

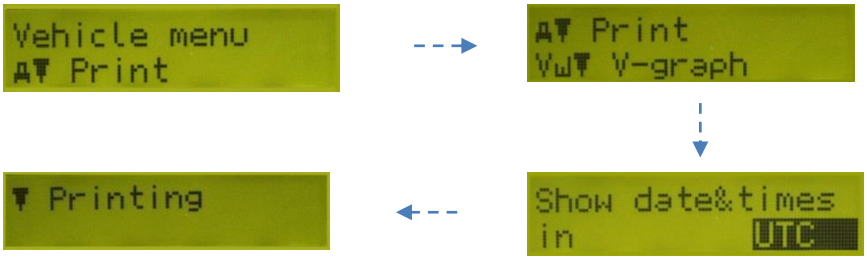
- Press **OK** to enter Main menu. Select **Vehicle menu>Print** by using control ◄ / ► keys. Press **OK** and navigate to **Print Technical Data**. Press **OK** to enter the menu. Select the date/time format (UTC or Local) to be printed and press **OK** to confirm. Printing message appears on the display and meanwhile printing job starts. Press the backspace ◄ button several times (press once to return one level back) to return standard display.



➔ *Please refer to "Vehicle Technical Data Printout" explanations given on page 104.*

5. To print **Vehicle menu > V-graph,** perform as follows:

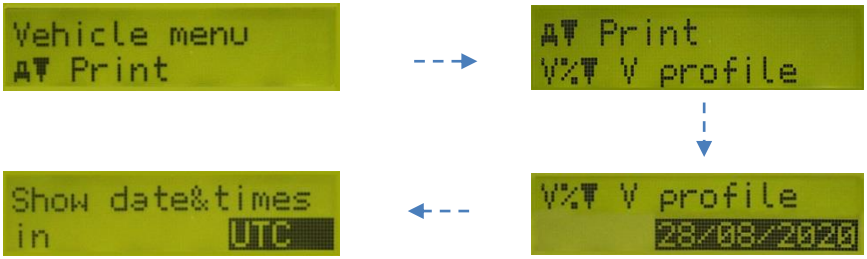
- Press **OK** to enter Main menu. Select **Vehicle menu>Print** by using control ◄ / ► keys. Press **OK** and navigate to **Print V-graph**. Press **OK** to enter the menu. Select the date/time format (UTC or Local) to be printed and press **OK** to confirm. Printing message appears on the display and meanwhile printing job starts. Press the backspace ◄| button several times (press once to return one level back) to return standard display.



➔ *Please refer to "V-graph Printout" explanations given on page 103.*

6. To print **Vehicle menu > V-profile,** perform as follows:

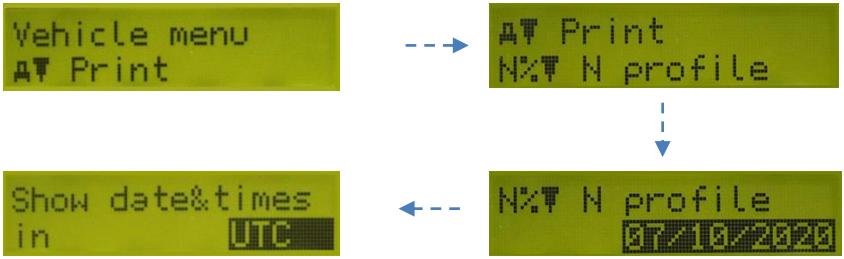
- Press **OK** to enter Main menu. Select **Vehicle menu>Print** by using control ◄ / ► keys. Press **OK** and navigate to **Print V-profile**. Press **OK** to enter the menu. Select the date/time format (UTC or Local) to be printed and press **OK** to confirm. Printing message appears on the display and meanwhile printing job starts. Press the backspace ◄| button several times (press once to return one level back) to return standard display.



➔ *Please refer to "V-profile printout" explanations given on page 100.*

7. To print **Vehicle menu > N-profile,** perform as follows:

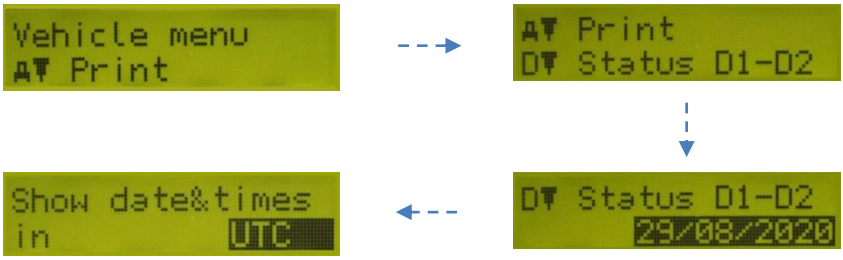
- Press **OK** to enter Main menu. Select **Vehicle menu>Print** by using control ◄ / ► keys. Press **OK** and navigate to **Print N-profile**. Press **OK** to enter the menu. Select the date/time format (UTC or Local) to be printed and press **OK** to confirm. Printing message appears on the display and meanwhile printing job starts. Press the backspace ◄| button several times (press once to return one level back) to return standard display.



➔ *Please refer to "N-profile printout" explanations given on page 99.*

8. To print **Vehicle menu > Status D1-D2,** perform as follows:

- Press **OK** to enter Main menu. Select **Vehicle menu>Print** by using control ◄ / ► keys. Press **OK** and navigate to **Print Status D1-D2**. Press **OK** to enter the menu. Select the date/time format (UTC or Local) to be printed and press **OK** to confirm. Printing message appears on the display and meanwhile printing job starts. Press the backspace ◄| button several times (press once to return one level back) to return standard display.



➔ *Please refer to "Status D1-D2 printout" explanations given on page 101.*

Downloading Vehicle Stored and Card Data

In control mode, control card should be in one of the card slots. Driver card or no cards should be in the other slot. Whether tachograph is locked by other company or not, control cards can access (display, print or download) to complete stored data (card insertion/withdrawals, driver activities and places) which is previously locked by any company.

Downloading procedure of vehicle stored data and card stored data in control mode is similar to the procedure that is explained in the company mode and operational mode, therefore will not be explained again.

➔ *Please refer to "Company mode – Downloading Mass Storage (Vehicle Data)" part on page 46 and "Operational mode – Downloading Driver Card Data" on page 30.*

Acknowledge Event or Fault messages

When certain events or faults occur, STC-8255 displays a warning message with an alert tone at the beginning of the warning message. While the warning message appears on the display, plain text of the message will flash for about **30 seconds** or until they are confirmed with **OK** button by the driver. After confirmation with **OK** button or 30 seconds timeout the flashing of the text message will stop

immediately. Press **OK** button again to acknowledge the message then the message disappears and the standard display will appear again.

Messages, Warnings and Faults of Control Mode/Role

There are four types of messages that can be shown on the display.

- **Messages –** give information on the process or remind the driver. Messages can not be stored and printed. Press the backspace button or **OK** button to clear the message.
- **Pre-warnings –** appear as early reminders to the warnings. Pre-warnings are not stored and can not be printed. Press **OK** button twice to clear the Pre-warning.
- **Warnings (Events) –** appear in the event of e.g. overspeed or violation of the law or if the tachograph can not be recording. Warnings are stored and can be printed. Press **OK** button twice to acknowledge the Warning.
- **Faults –** are more critical than warnings and are displayed if there is a fault detected in the tachograph, in the sensor or cards. On the other hand, fault messages are shown if the device detects a tampering. Faults are stored and can be printed. Press **OK** button twice to acknowledge the fault.

Table 10: List of Control Mode Messages

Pictogram & Message ID	Message	Explanation	User Action
 **1**	"non-valid card"	A non-valid or expired card has been inserted to one of the slots.	Acknowledge the message. Eject the non-valid or expired card if it is not ejected automatically.
 **134**	"Wrong card type"	Displayed if the inserted card into the tachograph is not a tachograph card. The card inserted is not accepted and ejected automatically.	Please insert a valid tachograph card.
 **136**	"Recording Inconsistent"	Displayed if there is an inconsistency with the driver or workshop card records while inserting the cards to any slot.	Acknowledge the message.
 	"Drive Cont." xxhxx	Displayed if the weekly allowable driving time is exceeded.	Acknowledge the message. Give a break.
 	"Rest Cont." xxhxx	Displays the remaining time for the upcoming weekly rest period.	Acknowledge the message. Prepare for the rest period.
 	"Drive Cont." xxhxx	Displayed if the daily allowable driving time is exceeded.	Acknowledge the message. Give a break.
 	"Rest Cont." xxhxx xxhxx	Displayed the remaining time for the upcoming daily rest period.	Acknowledge the message. Prepare for the rest period.
 	"Give a break" xxhxx xxhxx	Displayed after 4h15m or 4h30m continuous driving time.	Acknowledge the message. Give a break.

Pictogram & Message ID	Message	Explanation	User Action
	"Drive Cont."	Displayed if the 2 weekly allowable driving time is exceeded.	Acknowledge the message. Give a break.
	"xxhxx"		
171	"No personalization"	Displayed if the card is inserted into the not personalized tachograph.	Acknowledge the message. Have the STC-8255 checked by authorized workshop.
172	"Tachograph is not ready"	Displayed if the tachograph is not ready when the card is inserted into the tachograph.	Acknowledge the message. Try again after 1 minute. Have the STC-8255 checked by authorized workshop if this message is displayed frequently.
174	"Eject driver card"	Displayed if there is a valid driver card in any slots while the ignition is turning to OFF.	Acknowledge the message. Do not forget to withdraw your driver card from the tachograph.
174	"Veh. is not rdy to driv."	Displayed if the ignition is ON and there is no valid driver card in the first slot or there is a smart card rather than driver card in the second slot.	Acknowledge the message. Do not forget to insert your driver card to the first slot before you start your trip.
	"Power is insufficient"	Displayed if the power is insufficient when the printing process is started or during the printing.	Acknowledge the message. Have your vehicle battery checked in the authorized services.
	"Card is not ready"	Displayed if the card reading is not finished for both card slots before the printing process is started.	Acknowledge the message. Wait for a while for the card reading process.
	"Stop to print"	Appears when the vehicle moves while the printer is busy with printing.	Stop the vehicle and try to print again when the vehicle is stationary.
	"No paper"	Displayed when the printer is out of paper at the beginning of the printing task.	Load new roll of paper to the printer drawer.
	"No data"	Displayed when there is no data for the requested printout type.	Acknowledge the message.
	"Thermal stress"	Displayed if the printer is too hot and the user requests to print any report.	Acknowledge the message. Please wait for a while for the printer head to cool down and try again later.
	"Thermal stress"	Displayed if the printer has continuously printed and the printer head is too hot to continue. Disappears when the user pushes the backspace button or the print head cools down.	Wait for the printer head cools down and printer continues printing job or press backspace to cancel the job. Please do not touch the printer head with your hand when this message appears.
	"Cancel"		
	"Load paper"	Displayed when the printer is out of paper while printing and displayed until new printer paper roll is inserted (45 sec period).	Insert new printer paper roll.
	-Progress Bar-		
	"Printer is ready"	Displayed after the new roll of paper is loaded to the printer drawer.	Load new roll of paper to the printer drawer.
	-Progress Bar-		

Pictogram & Message ID	Message	Explanation	User Action
	"Cancel printing" Yes – No Dialog	Displayed when the printer is not loaded with a new roll of paper in a certain period (45sec).	Press YES to cancel the printing job.
	"not possible"	Displayed when the user is trying to enter the menu while the vehicle is moving.	Stop the vehicle if necessary to navigate in the menus.
	"Printing"	Appears when the printer is busy with printing.	No user action required.
	"Ejection is not possible"	Displayed if the cards inserted to the tachograph slots are requested while the battery supply level is too low.	Acknowledge the message. Have your vehicle battery checked in the authorized services.
	"Ejection is not possible"	Displayed if the cards inserted to the tachograph slots are requested while the vehicle is moving.	Please stop the vehicle and then request the card withdrawal from the tachograph.
	"UTC correct. not possible"	Displayed if you are trying to correct the UTC time which has already been corrected within the last 7 days.	No user action required.
	"Wait for ejection"	Displayed if the cards inserted to the tachograph slots are requested while the inserted activity is recorded to the tachograph cards.	Acknowledge the message. Please try again one minute later.

Table 11: List of Control Mode Pre-warnings

Pictogram & Message ID	Pre-Warning Messages	Explanation	User Action
>> 129	"Oversp. prev. warning"	The vehicle has exceeded the overspeed limit which is set for that vehicle. If exceeding continues for 30 sec. (or pre-defined value) the overspeed event will be recorded into tachograph memory.	Acknowledge the message. Reduce the vehicle speed.
 130	"Card will expire soon"	Means expiration prewarning day limit of card inserted to the tachograph slot is reached and this card will be expired soon.	Acknowledge the message. Contact the responsible authority to get new one.
 131	"Time for service"	Displayed if the calibration pre-warning date is passed and reminds the driver.	Acknowledge the message. Have the STC-8255 checked and calibrated by authorized workshop.
 146	"Download card"	Displayed if the card download pre-warning date is passed and reminds the driver to download his/her card to external media.	Acknowledge the message. Download the driver card to any external media.
 147	"Download vehicle unit"	Displayed if the tachograph download pre-warning date is passed and reminds the driver to download his/her tachograph to external media.	Acknowledge the message. Download the tachograph mass memory to any external media.

Table 12: List of Control Mode Warnings

Pictogram & Message ID	Warning (Event) Messages	Explanation	User Action
2	“Card conflict”	Displayed if the two tachograph cards incompatible with each other are inserted to the tachograph.	Acknowledge the message. Remove one of the cards (which is useless) from the card slot.
3	“Time overlap”	The last withdrawal date/time of the inserted driver card from previous tachograph is later than the date/time of recently inserted tachograph.	Acknowledge the message. Check the date/time of the tachographs used and determine the tachograph with the incorrect UTC time, make the tachograph time to be corrected by authorized workshop. Or, wait for the overlap period to be elapsed.
4	“Driving w/o valid card”	Driving started with an inappropriate card or with an inappropriate card combination.	Acknowledge the message. Stop vehicle and insert a valid driver card or remove inappropriate card from co-driver slot.
5	“Card insertion while driving”	A driver card was inserted after driving has begun.	Acknowledge the message. No further action required.
6	“Card closed incorrectly”	Card was not removed properly from the last tachograph or data could not be saved to the card correctly.	Acknowledge the message. No further action required.
7	“Over speeding”	Driver exceeded the permissible max speed limit for longer than 1 minute.	Acknowledge the message. Reduce vehicle speed.
8	“Power supply interruption”	The power supplied to the STC-8255 is below or above the certain limit more than 200 miliseconds or has been disconnected for more than 200 miliseconds. The event is not triggered in Calibration Mode.	Acknowledge the message. Have the STC-8255 checked by authorized workshop.
9	“Motion sensor data error”	The data link failure occurred between the motion sensor and the tachograph.	Acknowledge the message. Have the STC-8255 checked by authorized workshop.
10	“Motion conflict”	Displayed if the second source of motion data (GNSS or CAN-Bus) is not consistent with the primarily used motion sensor data.	Acknowledge the message. Have the STC-8255 checked by authorized workshop.
16	“No further details”	Displayed if an unexpected error occurs while authenticating the motion sensor.	Acknowledge the message. Have the STC-8255 checked by authorized workshop.
17	“Motion sensor auth. failure”	Displayed if the tachograph does not authenticate with the motion sensor	Acknowledge the message. Have the STC-8255 checked by authorized workshop.
	“Card auth. Failure”		

Pictogram & Message ID	Warning (Event) Messages	Explanation	User Action
18		Displayed if the tachograph does not authenticate with the smart card.	Acknowledge the message. Have the STC-8255 checked by authorized workshop.
! 19	"Unauthorised change of MS"	Displayed if unpaired motion sensor is linked to the tachograph paired with another motion sensor.	Acknowledge the message. Have the STC-8255 checked by authorized workshop.
! 20	"Card data integ. err."	Displayed if smart card data is corrupted.	Acknowledge the message. Have the STC-8255 checked by authorized workshop.
! 21	"Data integ. err."	Displayed if the tachograph data is corrupted.	Acknowledge the message. Have the STC-8255 checked by authorized workshop.
! 24	"Hardware sabotage"	Displayed if card(s) ejection by force is detected.	Acknowledge the message. Have the STC-8255 checked by authorized workshop.
! 32	"No further details"	Displayed if a generic error occurs in the motion sensor.	Acknowledge the message. Have the STC-8255 checked by authorized workshop.
! 33	"Auth. Failure"	Displayed when error related to authentication occurs in the motion sensor.	Acknowledge the message. Have the STC-8255 checked by authorized workshop if this fault occurs frequently.
! 34	"Data integ. err."	Displayed when error related to non-volatile memory occurs in the motion sensor.	Acknowledge the message. Have the STC-8255 checked by authorized workshop if this fault occurs frequently.
! 37	"Hardware sabotage"	Displayed when error related to sensor element or temperature occurs in the motion sensor and the tachograph is not in calibration mode.	Acknowledge the message. Have the STC-8255 checked by authorized workshop if this fault occurs frequently.

Table 13: List of Control Mode Faults

Pictogram & Message ID	Fault Messages	Explanation	User Action
50	"Printer fault"	Displayed when unexpected interrupt is received from printer.	Acknowledge the message. Repeat print request and if necessary switch off / on the ignition.
51	"Display fault"	Displayed if the display self-test fails.	Acknowledge the message. Have the STC-8255 checked by authorized workshop.
1 2	"No further details"	Displayed because of generic smart card fault.	

64			Acknowledge the message. Check if the card contact surface is clean. If it is not, clean the contacts and insert it again.
 157	“Internal fault”	Displayed if there is problem while the tachograph is rebooting.	Acknowledge the message. Have the STC-8255 checked by authorized workshop.
 158	“Sensor fault”	Displayed when error related to controller memory occurs in the motion sensor.	Acknowledge the message. Have the STC-8255 checked by authorized workshop if this fault occurs frequently.
 159	“Sensor fault”	Displayed when error related to controller instruction occurs in the motion sensor.	Acknowledge the message. Have the STC-8255 checked by authorized workshop if this fault occurs frequently.
 160	“Sensor fault”	Displayed when error related communication occurs in the motion sensor.	Acknowledge the message. Have the STC-8255 checked by authorized workshop if this fault occurs frequently.
 161	“Internal fault”	Displayed when there is motion sensor authentication failure.	Acknowledge the message. Have the STC-8255 checked by authorized workshop if this fault occurs frequently.
 162	“Internal fault”	Displayed when the smart card power-up voltage is invalid.	Acknowledge the message. Have the STC-8255 checked by authorized workshop if this fault occurs frequently.
 163	“Internal fault”	Displayed if the smart card protocol is unsupported.	Acknowledge the message. Have the STC-8255 checked by authorized workshop if this fault occurs frequently.
 164	“Internal fault”	Displayed if the smart card status is invalid slot.	Acknowledge the message. Have the STC-8255 checked by authorized workshop if this fault occurs frequently.
 165	“Internal fault”	Displayed if the smart card status is unexpected value.	Acknowledge the message. Have the STC-8255 checked by authorized workshop if this fault occurs frequently.
 166	“Internal fault”	Displayed if the user data storage is insufficient.	Acknowledge the message. Have the STC-8255 checked by authorized workshop.
 169	“Internal fault”	Displayed if the internal battery voltage is below critical level.	Acknowledge the message. Have the STC-8255 checked by authorized workshop.
 173	“Clock Failure”	Displayed when there is a fault in the real time clock (RTC) module of the tachograph and recorded to the device as a DTC record.	Acknowledge the message. Have the STC-8255 checked by authorized workshop.
 175	“Internal fault”	Displayed if software integrity failure occurs.	Acknowledge the message. Have the STC-8255 checked by authorized workshop.
 176	“Internal fault”	Displayed if software integrity failure occurs.	Acknowledge the message. Have the STC-8255 checked by authorized workshop.

177	"Internal fault"	Displayed if software integrity failure occurs.	Acknowledge the message. Have the STC-8255 checked by authorized workshop.

Acoustic Alerts

STC-8255 generates acoustic alerts (beep tone) to warn user about some warning situations (events) or irrelevant user entries. Acoustic alerts can be heard on these events:

1. Overspeed,
2. DAGS warning,
3. Pre-warnings,
4. Irrelevant user entries,
5. Buzzer test.

Calibration Mode (Workshop Role)

By inserting a workshop card and entering the respective PIN, the STC-8255 digital tachograph enters the CALIBRATION mode. This section is a brief description of the CALIBRATION mode and its access rights gained on this mode. By entering the calibration mode, the user has gained access rights to service menu, which is located under the main menu.

Note That!

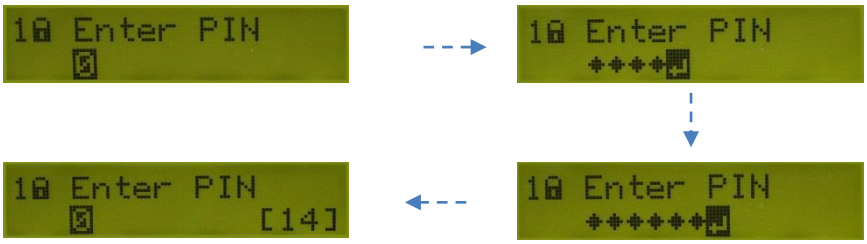
*Please refer to the "*Workshop (Preparation) Manual of the STC-8255*" for further information.*

PIN Entering

After insertion of a valid workshop card into one of the slots, the STC-8255 requests the entry of the respective PIN. The PIN may be 4 to 8 characters long and may consist of numbers (0-9), upper-case letters (A-Z), lower case letters (a-z) or special characters (@#$).

To enter the PIN,

1. Select the desired character by using control ◄ / ► keys. Then press **OK** to assign the character. After assigning the cursor moves to the next character and previously assigned character will be shown as "*". Repeat this procedure until the all charaecters in the PIN is entered. After 4 characters are entered, the carriage return symbol appears. Press **OK** to confirm the PIN or if it is more then 4 characters continue on selecting the desired characters by using control ◄ / ► keys.



2. To correct the entered character, use backspace ◄| key to move back to previous cursor. When the entered character becomes visible, select the desired character as described above. Press **OK** until the cursor stops on the carriage return symbol. Pressing **OK** again on the carriage return symbol confirms the whole PIN entry. The card will be read and the PIN will be checked.

3. If PIN entry is not validated, the PIN entry procedure starts again from beginning showing the number of remaining attempts.

Calibration

Vehicle specific parameters will be programmed into the STC-8255 data memory in the calibration procedure. Vehicle parameters include vehicle identification (VIN, VRN and Registering Member State) and vehicle characteristics (w, k, l, tyre size, speed limiting, device setting (if applicable), current UTC time, current odometer value). The front 6-Pin interface is used for calibration purposes in the Calibration Mode.

➔ The calibration procedure is explained in the STC-8255 Workshop (preparational) Manual in detail. Please refer to that document for more explanations.

Calibration of the STC-8255 digital tachograph requires calibration equipment (sold separately in the market and compliant with the protocol defined in the latest amended legislative document 3821/85 Annex-1B, Appendix-8). Aselsan recommends to use Digital Tachograph Calibration Device (STKC-8250) to calibrate the STC-8255.

Built-in Test

By entering the Calibration Mode, the user has gained access rights to Service Menu, which is located under the Main menu. Under the Service Menu, a test menu exists. Workshop staff can perform some built-in tests (e.g, backlight test, dead pixel test, keypad test, printing test, buzzer test, battery level test, data and SW integrity test, card reader test, data download tests and GNSS test if hardware configuration is suitable) by using this functionality of the STC-8255.

Test Drive

Workshop cards can be used like driver cards for test driving purposes besides their usual workshop (calibration and test) purposes. In that case, make sure that the workshop card is inserted to the driver slot. The same entries are required as for a driver card while insertion (manual entries) or withdrawal (end place entry) of the workshop cards.

Displaying Vehicle Stored and Card Data

Displaying procedure of vehicle stored data and card stored data in workshop mode is similar to the procedure that is explained in the control mode, therefore will not be explained again.

➔ *Please refer to "Control mode - Displaying Stored and Card Data" part on page 55.*

Printing Vehicle Stored and Card Data

Printing procedure of vehicle stored data and card stored data in workshop mode is similar to the procedure that is explained in the control mode, therefore will not be explained again.

➔ *Please refer to "Control mode – Printing Stored Data" part on page 59.*

Downloading Vehicle Stored and Card Data

Downloading procedure of vehicle stored data and card stored data in workshop mode is similar to the procedure that is explained in the company mode, therefore will not be explained again.

➔ *Please refer to "Company mode – Downloading Mass Storage (Vehicle Data)" part on page 46.*

Messages, Warnings and Faults of Calibration Mode

There are four types of messages that can be shown on the display.

- **Messages** – give information on the process or remind the driver. Messages can not be stored and printed. Press the backspace button or **OK** button to clear the message.
- **Pre-warnings** – appear as early reminders to the warnings. Pre-warnings are not stored and can not be printed. Press **OK** button twice to clear the Pre-warning.
- **Warnings (Events)** – appear in the event of e.g. overspeed or violation of the law or if the tachograph can not be recording. Warnings are stored and can be printed. Press **OK** button twice to acknowledge the Warning.
- **Faults** – are more critical than warnings and are displayed if there is a fault detected in the tachograph, in the sensor or cards. On the other hand, fault messages are shown if the device detects a tampering. Faults are stored and can be printed. Press **OK** button twice to acknowledge the fault.

Table 14: List of Calibration Mode Messages

Pictogram & Message ID	Message	Explanation	User Action
 1	"non-valid card"	A non-valid or expired card has been inserted to one of the slots.	Acknowledge the message. Eject the non-valid or expired card if it is not ejected automatically.
 134	"Wrong card type"	Displayed if the inserted card into the tachograph is not a tachograph card. The card inserted is not accepted and ejected automatically.	Please insert a valid tachograph card.
 136	"Recording Inconsistent"	Displayed if there is an inconsistency with the driver or workshop card records while inserting the cards to any slot.	Acknowledge the message.
 	"Drive Cont." xxhxx	Displayed if the weekly allowable driving time is exceeded.	Acknowledge the message. Give a break.
 	"Rest Cont." xxhxx	Displays the remaining time for the upcoming weekly rest period.	Acknowledge the message. Prepare for the rest period.
 	"Drive Cont." xxhxx	Displayed if the daily allowable driving time is exceeded.	Acknowledge the message. Give a break.
 	"Rest Cont." xxhxx xxhxx	Displayed the remaining time for the upcoming daily rest period.	Acknowledge the message. Prepare for the rest period.
 	"Give a break" xxhxx xxhxx	Displayed after 4h15m or 4h30m continuous driving time.	Acknowledge the message. Give a break.
 	"Drive Cont." "xxhxx"	Displayed if the 2 weekly allowable driving time is exceeded.	Acknowledge the message. Give a break.

Pictogram & Message ID	Message	Explanation	User Action
!  171	"No personalization"	Displayed if the card is inserted into the not personalized tachograph.	Acknowledge the message. Have the STC-8255 checked by authorized workshop.
!  172	"Tachograph is not ready"	Displayed if the tachograph is not ready when the card is inserted into the tachograph.	Acknowledge the message. Try again after 1 minute. Have the STC-8255 checked by authorized workshop if this message is displayed frequently.
!  174	"Eject driver card"	Displayed if there is a valid driver card in any slots while the ignition is turning to OFF.	Acknowledge the message. Do not forget to withdraw your driver card from the tachograph.
!  174	"Veh. is not rdy to driv."	Displayed if the ignition is ON and there is no valid driver card in the first slot or there is a smart card rather than driver card in the second slot.	Acknowledge the message. Do not forget to insert your driver card to the first slot before you start your trip.
! 	"Power is insufficient"	Displayed if the power is insufficient when the printing process is started or during the printing.	Acknowledge the message. Have your vehicle battery checked in the authorized services.
! 	"Card is not ready"	Displayed if the card reading is not finished for both card slots before the printing process is started.	Acknowledge the message. Wait for a while for the card reading process.
	"Stop to print"	Appears when the vehicle moves while the printer is busy with printing.	Stop the vehicle and try to print again when the vehicle is stationary.
	"No paper"	Displayed when the printer is out of paper at the beginning of the printing task.	Load new roll of paper to the printer drawer.
	"No data"	Displayed when there is no data for the requested printout type.	Acknowledge the message.
	"Thermal stress"	Displayed if the printer is too hot and the user requests to print any report.	Acknowledge the message. Please wait for a while for the printer head to cool down and try again later.
	"Thermal stress" "Cancel"	Displayed if the printer has continuously printed and the printer head is too hot to continue. Disappears when the user pushes the backspace button or the print head cools down.	Wait for the printer head cools down and printer continues printing job or press backspace to cancel the job. Please do not touch the printer head with your hand when this message appears.
	"Load paper" -Progress Bar-	Displayed when the printer is out of paper while printing and displayed until new printer paper roll is inserted (45 sec period).	Insert new printer paper roll.
	"Printer is ready" -Progress Bar-	Displayed after the new roll of paper is loaded to the printer drawer.	Load new roll of paper to the printer drawer.
	"Cancel printing" Yes – No Dialog	Displayed when the printer is not loaded with a new roll of paper in a certain period (45sec).	Press YES to cancel the printing job.

Pictogram & Message ID	Message	Explanation	User Action
	“not possible”	Displayed when the user is trying to enter the menu while the vehicle is moving.	Stop the vehicle if necessary to navigate in the menus.
	“Printing”	Appears when the printer is busy with printing.	No user action required.
	“Ejection is not possible”	Displayed if the cards inserted to the tachograph slots are requested while the battery supply level is too low.	Acknowledge the message. Have your vehicle battery checked in the authorized services.
	“Ejection is not possible”	Displayed if the cards inserted to the tachograph slots are requested while the vehicle is moving.	Please stop the vehicle and then request the card withdrawal from the tachograph.
	“UTC correct. not possible”	Displayed if you are trying to correct the UTC time which has already been corrected within the last 7 days.	No user action required.
	“Wait for ejection”	Displayed if the cards inserted to the tachograph slots are requested while the inserted activity is recorded to the tachograph cards.	Acknowledge the message. Please try again one minute later.

Table 15: List of Calibration Mode Pre-warnings

Pictogram & Message ID	Pre-Warning Messages	Explanation	User Action
>> 129	“Oversp. prev. warning”	The vehicle has exceeded the overspeed limit which is set for that vehicle. If exceeding continues for 30 sec. (or pre-defined value) the overspeed event will be recorded into tachograph memory.	Acknowledge the message. Reduce the vehicle speed.
130	“Card will expire soon”	Means expiration prewarning day limit of card inserted to the tachograph slot is reached and this card will be expired soon.	Acknowledge the message. Contact the responsible authority to get new one.
131	“Time for service”	Displayed if the calibration pre-warning date is passed and reminds the driver.	Acknowledge the message. Have the STC-8255 checked and calibrated by authorized workshop.
146	“Download card”	Displayed if the card download pre-warning date is passed and reminds the driver to download his/her card to external media.	Acknowledge the message. Download the driver card to any external media.
147	“Download vehicle unit”	Displayed if the tachograph download pre-warning date is passed and reminds the driver to download his/her tachograph to external media.	Acknowledge the message. Download the tachograph mass memory to any external media.

Table 16: List of Calibration Mode Warnings

Pictogram & Message ID	Warning (Event) Messages	Explanation	User Action
	“Card conflict”		

Pictogram & Message ID	Warning (Event) Messages	Explanation	User Action
2		Displayed if the two tachograph cards incompatible with each other are inserted to the tachograph.	Acknowledge the message. Remove one of the cards (which is useless) from the card slot.
 3	“Time overlap”	The last withdrawal date/time of the inserted driver card from previous tachograph is later than the date/time of recently inserted tachograph.	Acknowledge the message. Check the date/time of the tachographs used and determine the tachograph with the incorrect UTC time, make the tachograph time to be corrected by authorized workshop. Or, wait for the overlap period to be elapsed.
 4	“Driving w/o valid card”	Driving started with an inappropriate card or with an inappropriate card combination.	Acknowledge the message. Stop vehicle and insert a valid driver card or remove inappropriate card from co-driver slot.
 5	“Card insertion while driving”	A driver card was inserted after driving has begun.	Acknowledge the message. No further action required.
 6	“Card closed incorrectly”	Card was not removed properly from the last tachograph or data could not be saved to the card correctly.	Acknowledge the message. No further action required.
 7	“Over speeding”	Driver exceeded the permissible max speed limit for longer than 1 minute.	Acknowledge the message. Reduce vehicle speed.
 9	“Motion sensor data error”	The data link failure occurred between the motion sensor and the tachograph.	Acknowledge the message. Have the STC-8255 checked by authorized workshop.
 10	“Motion conflict”	Displayed if the second source of motion data (GNSS or CAN-Bus) is not consistent with the primarily used motion sensor data.	Acknowledge the message. Have the STC-8255 checked by authorized workshop.
 16	“No further details”	Displayed if an unexpected error occurs while authenticating the motion sensor.	Acknowledge the message. Have the STC-8255 checked by authorized workshop.
 17	“Motion sensor auth. failure”	Displayed if the tachograph does not authenticate with the motion sensor	Acknowledge the message. Have the STC-8255 checked by authorized workshop.
 18	“Card auth. Failure”	Displayed if the tachograph does not authenticate with the smart card.	Acknowledge the message. Have the STC-8255 checked by authorized workshop.
 20	“Card data integ. err.”	Displayed if smart card data is corrupted.	Acknowledge the message. Have the STC-8255 checked by authorized workshop.
	“Data integ. err.”		

Pictogram & Message ID	Warning (Event) Messages	Explanation	User Action
21		Displayed if the tachograph data is corrupted.	Acknowledge the message. Have the STC-8255 checked by authorized workshop.
! 32	"No further details"	Displayed if a generic error occurs in the motion sensor.	Acknowledge the message. Have the STC-8255 checked by authorized workshop.
! 33	"Auth. Failure"	Displayed when error related to authentication occurs in the motion sensor.	Acknowledge the message. Have the STC-8255 checked by authorized workshop if this fault occurs frequently.
! 34	"Data integ. err."	Displayed when error related to non-volatile memory occurs in the motion sensor.	Acknowledge the message. Have the STC-8255 checked by authorized workshop if this fault occurs frequently.

Table 17: List of Calibration Mode Faults

Pictogram & Message ID	Fault Messages	Explanation	User Action
50	"Printer fault"	Displayed when unexpected interrupt is received from printer.	Acknowledge the message. Repeat print request and if necessary switch off / on the ignition.
51	"Display fault"	Displayed if the display self-test fails.	Acknowledge the message. Have the STC-8255 checked by authorized workshop.
64	"No further details"	Displayed because of generic smart card fault.	Acknowledge the message. Check if the card contact surface is clean. If it is not, clean the contacts and insert it again.
157	"Internal fault"	Displayed if there is problem while the tachograph is rebooting.	Acknowledge the message. Have the STC-8255 checked by authorized workshop.
158	"Sensor fault"	Displayed when error related to controller memory occurs in the motion sensor.	Acknowledge the message. Have the STC-8255 checked by authorized workshop if this fault occurs frequently.
159	"Sensor fault"	Displayed when error related to controller instruction occurs in the motion sensor.	Acknowledge the message. Have the STC-8255 checked by authorized workshop if this fault occurs frequently.
160	"Sensor fault"	Displayed when error related communication occurs in the motion sensor.	Acknowledge the message. Have the STC-8255 checked by authorized workshop if this fault occurs frequently.
	"Internal fault"		

161		Displayed when there is motion sensor authentication failure.	Acknowledge the message. Have the STC-8255 checked by authorized workshop if this fault occurs frequently.
162	"Internal fault"	Displayed when the smart card power-up voltage is invalid.	Acknowledge the message. Have the STC-8255 checked by authorized workshop if this fault occurs frequently.
163	"Internal fault"	Displayed if the smart card protocol is unsupported.	Acknowledge the message. Have the STC-8255 checked by authorized workshop if this fault occurs frequently.
164	"Internal fault"	Displayed if the smart card status is invalid slot.	Acknowledge the message. Have the STC-8255 checked by authorized workshop if this fault occurs frequently.
165	"Internal fault"	Displayed if the smart card status is unexpected value.	Acknowledge the message. Have the STC-8255 checked by authorized workshop if this fault occurs frequently.
166	"Internal fault"	Displayed if the user data storage is insufficient.	Acknowledge the message. Have the STC-8255 checked by authorized workshop.
169	"Internal fault"	Displayed if the internal battery voltage is below critical level.	Acknowledge the message. Have the STC-8255 checked by authorized workshop.
173	"Clock Failure"	Displayed when there is a fault in the real time clock (RTC) module of the tachograph and recorded to the device as a DTC record.	Acknowledge the message. Have the STC-8255 checked by authorized workshop.
175	"Internal fault"	Displayed if software integrity failure occurs.	Acknowledge the message. Have the STC-8255 checked by authorized workshop.
176	"Internal fault"	Displayed if software integrity failure occurs.	Acknowledge the message. Have the STC-8255 checked by authorized workshop.
177	"Internal fault"	Displayed if software integrity failure occurs.	Acknowledge the message. Have the STC-8255 checked by authorized workshop.

Acoustic Alerts

STC-8255 generates acoustic alerts (beep tone) to warn user about some warning situations (events) or irrelevant user entries. Acoustic alerts can be heard on these events:

1. Overspeed,
2. DAGS warning,
3. Pre-warnings,
4. Irrelevant user entries,
5. Buzzer test.

Unknown Equipment

Unknown equipment refers to a tachograph card or a motion sensor, which can't be authenticated when inserted into the VU.

If a tachograph card is not authenticated after inserted into the VU, STC-8255 automatically ejects the card after five authentication attempts and the VU user becomes UNKNOWN.

In such cases the user needs to apply to card issuing authority for the card renewal at *www.staum.org.tr*.

The messages, warnings and faults of unknown equipment are the same as unknown role.

Motion Sensor

Motion sensor is a part of the recording equipment, providing a signal reprensentative of vehicle speed and/or distance travelled. A motion sensor possesses valid credentials for its authentication and their validity is verifiable. If the motion sensor is not authenticated to the VU, the pairing process can not be completed successfully. In this situation, the connected motion sensor, if not paired with vehicle unit, will fail for the subsequent authentication checks resulting with "motion data error" or "unauthorized change of motion sensor" (if VU is already paired with another motion sensor). These warnings are important and recorded to the VU memory for the controller inspections.

In such a situation, it is recomended to bring STC-8255 to authorized workshops.

The messages, warnings and faults of Motion Sensor are the same as Unknown User role.

Unknown User (Role)

Unknown user refers to the role of the VU user when no card inserted, the card inserted is not authenticated by the VU or the card inserted is expired. If no card is inserted into the VU, the mode of operation becomes operational mode. However, since the card is absent in the VU, the VU can not read/write, print or download card data. In addition, the vehicle data can not be downloaded in this mode.

When an expired tachograph card is inserted into the VU, only the card information can be shown, printed or downloaded while the VU user role becomes UNKNOWN.

➔ *Please refer to the "Downloading Card Data" explanations given on page 30.*
➔ *Please refer to the "Displaying Card Data" explanations given on page 31.*
➔ *Please refer to the "Printing Card Data" explanations given on page 32.*

In such cases the user needs to apply to card issuing authority for the card renewal at *www.staum.org.tr.*

➔ Please refer to "Operational Mode" on page 19, for the other operations except from card operations.

Messages, Warnings and Faults of Unknown Role

There are four types of messages that can be shown on the display.

- **Messages –** give information on the process or remind the driver. Messages can not be stored and printed. Press the backspace button or **OK** button to clear the message.
- **Pre-warnings –** appear as early reminders to the warnings. Pre-warnings are not stored and can not be printed. Press **OK** button twice to clear the Pre-warning.
- **Warnings (Events) –** appear in the event of e.g. overspeed or violation of the law or if the tachograph can not be recording. Warnings are stored and can be printed. Press **OK** button twice to acknowledge the Warning.
- **Faults –** are more critical than warnings and are displayed if there is a fault detected in the tachograph, in the sensor or cards. On the other hand, fault messages are shown if the device detects a tampering. Faults are stored and can be printed. Press **OK** button twice to acknowledge the fault.

Table 18: List of Unknown User Mode Messages

Pictogram & Message ID	Message	Explanation	User Action
 1	“non-valid card”	A non-valid or expired card has been inserted to one of the slots.	Acknowledge the message. Eject the non-valid or expired card if it is not ejected automatically.
 134	“Wrong card type”	Displayed if the inserted card into the tachograph is not a tachograph card. The card inserted is not accepted and ejected automatically.	Please insert a valid tachograph card.
 135	“PIN is blocked”	Displayed when workshop card is blocked because of wrong card pin entries.	Acknowledge the message. Apply for the new service card to “STAUM”. https://staum.org.tr.
 136	“Recording Inconsistent”	Displayed if there is an inconsistency with the driver or workshop card records while inserting the cards to any slot.	Acknowledge the message.
 144	“Upgrade failed”	Displayed when software upgrade is failed.	Acknowledge the message.
 	“Drive Cont.” xxhxx	Displayed if the weekly allowable driving time is exceeded.	Acknowledge the message. Give a break.
 	“Rest Cont.” xxhxx	Displays the remaining time for the upcoming weekly rest period.	Acknowledge the message. Prepare for the rest period.
 	“Drive Cont.” xxhxx	Displayed if the daily allowable driving time is exceeded.	Acknowledge the message. Give a break.
 	“Rest Cont.” xxhxx xxhxx	Displays the remaining time for the upcoming daily rest period.	Acknowledge the message. Prepare for the rest period.
 	“Give a break” xxhxx xxhxx	Displayed after 4h15m or 4h30m continuous driving time.	Acknowledge the message. Give a break.
 	“Drive Cont.” “xxhxx”	Displayed if the 2 weekly allowable driving time is exceeded.	Acknowledge the message. Give a break.
 171	“No personalization”	Displayed if the card is inserted into the not personalized tachograph.	Acknowledge the message. Have the STC-8255 checked by authorized workshop.
 172	“Tachograph is not ready”	Displayed if the tachograph is not ready when the card is inserted into the tachograph.	Acknowledge the message. Try again after 1 minute. Have the STC-8255 checked by authorized workshop if this message is displayed frequently.

Pictogram & Message ID	Message	Explanation	User Action
! 174	“Eject driver card”	Displayed if there is a valid driver card in any slots while the ignition is turning to OFF.	Acknowledge the message. Do not forget to withdraw your driver card from the tachograph.
! 174	“Veh. is not rdy to driv.”	Displayed if the ignition is ON and there is no valid driver card in the first slot or there is a smart card rather than driver card in the second slot.	Acknowledge the message. Do not forget to insert your driver card to the first slot before you start your trip.
!	“Power is insufficient”	Displayed if the power is insufficient when the printing process is started or during the printing.	Acknowledge the message. Have your vehicle battery checked in the authorized services.
!	“Card is not ready”	Displayed if the card reading is not finished for both card slots before the printing process is started.	Acknowledge the message. Wait for a while for the card reading process.
	“Stop to print”	Appears when the vehicle moves while the printer is busy with printing.	Stop the vehicle and try to print again when the vehicle is stationary.
	“No paper”	Displayed when the printer is out of paper at the beginning of the printing task.	Load new roll of paper to the printer drawer.
	“No data”	Displayed when there is no data for the requested printout type.	Acknowledge the message.
	“Thermal stress”	Displayed if the printer is too hot and the user requests to print any report.	Acknowledge the message. Please wait for a while for the printer head to cool down and try again later.
 	“Thermal stress” “Cancel”	Displayed if the printer has continuously printed and the printer head is too hot to continue. Disappears when the user pushes the backspace button or the print head cools down.	Wait for the printer head cools down and printer continues printing job or press backspace to cancel the job. Please do not touch the printer head with your hand when this message appears.
	“Load paper” -Progress Bar-	Displayed when the printer is out of paper while printing and displayed until new printer paper roll is inserted (45 sec period).	Insert new printer paper roll.
	“Printer is ready” -Progress Bar-	Displayed after the new roll of paper is loaded to the printer drawer.	Load new roll of paper to the printer drawer.
	“Cancel printing” Yes – No Dialog	Displayed when the printer is not loaded with a new roll of paper in a certain period (45sec).	Press YES to cancel the printing job.
	“not possible”	Displayed when the user is trying to enter the menu while the vehicle is moving.	Stop the vehicle if necessary to navigate in the menus.
	“Printing”	Appears when the printer is busy with printing.	No user action required.
!	“Ejection is not possible”	Displayed if the cards inserted to the tachograph slots are requested while the battery supply level is too low.	Acknowledge the message. Have your vehicle battery checked in the authorized services.
1 2	“Ejection is not possible”	Displayed if the cards inserted to the tachograph slots are requested while the vehicle is moving.	Please stop the vehicle and then request the card withdrawal from the tachograph.

Pictogram & Message ID	Message	Explanation	User Action
!	“UTC correct. not possible”	Displayed if you are trying to correct the UTC time which has already been corrected within the last 7 days.	No user action required.
⏏	“Wait for ejection”	Displayed if the cards inserted to the tachograph slots are requested while the inserted activity is recorded to the tachograph cards.	Acknowledge the message. Please try again one minute later.

Table 19: List of Unknown User Mode Pre-warnings

Pictogram & Message ID	Pre-Warning Messages	Explanation	User Action
>> 129	“Oversp. prev. warning”	The vehicle has exceeded the overspeed limit which is set for that vehicle. If exceeding continues for 30 sec. (or pre-defined value) the overspeed event will be recorded into tachograph memory.	Acknowledge the message. Reduce the vehicle speed.
 130	“Card will expire soon”	Means expiration prewarning day limit of card inserted to the tachograph slot is reached and this card will be expired soon.	Acknowledge the message. Contact the responsible authority to get new one.
 131	“Time for service”	Displayed if the calibration pre-warning date is passed and reminds the driver.	Acknowledge the message. Have the STC-8255 checked and calibrated by authorized workshop.
 146	“Download card”	Displayed if the card download pre-warning date is passed and reminds the driver to download his/her card to external media.	Acknowledge the message. Download the driver card to any external media.
 147	“Download vehicle unit”	Displayed if the tachograph download pre-warning date is passed and reminds the driver to download his/her tachograph to external media.	Acknowledge the message. Download the tachograph mass memory to any external media.

Table 20: List of Unknown User Mode Warnings

Pictogram & Message ID	Warning (Event) Messages	Explanation	User Action
!  2	“Card conflict”	Displayed if the two tachograph cards incompatible with each other are inserted to the tachograph.	Acknowledge the message. Remove one of the cards (which is useless) from the card slot.
! 3	“Time overlap”	The last withdrawal date/time of the inserted driver card from previous tachograph is later than the date/time of recently inserted tachograph.	Acknowledge the message. Check the date/time of the tachographs used and determine the tachograph with the incorrect UTC time, make the tachograph time to be corrected by authorized workshop.

Pictogram & Message ID	Warning (Event) Messages	Explanation	User Action
			Or, wait for the overlap period to be elapsed.
 4	“Driving w/o valid card”	Driving started with an inappropriate card or with an inappropriate card combination.	Acknowledge the message. Stop vehicle and insert a valid driver card or remove inappropriate card from co-driver slot.
 5	“Card insertion while driving”	A driver card was inserted after driving has begun.	Acknowledge the message. No further action required.
 6	“Card closed incorrectly”	Card was not removed properly from the last tachograph or data could not be saved to the card correctly.	Acknowledge the message. No further action required.
 7	“Over speeding”	Driver exceeded the permissible max speed limit for longer than 1 minute.	Acknowledge the message. Reduce vehicle speed.
 8	“Power supply interruption”	The power supplied to the STC-8255 is below or above the certain limit more than 200 miliseconds or has been disconnected for more than 200 miliseconds. The event is not triggered in Calibration Mode.	Acknowledge the message. Have the STC-8255 checked by authorized workshop.
 9	“Motion sensor data error”	The data link failure occurred between the motion sensor and the tachograph.	Acknowledge the message. Have the STC-8255 checked by authorized workshop.
 10	“Motion conflict”	Displayed if the second source of motion data (GNSS or CAN-Bus) is not consistent with the primarily used motion sensor data.	Acknowledge the message. Have the STC-8255 checked by authorized workshop.
 16	“No further details”	Displayed if an unexpected error occurs while authenticating the motion sensor.	Acknowledge the message. Have the STC-8255 checked by authorized workshop.
 17	“Motion sensor auth. failure”	Displayed if the tachograph does not authenticate with the motion sensor	Acknowledge the message. Have the STC-8255 checked by authorized workshop.
 18	“Card auth. Failure”	Displayed if the tachograph does not authenticate with the smart card.	Acknowledge the message. Have the STC-8255 checked by authorized workshop.
 19	“Unauthorised change of MS”	Displayed if unpaired motion sensor is linked to the tachograph paired with another motion sensor.	Acknowledge the message. Have the STC-8255 checked by authorized workshop.
 20	“Card data integ. err.”	Displayed if smart card data is corrupted.	Acknowledge the message. Have the STC-8255 checked by authorized workshop.
 21	“Data integ. err.”	Displayed if the tachograph data is corrupted.	Acknowledge the message. Have the STC-8255 checked by authorized workshop.
	“Hardware sabotage”		

Pictogram & Message ID	Warning (Event) Messages	Explanation	User Action
24		Displayed if card(s) ejection by force is detected.	Acknowledge the message. Have the STC-8255 checked by authorized workshop.
! 32	“No further details”	Displayed if a generic error occurs in the motion sensor.	Acknowledge the message. Have the STC-8255 checked by authorized workshop.
! 33	“Auth. Failure”	Displayed when error related to authentication occurs in the motion sensor.	Acknowledge the message. Have the STC-8255 checked by authorized workshop if this fault occurs frequently.
! 34	“Data integ. err.”	Displayed when error related to non-volatile memory occurs in the motion sensor.	Acknowledge the message. Have the STC-8255 checked by authorized workshop if this fault occurs frequently.
! 37	“Hardware sabotage”	Displayed when error related to sensor element or temperature occurs in the motion sensor and the tachograph is not in calibration mode.	Acknowledge the message. Have the STC-8255 checked by authorized workshop if this fault occurs frequently.

Table 21: List of Unknown User Mode Faults

Pictogram & Message ID	Fault Messages	Explanation	User Action
 50	“Printer fault”	Displayed when unexpected interrupt is received from printer.	Acknowledge the message. Repeat print request and if necessary switch off / on the ignition.
 51	“Display fault”	Displayed if the display self-test fails.	Acknowledge the message. Have the STC-8255 checked by authorized workshop.
 64	"No further details"	Displayed because of generic smart card fault.	Acknowledge the message. Check if the card contact surface is clean. If it is not, clean the contacts and insert it again.
 157	“Internal fault”	Displayed if there is problem while the tachograph is rebooting.	Acknowledge the message. Have the STC-8255 checked by authorized workshop.
 158	“Sensor fault”	Displayed when error related to controller memory occurs in the motion sensor.	Acknowledge the message. Have the STC-8255 checked by authorized workshop if this fault occurs frequently.
 159	“Sensor fault”	Displayed when error related to controller instruction occurs in the motion sensor.	Acknowledge the message. Have the STC-8255 checked by authorized workshop if this fault occurs frequently.
 160	“Sensor fault”	Displayed when error related communication occurs in the motion sensor.	Acknowledge the message. Have the STC-8255 checked by authorized workshop if this fault occurs frequently.
 161	“Internal fault”	Displayed when there is motion sensor authentication failure.	Acknowledge the message. Have the STC-8255 checked by authorized workshop if this fault occurs frequently.
 162	“Internal fault”	Displayed when the smart card power-up voltage is invalid.	Acknowledge the message. Have the STC-8255 checked by authorized workshop if this fault occurs frequently.
 163	“Internal fault”	Displayed if the smart card protocol is unsupported.	Acknowledge the message. Have the STC-8255 checked by authorized workshop if this fault occurs frequently.
 164	“Internal fault”	Displayed if the smart card status is invalid slot.	Acknowledge the message. Have the STC-8255 checked by authorized workshop if this fault occurs frequently.
 165	“Internal fault”	Displayed if the smart card status is unexpected value.	Acknowledge the message. Have the STC-8255 checked by authorized workshop if this fault occurs frequently.
 166	“Internal fault”	Displayed if the user data storage is insufficient.	Acknowledge the message. Have the STC-8255 checked by authorized workshop.

 169	"Internal fault"	Displayed if the internal battery voltage is below critical level.	Acknowledge the message. Have the STC-8255 checked by authorized workshop.
 173	"Clock Failure"	Displayed when there is a fault in the real time clock (RTC) module of the tachograph and recorded to the device as a DTC record.	Acknowledge the message. Have the STC-8255 checked by authorized workshop.
 175	"Internal fault"	Displayed if software integrity failure occurs.	Acknowledge the message. Have the STC-8255 checked by authorized workshop.
 176	"Internal fault"	Displayed if software integrity failure occurs.	Acknowledge the message. Have the STC-8255 checked by authorized workshop.
 177	"Internal fault"	Displayed if software integrity failure occurs.	Acknowledge the message. Have the STC-8255 checked by authorized workshop.

Acoustic Alerts

STC-8255 generates acoustic alerts (beep tone) to warn user about some warning situations (events) or irrelevant user entries. Acoustic alerts can be heard on these events:

1. Overspeed,
2. DAGS warning,
3. Pre-warnings,
4. Irrelevant user entries,
5. Buzzer test.

GENERAL SETTINGS

Language

The STC-8255 supports 17 (seventeen) languages at the moment. The language which is used for the display depends on the tachograph card which is inserted to the driver card slot. The language is automatically adjusted upon card insertion or withdrawal. However, it can be selected manually from the vehicle menu.

To perform manual language selection:

1. Press **OK** to enter main menu. Select **Vehicle menu Language** by using control ‹ / › keys. Press **OK** to enter the menu. Select the desired language you want to set. Press **OK** to activate the current language selection. After confirmation STC-8255 returns one level back.



Display Backlight Level

STC-8255 backlight level can be adjusted manually from the menu if the backlight source is selected as Menu from Service Configuration settings. Backlight level can also be adjusted from the A2 illumination Pin or from the Cabin Illumination of the vehicle. These settings will be explained in the "*Workshop Manual*" in detail. Please refer to that documentation for further information.

To adjust display backlight level:

1. Press **OK** to enter main menu. Select **Vehicle menu>Entry** by using control ‹ / › keys. Press **OK** to enter the menu. Select **Entry Backlight lev.** by using control ‹ / › keys. Press **OK** to enter the menu. The current backlight level is displayed. Adjust the backlight level to the desired level by using control ‹ / › keys. After adjusting the desired LCD backlight level, press **OK** to adjust the Keypad backlight level to the desired level by using control ‹ / › keys. Press **OK** to save current settings. After confirmation, STC-8255 returns one level back.

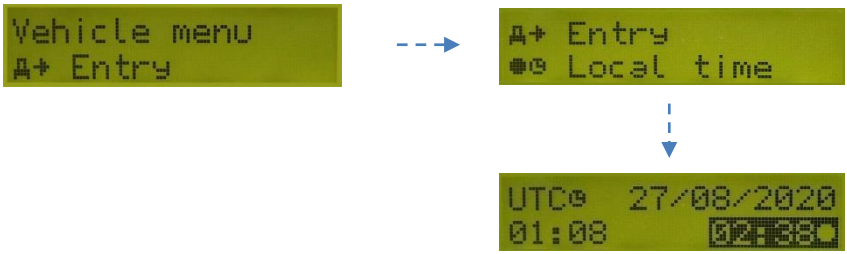


Local Time

In STC-8255 standart (default) display, local time is only shown as information and may be adjusted according to the current time in a specific country. Local time can be adjusted manualy in steps of ±30 minutes from the menu.

To adjust the **local time**, do as follows:

1. Press **OK** to enter main menu. Select **Vehicle menu>Entry** by using control ◄ / ► keys. Press **OK** to enter the menu. Select **Entry Local time** by using control ◄ / ► keys. Press **OK** to enter the menu. The current UTC and Local time is displayed. Adjust the local time to the desired value in steps of ±30 minutes by using control ◄ / ► keys. Press **OK** to save current settings. After confirmation, STC-8255 returns one level back.



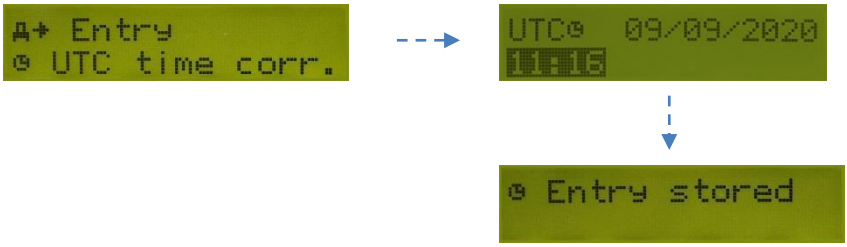
➔ *Please refer to "National / Regional abbreviations and European time zones" on page 106.*

UTC Correction

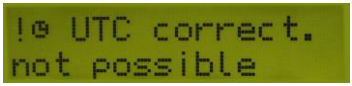
All of the tachograph activities are registered to the memory in UTC time (Universal Time Coordinated). You can correct the UTC time by ±1 minute per week. Greater deviations can only be changed by an authorized workshop with calibration process.

To make **UTC correction**, follow the steps:

1. Press **OK** to enter main menu. Select **Vehicle menu Entry** by using control ◄ / ► keys. Press **OK** to enter the menu. Select **Entry UTC time corr.** by using control ◄ / ► keys. Press **OK** to enter the menu. The current UTC time is displayed. Adjust the UTC time to the desired value in steps of ±1 minute by using control ◄ / ► keys. Press **OK** to save current settings. After confirmation STC-8255 returns one level back.



If you try to correct the UTC time within 7 days, and you will get the following message on the display.



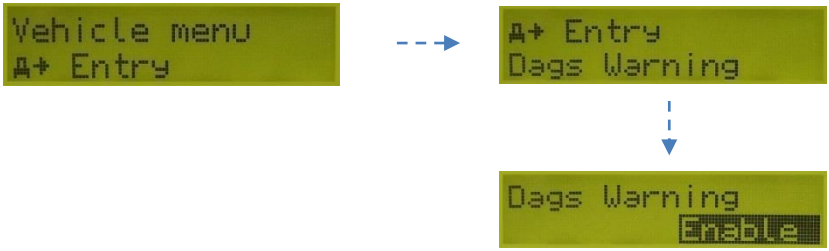
DAGS Warning

Driver Activity Guidance System (DAGS) warning, reminds the driver to give a break after continuous driving time of 4h15min, 4h30min, daily, weekly and fortnightly driving and upcoming resting times . DAGS Warning of the STC-8255 can be enabled or disabled manually from this menu. If the DAGS buzzer control is enabled from **Service menu>General Configurations>DAGS buzzer cnt**. settings, it gives an audible alert to the user when DAGS warning system is enabled.

➔ Please refer to "Section 8.2.1.19 DAGS Buzzer Control" for adjusting buzzer control in Preperation (Workshop) Manual.

To adjust ***Vehicle menu>Entry>DAGS warning***, do as follows:

1. Press **OK** to enter main menu. Select ***Vehicle menu>Entry*** by using control ‹ / › keys. Press **OK** to enter the menu. Select ***DAGS warning*** by using control ‹ / › keys. Press **OK** to enter the menu. ***Enable*** or ***Disable*** the DAGS warning to the desired status by using control ‹ / › keys. Press **OK** to save current settings. After confirmation, STC-8255 returns one level back.

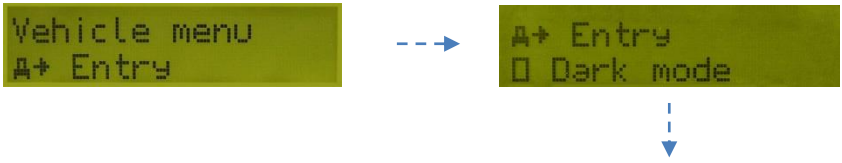


Dark Mode

The STC-8255's contrast of LCD can be altered by using Dark mode menu.

To enable ***Vehicle menu>Entry>Dark Mode***, do as follows:

1. Press **OK** to enter main menu. Select ***Vehicle menu>Entry*** by using control ‹ / › keys. Press **OK** to enter the menu. Select ***Dark Mode*** by using control ‹ / › keys. Press **OK** to enter the menu. ***Enable*** or ***Disable*** the desired mode by using control ‹ / › keys. Press **OK** to save current settings. After confirmation, STC-8255 returns one level back.



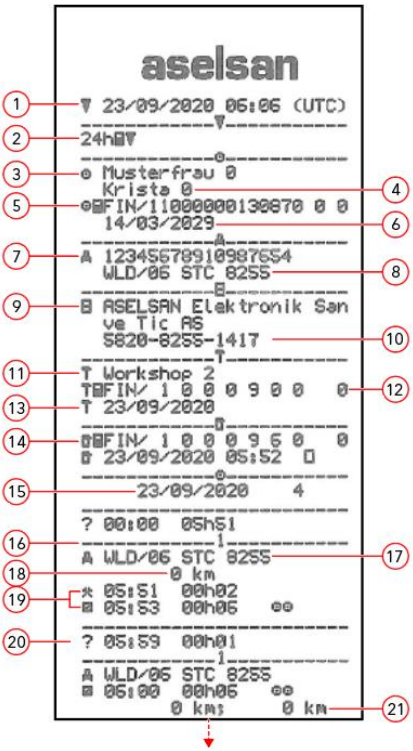


PICTOGRAM COMBINATIONS

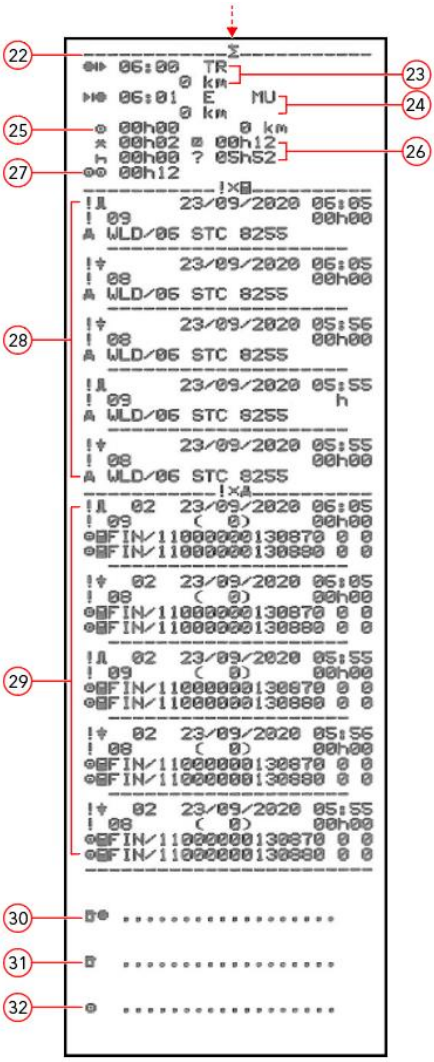
Symbol	Meaning / Description
	Control Place
	Location start of daily work period
	Location end of daily work period
	From time
	To time
	From vehicle
OUT	Out of scope begin
OUT	Out of scope end
	Driver card
	Company card
	Control card
	Workshop card
	No card
	Crew driving
	Driving time for one week
	Driving time for two weeks
24h	Daily driver activity printout from smartcard
24h	Daily driver activity printout from tachograph memory
!	Printouts of events and faults from smartcard
!	Printouts of events and faults from tachograph memory
	Technical data printout
>>	Over speeding (speed infringment) printout
!	Insertion of a non valid smartcard
!	Card conflict
!	Time overlap
!	Driving without appropriate card
!	Card insertion while driving
!	Last card session not correctly closed

Symbol	Meaning / Description
	Over speeding
	Power supply interrruption
	Motion data error
	Security breach
	Time adjustment (by workshop)
	Over speeding control
	Vehicle motion conflict
	Card fault (driver slot)
	Card fault (co-driver slot)
	Display fault
	Downloading fault
	Printer fault
	Sensor fault
	Vu internal fault
	Still the same daily work period?
	End of previous work period?
	Confirm or enter location of end of work period
	Enter start time
	Enter location of start of work period
	LSE Clock Failure Warning
	Company Entry
	Display GNSS
	Print D1/D2
	Display D1/D2
	Print V-Profile
	Display V-Profile
	Print N-Profile
	Display N-Profile
	Print Vehicle V-Graph
	Display Card Activity Graph
	Daily Rest Counter Warning
	Two Week Driving Warning
	Display Version Info

24h Driver Card Printout



1 Date and time of printout (UTC).
2 Type of printout (24h, card).
3 Card holder's surname.
4 Card holder's first name.
5 Country and card identification number.
6 Expiry date of the card inserted.
7 Vehicle identification, VIN.
8 Registering member state and Vehicle Registration Number (VRN).
9 Tachograph manufacturer's name.
10 Tachograph part number.
11 Workshop name where the last calibration is made.
12 Workshop card identification number.
13 Date of the last calibration.
14 Last control the inspected driver has been subjected to.
15 Enquiry date and daily card presence counter.
16 Slot where card was inserted.
17 Vehicle Registration Number (VRN), for the vehicle where the driver card was inserted.
18 Vehicle odometer at card insertion.
19 Activities while driver card inserted, start and duration time, crew status.
20 Activity while driver card not inserted to any tachograph, start and duration time.
21 Card withdrawal: Vehicle odometer and distance travelled since last insertion for which odometer is known.



22 Daily summary of activities.

23 Begin country entering time, country (if selected country is Spain, region is placed after country) and odometer value on enter.

24 End country entering time, country (if selected country is Spain, region is placed after country) and odometer value on enter.

25 Driving activity duration and trip distance.

26 Other activities' durations.

27 Driving crew status duration.

28 Last five events and faults stored in the driver card.

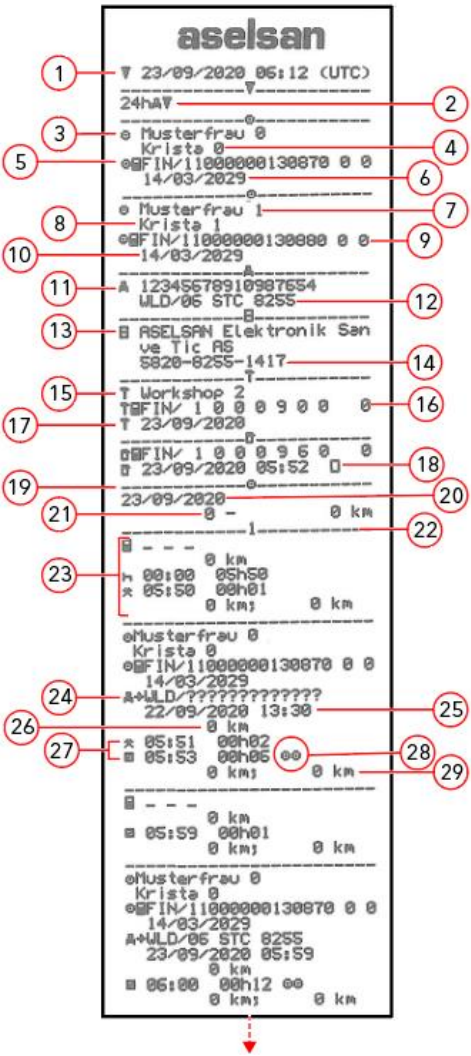
29 Last five events and faults stored in the vehicle unit (VU).

30 Control place.

31 Controller´s signature.

32 Driver´s signature.

24h Vehicle Unit Printout



1 Date and time of printout (UTC).

2 Type of printout (24h, VU).

3 Card holder's surname (driver).

4 Card holder's first name (driver).

5 Country and card identification number.

6 Expiry date of the driver card.

7 Card holder's surname (co-driver).

8 Card holder's first name (co-driver).

9 Country and card identification number (co-driver).

10 Expiry date of the co-driver card.

11 Vehicle identification, VIN.

12 Registering member state and Vehicle Registration Number (VRN).

13 Tachograph manufacturer.

14 Tachograph part number.

15 Workshop name where the last calibration is made.

16 Workshop card identification number.

17 Date of the last calibration.

18 Date and time, and type of the last control.

19 Drivers' activities stored in the VU per slot in chronological order.

20 Enquiry date.

21 Vehicle odometer at 00:00 and 23:59 (if the printout refers records of present day, right-hand odometer value is for present time).

22 Driver slot indicator.

23 Driver's activities when there is no card in driver slot.

24 Registration member state and vehicle registration number of previous vehicle used.

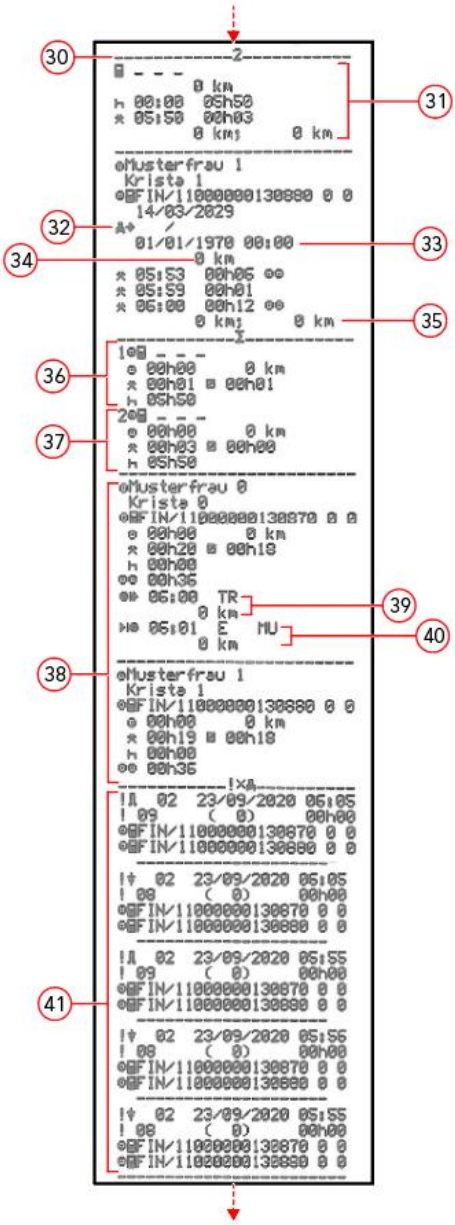
25 Date and time of card withdrawal from previous vehicle.

26 Vehicle odometer at card insertion.

27 Activities with start and duration time.

28 Activity status as crew.

29 Vehicle odometer at card withdrawal and trip distance during card is inserted in the slot.



30 Co-driver slot indicator.

31 Co-driver's activities when there is no card in co-driver slot.

32 Registration member state and vehicle registration number of previous vehicle used.

33 Date and time of card withdrawal from previous vehicle.

34 Vehicle odometer at card insertion.

35 Vehicle odometer at card withdrawal and trip distance during card is inserted in the slot.

36 Summary of periods without card in driver slot.

37 Summary of periods without card in co-driver slot.

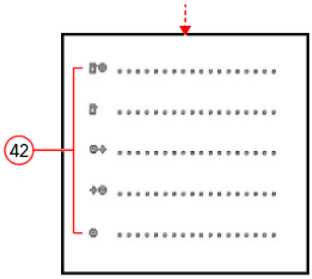
38 Daily summary of activities with card(s) is inserted.

39 Begin country entering time, country (if selected country is Spain, region is placed after country) and odometer value on enter.

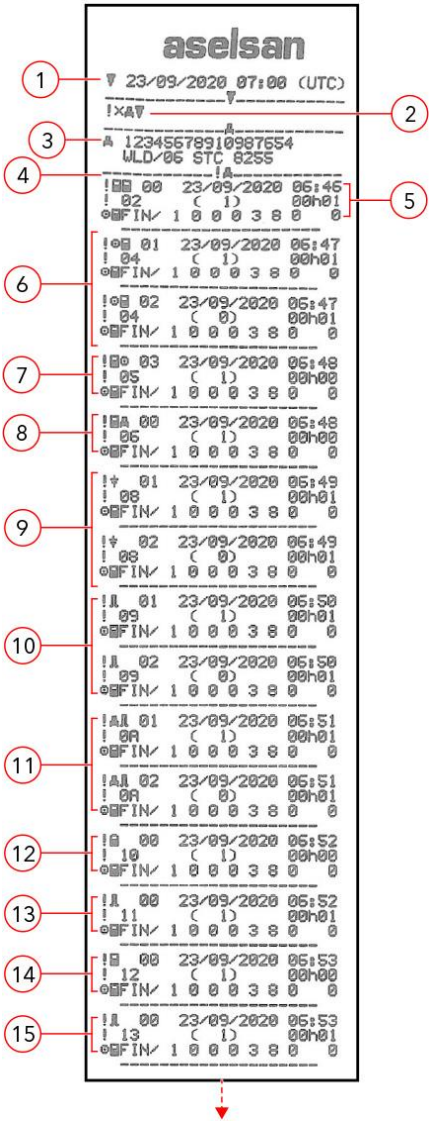
40 End country entering time, country (if selected country is Spain, region is placed after country) and odometer value on enter.

41 Last five events and faults stored in the vehicle unit (VU).

42 Hand-written and signature part.



Events & Faults Vehicle Unit Printout



1 Date and time of printout (UTC).

2 Type of printout (Events, VU).

3 Vehicle Identification Number (VIN), Vehicle Registration Number (VRN) and Registering Member State.

4 Block identifier of events.

5 Card conflict event.

6 Not appropriate card event.

7 Card inserted while driving event.

8 Card session close event.

9 Power supply interruption event.

10 Motion data error event.

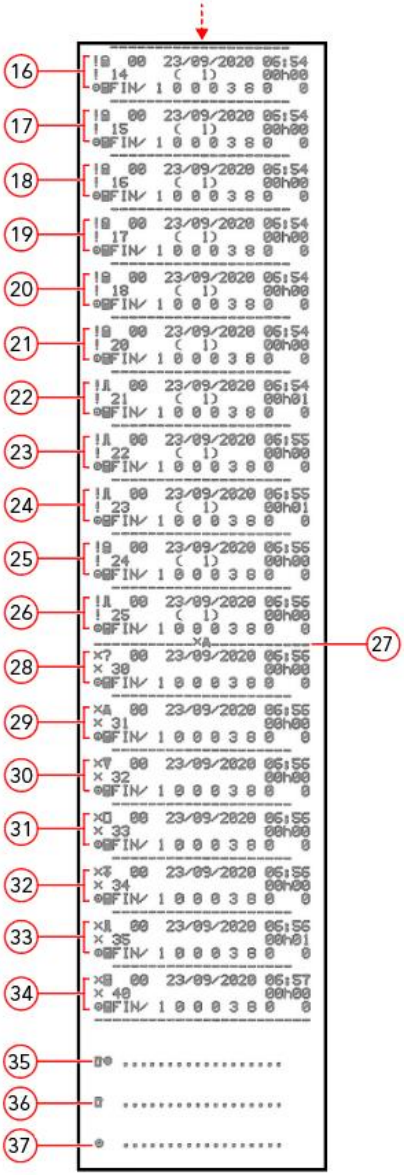
11 Vehicle motion conflict event.

12 No further details event.

13 VU MS auth failure event.

14 Card auth failure event.

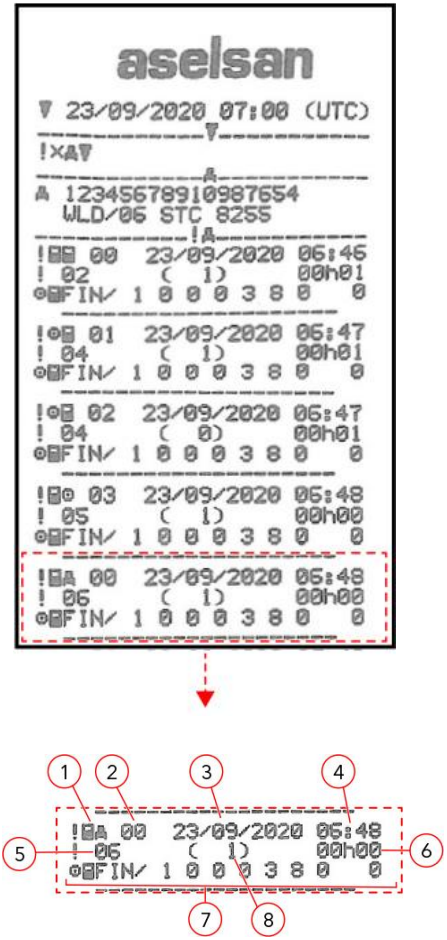
15 Unauth MS change event.



16 Card integrity error event.
17 VU integrity error event.
18 VU internal transfer error event.
19 VU unauth case opening event.
20 VU hardware sabotage event.
21 MS general event.
22 MS auth failure event.
23 MS integrity error event.
24 MS internal transfer error event.
25 MS unauth case opening event.
26 MS hardware sabotage event.
27 Block identifier of faults.
28 VU general fault.
29 VU internal fault.
30 Printer fault.
31 Display fault.
32 Downloading fault.
33 Sensor fault.
34 Card general fault.
35 Control place.
36 Controller's signature.
37 Driver's signature.

➔ *Refer to Workshop Manual (DTVU-PRE) for "List of Event and Fault Codes and Diagnostic Trouble Codes (DTCs)"*

Detailed View of Events & Faults VU Printout



1 Type of event or fault.

2 Event or fault record purpose:

00 - one of the 10 most recent (or last) events or faults.

01 - the longest event for one of the last 10 days of occurrence.

02 - one of the 5 longest events over the last 365 days.

03 - the last event for one of the last 10 days of occurrence.

04 - the most serious event for one of the last 10 days of occurrence.

05 - one of the 5 most serious events over the last 365 days.

06 - the first event or fault having occurred after the last calibration.

07 - an active/on-going event or fault.

3 Date of event or fault.

4 Start time of event or fault.

5 Event or fault code.

6 Duration of event or fault.

7 Card inserted in slot 1 at the beginning of event or fault (Driver card).

8 Number of events of the same type during the day.

0 – For this event, it is not necessary to save "Number of similar events"

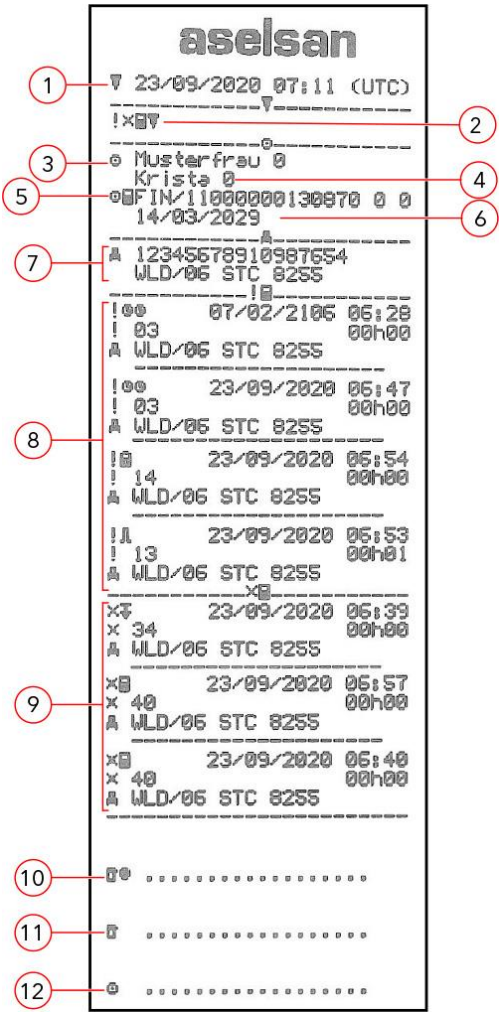
1 – One event of this type occurred on this day.

2 – Two event of this type occurred on this day, but only the one was saved.

n – "n" events of this type occurred on this day, but only the one was saved.

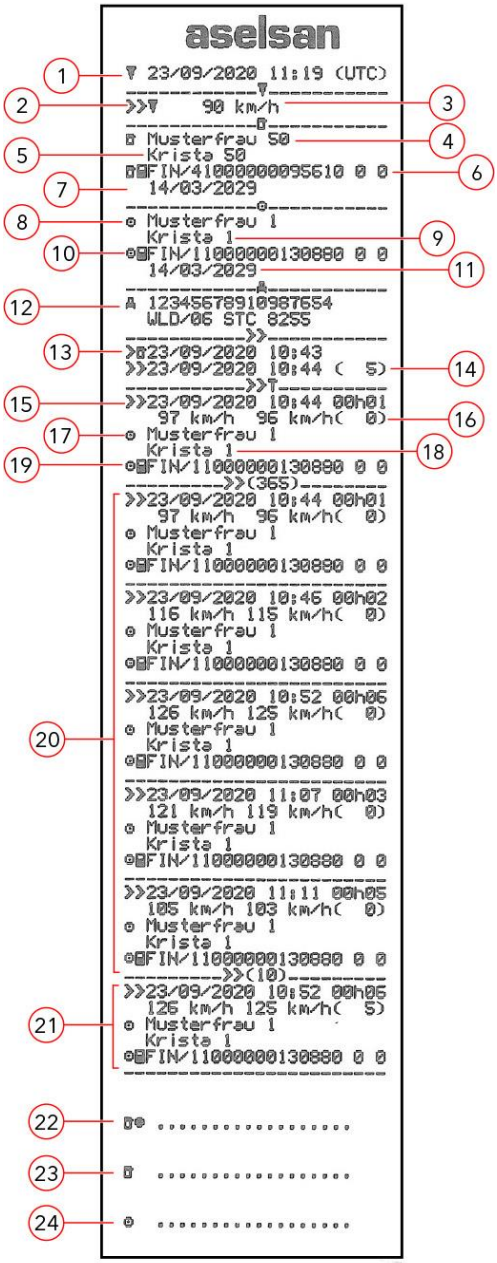
➔ *Refer to Workshop Manual (DTVU-PRE) for "List of Event and Fault Codes and Diagnostic Trouble Codes (DTCs)"*

Events & Faults Card Printout



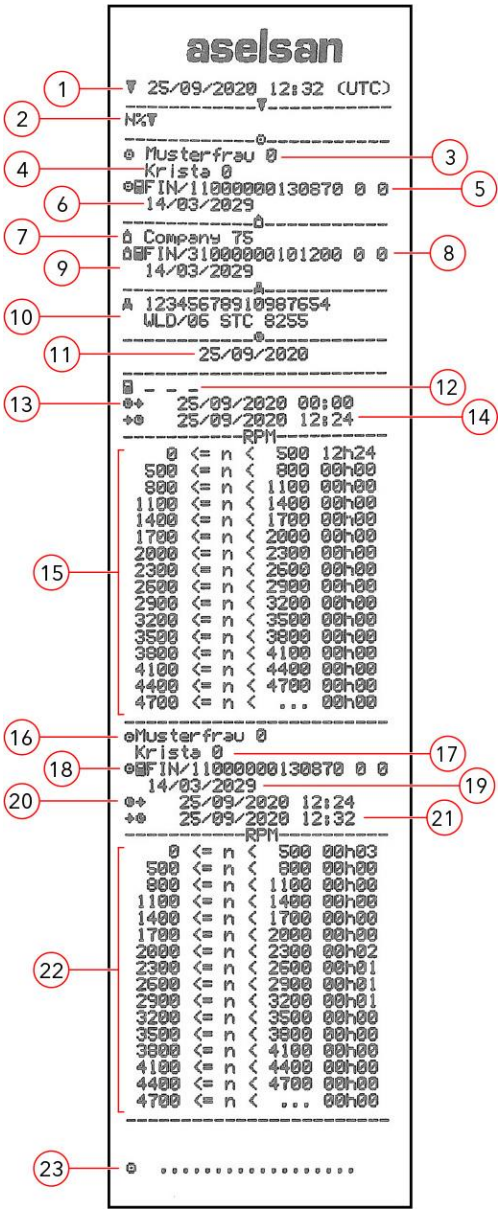
1 Date and time of printout (UTC).
2 Type of printout (Events, card).
3 Card holder's surname.
4 Card holder's first name.
5 Country and card identification number.
6 Expiry date of the card.
7 Vehicle identification. VIN, registering member state and VRN.
8 List of all events stored on the card.
9 List of all faults stored on the card.
10 Control place.
11 Controller´s signature.
12 Driver´s signature.

Overspeeding Printout



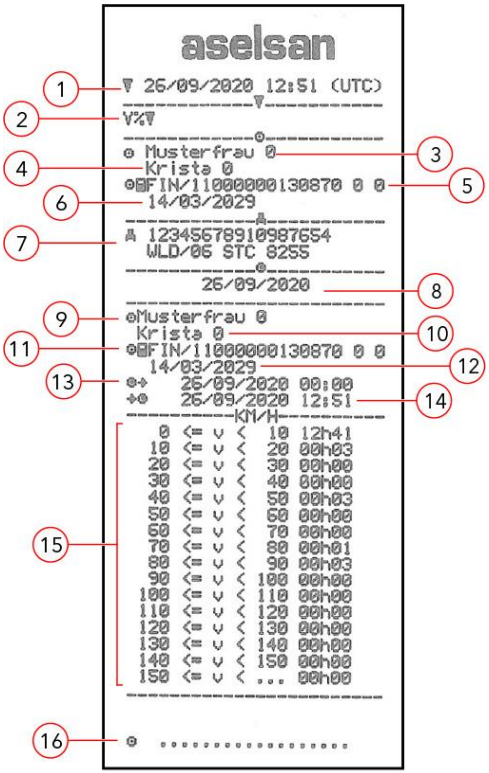
1 Date and time of printout (UTC).
2 Type of printout (overspeeding).
3 Speed limiting setting.
4 Card holder's surname (driver).
5 Card holder's first name (driver).
6 Country and card identification number.
7 Expiry date of the driver card.
8 Card holder's surname (co-driver).
9 Card holder's first name (co-driver).
10 Card and country identification number.
11 Expiry date of the co-driver card.
12 Vehicle Identification Number (VIN), Vehicle Registration Number (VRN) and Registering Member State (RMS).
13 Date and time of first overspeeding control.
14 Date and time of the last overspeeding control and number of overspeeding events.
15 First overspeeding after the last calibration, date time and duration.
16 Max and average speed.
17 Driver card holder's surname.
18 Driver card holder's name.
19 Card and country identification number.
20 Five most serious overspeeding over the last 365 days: Date time and duration, max and average speed, driver and drivers card identification.
21 Five most serious overspeeding over the last ten days: Date time and duration, max and average speed, driver and drivers card identification.
22 Control place.
23 Controller´s signature.
24 Driver´s signature.

N-profile Printout



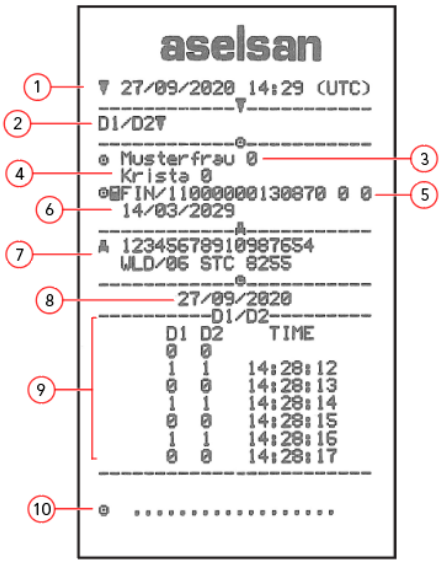
1 Date and time of printout (UTC).
2 Type of printout (N-profile).
3 Card holder's surname (driver).
4 Card holder's first name (driver).
5 Country and card identification number.
6 Expiry date of the driver card.
7 Card holder's name (company).
8 Country and card identification number (company).
9 Expiry date of the company card.
10 Vehicle Identification Number (VIN), Vehicle Registration Number (VRN) and Registering Member State (RMS).
11 Enquiry date.
12 First card slot info during profile recording (there is no card during recording, this area is shown as dashed line).
13 Begin date and time of the recording of rpm profile.
14 End date and time of the recording of rpm profile.
15 List of defined engine revolution levels and durations.
16 Card holder's surname (driver).
17 Card holder's first name (driver).
18 Card and country identification number.
19 Expiry date of the driver card.
20 Begin date and time of the recording of rpm profile.
21 End date and time of the recording of rpm profile.
22 List of defined engine revolution levels and durations.
23 Driver´s signature.

V-profile Printout



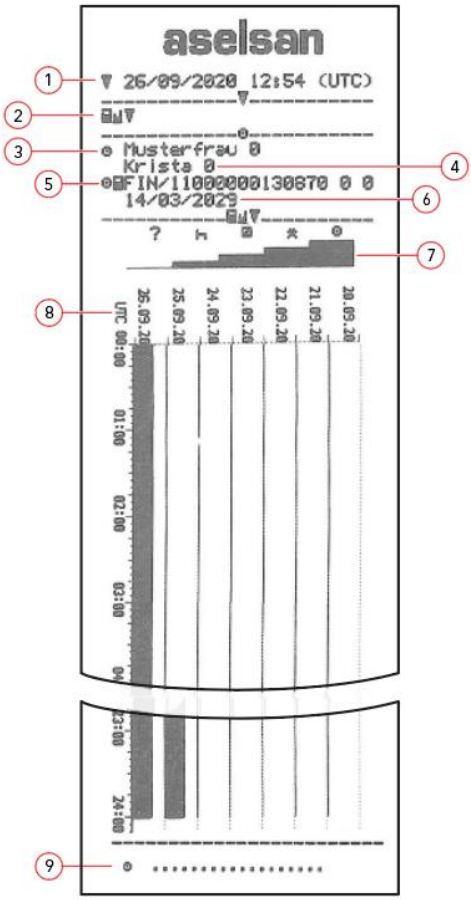
1 Date and time of printout (UTC).
2 Type of printout (V-profile).
3 Card holder's surname (driver).
4 Card holder's first name (driver).
5 Country and card identification number.
6 Expiry date of the driver card.
7 Vehicle Identification Number (VIN), Vehicle Registration Number (VRN) and Registering Member State (RMS).
8 Enquiry date.
9 Card holder's surname (driver).
10 Card holder's first name (driver).
11 Card and country identification number.
12 Expiry date of the driver card.
13 Begin date and time of the recording of speed profile.
14 End date and time of the recording of speed profile.
15 List of defined speed ranges and durations.
16 Driver´s signature.

Status D1-D2 Printout



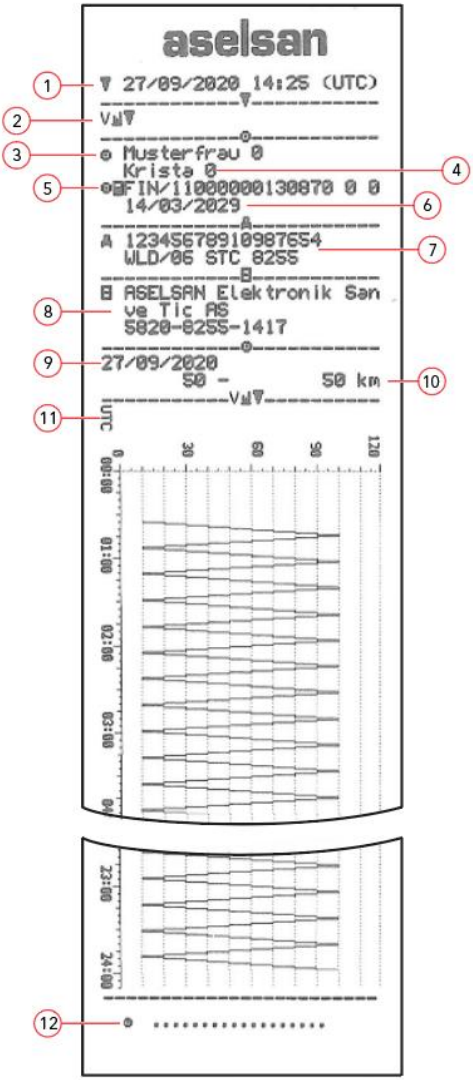
1 Date and time of printout (UTC).
2 Type of printout (Status D1-D2).
3 Card holder's surname (driver).
4 Card holder's first name (driver).
5 Country and card identification number.
6 Expiry date of the driver card.
7 Vehicle Identification Number (VIN), Vehicle Registration Number (VRN) and Registering Member State.
8 Enquiry date.
9 List of status D1-D2.
10 Driver´s signature.

Activitiy Graph Printout



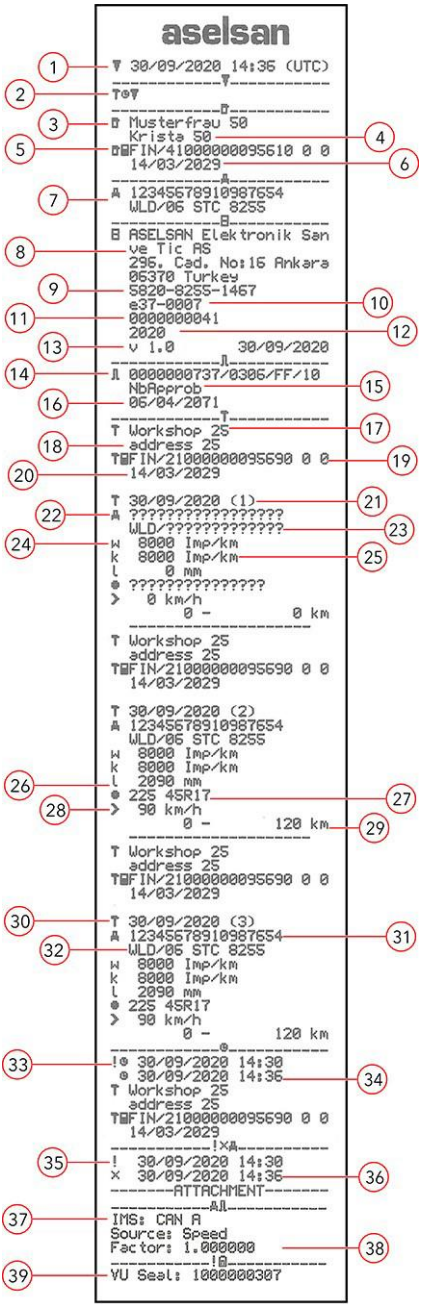
1 Date and time of printout (UTC).
2 Type of printout (Activities).
3 Card holder's surname (driver).
4 Card holder's first name (driver).
5 Country and card identification number.
6 Expiry date of the driver card.
7 Legend of the activity sembols.
8 Activities diagram of the last seven calendar days from the selected day.
9 Driver´s signature.

V-Graph Printout



1 Date and time of printout (UTC).
2 Type of printout (V-graph).
3 Card holder's surname (driver).
4 Card holder's first name (driver).
5 Country and card identification number.
6 Expiry date of the driver card.
7 Vehicle Identification Number (VIN), Vehicle Registration Number (VRN) and Registering Member State.
8 Tachograph manufacturer's name and tachograph part number.
9 Selected day.
10 Odometer values of the beginning (00:00) and end (23:59) of the day (if the printout refers records of present day, right-hand odometer value is for present time).
11 Recording of speed profile of the selected day.
12 Driver´s signature.

Technical Data Printout



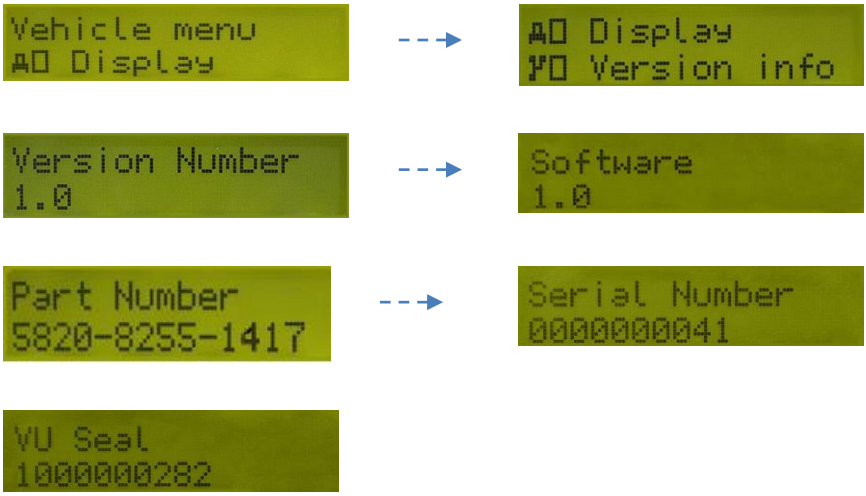
1 Date and time of printout (UTC).
2 Type of printout (technical data).
3 Card holder's surname (driver).
4 Card holder's first name (driver).
5 Country and card identification number.
6 Expiry date of the driver card.
7 Vehicle Identification Number (VIN), Vehicle Registration Number (VRN) and Registering Member State (RMS).
8 Tachograph manufacturer.
9 Tachograph part number.
10 Tachograph approval number.
11 Tachograph serial number.
12 Year of manufacture.
13 Software version and installation date.
14 Motion sensor serial number.
15 Motion sensor approval number.
16 Motion sensor first installation date.
17 Workshop name having performed the calibration.
18 Workshop address.
19 Workshop card identification.
20 Workshop card expiry date.
21 Calibration date and purpose.
22 VIN (shown as question marks before registered).
23 VRN (shown as question marks before registered).
24 Characteristic coefficient of vehicle.
25 Constant of the recording equipment.
26 Effective circumference of wheel tyres.
27 Size of tyres.
28 Authorized speed setting.
29 Old and new odometer values.
30 Calibration date and purpose.
31 VIN.
32 VRN and country of registration.
33 Old date and time.
34 New date and time.
35 Most recent event date time.
36 Most recent fault date time.
37 IMS source (if IMS source is selected as CAN, PGN type is printed under this line).
38 IMS factor.
39 VU seal number.

SOFTWARE INFORMATION

You can display the software & VU version numbers, serial & part number and sealing label serial number defined on the STC-8255.

To display software and VU version, follow the steps:

1. Press **OK** to enter main menu. Select **Vehicle menu>Display** by using control ‹/› keys. Press **OK** to enter the menu. Select **Version info.** by using control ‹/› keys. Press **OK** to display the version info. Use control ‹/› keys to see the software, VU versions, part number, serial number and seal number. Press **OK** or **Backspace** ⌫ to navigate one level back.



NATIONAL / REGIONAL ABBREVIATIONS AND EUROPEAN TIME ZONES

The following table provides an overview of national and regional abbreviations, as well as time zones:

Nation	Abbreviation	Time zone offset
Austria	A	+01:00h
Albania	AL	+01:00h
Andorra	AND	+01:00h
Armenia	ARM	+04:00h
Azerbaijan	AZ	+03:00h
Belgium	B	+01:00h
Bulgaria	BG	+02:00h
Bosnia and Herzegovina	BIH	+01:00h
Belarus	BY	+02:00h
Croatia	HR	+01:00h
Cyprus	CY	+01:00h
Czech Republic	CZ	+01:00h
Denmark	DK	+01:00h
Estonia	EST	+02:00h
Faeroe Islands	FR	+00:00h
Finland	FIN	+02:00h
France	F	+01:00h
Georgia	GE	+02:00h
Germany	D	+01:00h
Greece	GR	+02:00h
Hungary	H	+01:00h
Iceland	IS	+00:00h
Ireland	IRL	+00:00h
Italy	I	+01:00h
Kazakhstan	KZ	+04:00h to +06:00h
Latvia	LV	+02:00h
Lichtenstein	FL	+01:00h
Lithuania	LT	+02:00h
Luxembourg	L	+01:00h
Malta	M	+01:00h
Macedonia	MK	+01:00h
Monaco	MC	+01:00h
Montenegro	MNE	+01:00h
Netherlands	NL	+01:00h
Norway	N	+01:00h

Poland	PL	+01:00h
Portugal	P	+00:00h
Republic of Moldova	MD	+02:00h
Romania	RO	+02:00h
Russian Federation	RUS	+03:00h
San Marino	RSM	+01:00h
Serbia	SRB	+01:00h
Slovakia	SK	+01:00h
Slovenia	SLO	+01:00h
Spain	E	+01:00h
Sweden	S	+01:00h
Switzerland	CH	+01:00h
Tajikistan	TJ	+05:00h
Turkey	TR	+02:00h
Turkmenistan	TM	+05:00h
Ukraine	UA	+02:00h
United Kingdom	UK	+00:00h
Uzbekistan	UZ	+05:00h
Vatican City	V	+01:00h
Yugoslavia	YU	+01:00h
European Community	EC	
Rest of Europe	EUR	
Rest of World	WLD	

REGION ABRREVIATIONS OF SPAIN

Region	Abbreviation
Andalusia	AN
Aragon	AR
Asturias	AST
Cantabria	C
Catalonia	CAT
Castil-Leon	CL
Castile-La Mancha	CM
Valencia	CV
Extremadura	EXT
Galicia	G
Balearic Islands	IB
Canary Islands	IC
La Rioja	LR
Madrid	M
Murcia	MU
Navarra	NA
Basque Community	PV



Before starting to use the STC-8255 digital tachograph, read all related guides and manuals.



Keep the STC-8255 digital tachograph away from dust and humidity. Do not leave the tachograph under direct sun exposure.



Do not drop or force the STC-8255 digital tachograph at the extent, which could make physical damage.



If the STC-8255 digital tachograph is damaged by an accident, inform the company or related authorised bodies providing the serial number and vehicle identification number of the device!

Care of Tachograph

To increase the life-time of your tachograph please be aware of the following:

- Try to keep the tachograph and the environment where it is installed clean and dust-free.
- Clean the plastic box, display and the function keys with a slightly moistened cloth.
- Do not use any alcohol or chemical cleaning solvents to clean any parts of the tachograph.
- Do not insert objects or other cards (rather than tachograph cards, e.g. credit cards) to the card slots otherwise the cards and the tachograph may be damaged.
- Protect the tachograph from direct sunlight.

Care of Cards

Please take care of the cards and keep in mind the following:

- Try to keep the card contacts free from dirt and dust.
- Clean the cards with a slightly moistened soft cleaning cloth if the contacts have grease or oil.
- Do not bend or fold the cards and do not use it for anything other than their primarily usage.
- Protect it from damage and do not use damaged cards in the tachograph.
- Do not let the cards to be under strong electromagnetic field.

WARRANTY AND PRODUCER INFORMATION

The warranty period for STC-8255 digital tachograph is two years.

To get the most out of your purchase, be sure to carefully read this manual and keep it on hand for later reference when needed.



Before using your STC-8255 digital tachograph, please read this operation manual.

Should you encounter any problems with this product, or its features, please refer to this operation manual. If you require further assistance after reading this manual, please contact your local dealer.

For warranty, product service and accessory information, please contact your local dealer or distributor.



Do not attempt to change battery or repair any internal parts by yourself.

This STC-8255 digital tachograph should be opened by authorized personnel only.

Otherwise, the warranty VOIDS.

A tachograph case must never be opened.

No tampering with or modifications to the tachograph system are permitted.

Persons that modify this equipment are committing a punishable offence.

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http://www.aselsan.com.tr

DOCUMENT REVISION HISTORY

Rev. No	Revision Reason	Date of Revision
A	First Release	20/10/2021

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